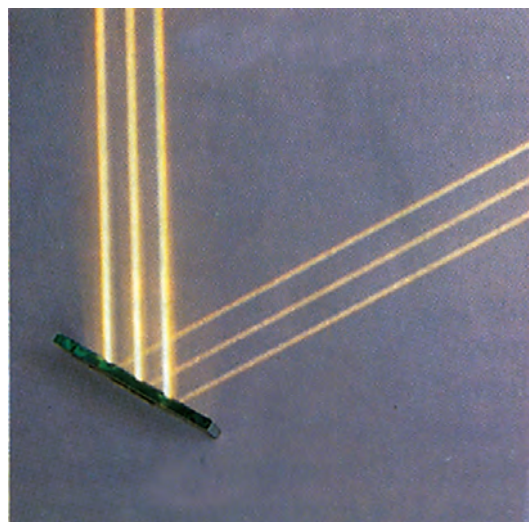
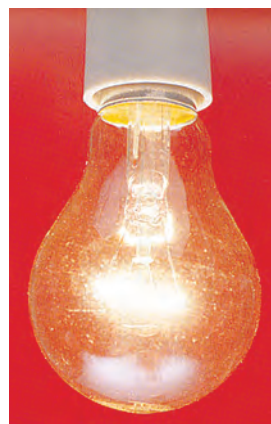




Reflection of Light

Light is a form of energy. Light is needed to see things around us. We are able to see the beautiful world around us because of light. We can read a book, see pictures in a magazine, and watch television and movies due to the existence of light. And it is light which makes us see our image in a looking mirror. **We detect light with our eyes.** Though we see the various objects (or things) around us with our eyes but eyes alone cannot see any object. We also need a source of light to make objects visible. For example, we cannot see any object in a dark room or in the darkness of night because there is no light in a dark room or in the night to make objects visible. But as soon as an electric bulb (a tube-light or a torch, etc.) is switched on and light falls on the objects, we are able to see them clearly. *It is only when light coming from an object enters our eyes that we are able to see that object.* This light may have been *emitted* by the object itself or it may have been *reflected* by the object.

Light enables us to see objects from which it comes or from which it is reflected. For example, the sun gives out light. We can see the sun because the light coming from the sun enters our eyes. **The objects like the sun, other stars, electric bulb, tube-light, torch, candle and fire, etc., which emit their own light are called luminous objects** (see Figure 1). We can see the luminous objects due to the light emitted by them. Though luminous objects are very small in number but they help us to see a large variety of non-luminous objects around us. The objects like a flower, a chair or a table do not have light of their own but even then we are able to see them. This can be explained as follows: Though the objects like a flower, a chair or a table, etc., do not emit light themselves, we can see them by the light which they reflect (or scatter) by taking it from a luminous object like the sun or an electric bulb, etc. (see Figure 2). So, when the sunlight or bulb light falls on a flower or chair (or any other object), some of this light is reflected towards us. And

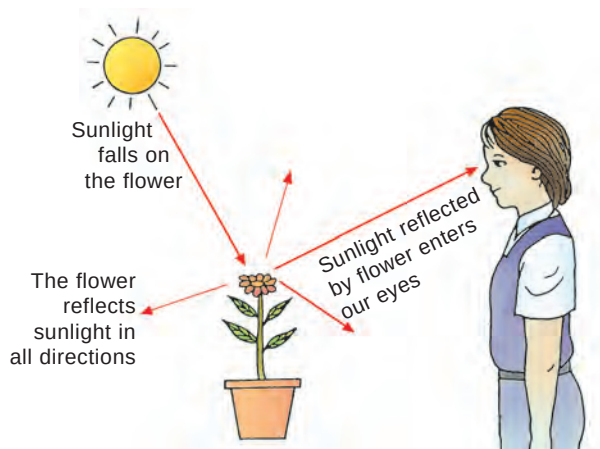


(a) An electric bulb gives its own light. It is a luminous object.

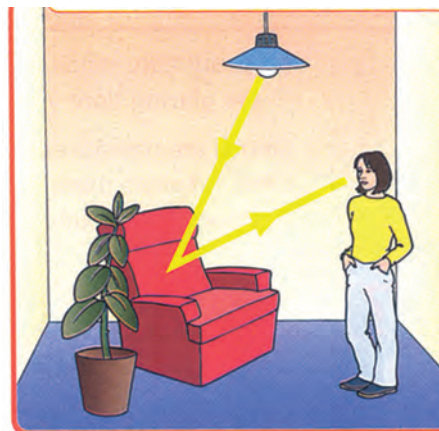


(b) A flower does not give its own light. It is a non-luminous object.

Figure 1. Luminous and non-luminous objects.



(a) We can see the flower during day time because it reflects sunlight falling on it into our eyes.



(b) We can see the chair at night because it reflects bulb light falling on it into our eyes.

Figure 2. We see most of the objects around us by the light reflected from them.

when this reflected light enters our eyes, then we are able to see the flower or chair (because to our eyes, this light appears to be coming from the flower or chair) (see Figure 2). **Those objects which do not emit light themselves but only reflect (or scatter) the light which falls on them, are called non-luminous objects** (see Figure 2). A flower, chair, table, book, trees, other plants, human beings, fan, bed, mirror, diamond, walls, floor, and road, etc., are all non-luminous objects. In fact, *most of the objects around us are non-luminous objects*. **We can see the non-luminous objects because they reflect light (received from a luminous object) into our eyes.** Even the moon is a non-luminous object (because it does not have its own light). We can see the moon because it reflects the sunlight falling on its surface towards us (on the earth).

From the above discussion we conclude that light is a form of energy which causes in us the sensation of sight. **Light travels in straight lines.** The fact that a small source of light casts a sharp shadow of an opaque object tells us that light travels in a straight line path. This is because if light could bend easily and go behind the opaque object, then no shadow could be formed. We will now discuss the nature of light.

Nature of Light

There are two theories about the nature of light : wave theory of light and particle theory of light. **According to wave theory : Light consists of electromagnetic waves which do not require a material medium (like solid, liquid or gas) for their propagation.** The wavelength of visible light waves is very small (being only about 4×10^{-7} m to 8×10^{-7} m). The speed of light waves is very high (being about 3×10^8 metres per second in vacuum). **According to particle theory : Light is composed of particles which travel in a straight line at very high speed.** The elementary particle that defines light is the 'photon'.

Some of the phenomena of light can be explained only if light is considered to be made up of waves whereas others can be explained only if light is thought to be made up of particles. For example, the phenomena of *diffraction* (bending of light around the corners of tiny objects), *interference* and *polarization* of light can only be explained if light is considered to be of wave nature. The particle theory of light cannot explain these phenomena. On the other hand, the phenomena of *reflection* and *refraction* of light, and *casting of shadows* of objects by light, can be explained only if light is thought to be made of particles. Wave theory of light cannot explain these phenomena. Thus, there is evidence for the wave nature of light as well as for particle nature of light.

Physics experiments over the past hundred years or so have demonstrated that **light has a dual nature (double nature) : light exhibits the properties of both waves and particles (depending on the situation it is in).** The modern theory of light called 'Quantum Theory of Light' combines both the wave and particle models of light.

REFLECTION OF LIGHT

When light falls on the surface of an object, some of it is sent back. **The process of sending back the light rays which fall on the surface of an object, is called reflection of light.** The reflection of light is shown in Figure 3. When a beam of light AO falls on a mirror at point O , it is sent back by the mirror in another direction OB (see Figure 3). And we say that the mirror has reflected the beam of light falling on it. We can compare the reflection of light to the bouncing back of a tennis ball on hitting a wall. For example, if we throw a tennis ball at a wall, the ball bounces back. This means that the wall sends it back. Similarly, when light falls on the surface of an object, the object sends the light back. And we say that the object reflects the light. Most of the objects reflect light which falls on them. Some objects reflect more light whereas other objects reflect less light. **The objects having polished, shining surfaces reflect more light than objects having unpolished, dull surfaces.**

We know that an object (say, a chair) kept in a room can be seen from all the parts of the room. This is due to the fact that usually, because of its rough surface, an object reflects light (or scatters light) in all the directions. Since the reflected light reaches all the parts of the room, the object can be seen from all the parts of the room. If, however, the surface of an object is smooth (like that of a mirror), then the light falling on it is reflected in only one direction (as shown in Figure 3).

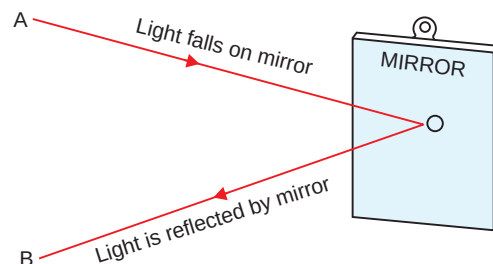
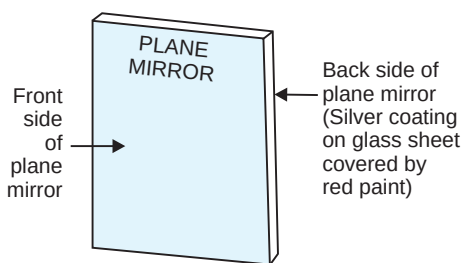
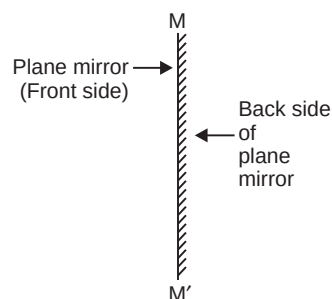


Figure 3. A mirror reflects the light falling on it.

Silver metal is one of the best reflectors of light. For example, a polished block of silver metal reflects almost all the light falling on it and does not transmit any light through it. But the surface of silver metal is easily scratched and soon becomes rough. So, **ordinary mirrors are made by depositing a thin layer of silver metal on the back side of a plane glass sheet.** The silver layer is then protected by a coat of red paint. **The reflection of light in a plane mirror (or any other mirror) takes place at the silver surface in it.** Thus, a plane mirror is a thin, flat and smooth sheet of glass having a shining coating of silver metal on one side. The silver coating is protected by a red paint. These days mirrors are being made increasingly by depositing a thin coating of aluminium metal at the back of a glass sheet (instead of silver coating). This is because aluminium is much cheaper than silver and it reflects light very well. A plane mirror is shown in Figure 4(a). The mirror on our dressing table in which we see our face is a plane mirror.



(a) A plane mirror.



(b) This is how a plane mirror is represented in diagrams.

Figure 4.

In our everyday life, we use plane mirrors of different shapes and sizes depending on where they are being used. But in a science laboratory, a small rectangular strip of plane mirror is usually used for performing experiments on the reflection of light. **In our diagrams, a plane mirror is represented by a straight line having a number of short, oblique lines on one side** [see Figure 4(b)]. The plane side of straight line is the front side of the mirror (where the reflection of light takes place). And the side having short, oblique lines represents the back side of the plane mirror. In Figure 4(b), the straight line MM' (read as : MM -dash) represents a plane mirror.

We will also be using rays of light in constructing the ray-diagrams. **A ray of light is the straight line along which light travels.** The arrow head put on the straight line tells us the direction in which the light is travelling. **A 'bundle of light rays' is called a 'beam of light'.** We will now study the reflection of light from the plane surfaces like that of a plane mirror. This is necessary to understand the reflection of light from spherical mirrors.

REFLECTION OF LIGHT FROM PLANE SURFACES : PLANE MIRROR

When a ray of light falls on a plane mirror (or any other plane surface), it is reflected according to some laws, called the laws of reflection of light. In order to understand the laws of reflection of light, we should first know the meaning of the terms : incident ray, point of incidence, reflected ray, normal (at the point of incidence), angle of incidence, and angle of reflection. So, we will discuss these terms first.

In Figure 5, we have a plane mirror MM' . **The ray of light which falls on the mirror surface is called the incident ray.** In Figure 5, AO is the incident ray of light. The incident ray gives the direction in which light falls on the mirror. **The point at which the incident ray falls on the mirror is called the point of incidence.** In Figure 5, point O on the surface of the mirror is the point of incidence (because the incident ray AO touches the mirror surface at this point). We know that when a ray of light falls on a mirror, the mirror sends it back in another direction and we say that the mirror has reflected the ray of light. **The ray of light which is sent back by the mirror is called the reflected ray.** In Figure 5, OB is the reflected ray of light. The reflected ray of light shows the direction in which the light goes after reflection from the mirror.

Another term that we will be using is the 'normal'. **The 'normal' is a line at right angle to the mirror surface at the point of incidence.** In other words, 'normal' is a line which is perpendicular to the mirror at the point of incidence. In Figure 5, the mirror is MM' and the point of incidence is O . So, the line ON is the normal to the mirror surface at point O . The normal has been represented by a dotted line to distinguish it from the incident ray and the reflected ray. Please note that 'normal' is just a 'perpendicular line' to the mirror, and it should not be called the 'normal ray' like the incident ray or reflected ray. The normal is a line at right angles to the mirror surface. In other words, normal makes an angle of 90° with the surface of the plane mirror. We will now discuss the angle of incidence and the angle of reflection.

The angle of incidence is the angle made by the incident ray with the normal at the point of incidence. In other words, the angle between the incident ray and normal is called the angle of incidence. In Figure 5, the incident ray is AO and the normal is ON , so the angle AON is the angle of incidence. The angle of incidence is denoted by the letter i .

The angle of reflection is the angle made by the reflected ray with the normal at the point of incidence. In other words, the angle between the reflected ray and normal is called the angle of reflection. In Figure 5, the reflected ray is OB and the normal is ON , so the angle NOB is the angle of reflection. The angle of reflection is denoted by the letter r . Keeping these points in mind, we will now describe the laws of reflection of light.

Laws of Reflection of Light

The reflection of light from a plane surface (like that of a plane mirror) or from a spherical surface (like that of a concave mirror or convex mirror) takes place according to two laws, which are known as the laws of reflection of light. The laws of reflection of light are given below :

1. First Law of Reflection. According to the first law of reflection of light : **The incident ray, the reflected ray, and the normal (at the point of incidence), all lie in the same plane.** For example, in Figure 5, the incident ray AO , the reflected ray OB and the normal ON , all lie in

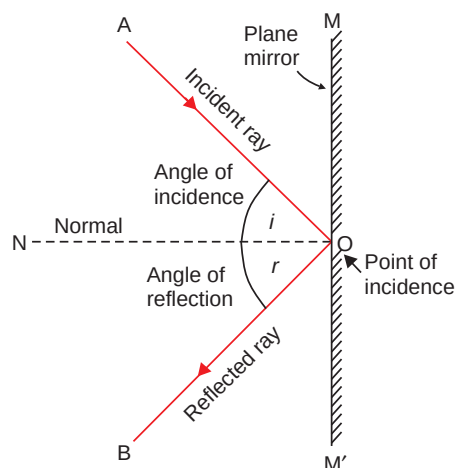


Figure 5. Reflection of light from a plane mirror.

the same plane, the plane of paper. They are neither coming up out of the paper; nor going down into the paper.

2. Second Law of Reflection. According to the second law of reflection of light : **The angle of reflection is always equal to the angle of incidence.** We can also state the second law of reflection of light as follows by writing the angle of incidence first : *The angle of incidence is equal to the angle of reflection.* If the angle of incidence is i and the angle of reflection is r , then according to the second law of reflection :

$$\angle i = \angle r$$

For example, if we measure the angle of reflection NOB in Figure 5, we will find that it is exactly equal to the angle of incidence AON .

The second law of reflection of light means that if the angle of incidence is 35° , then the angle of reflection will also be 35° (see Figure 6). And if the angle of incidence is changed to 45° , then the angle of reflection will also change and become 45° (see Figure 7). *In every case, the angle of reflection remains equal to the angle of incidence.*

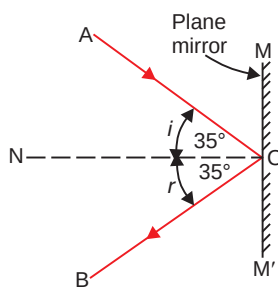


Figure 6. If the angle of incidence is 35° , then the angle of reflection is also 35° .

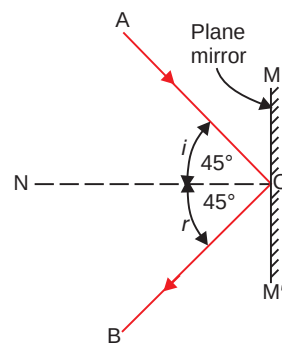


Figure 7. When the angle of incidence is changed to 45° , the angle of reflection also becomes 45° .

We will now describe **what happens when a ray of light falls normally (or perpendicularly) on the surface of a mirror.** If a ray of

light is incident normally on a mirror, it means that it is travelling along the normal to the mirror. So, the angle of incidence (i) for such a ray of light is zero (0). And since the angle of incidence is zero, therefore, according to the second law of reflection, its angle of reflection (r) will also be zero (0). This means that **a ray of light which is incident normally (or perpendicularly) on a mirror, is reflected back along the same path (because the angle of incidence as well as the angle of reflection for such a ray of light are zero).** For example, in Figure 8, a ray of light falls on the plane mirror along the normal NO , therefore, it will be reflected along the same path ON . So, in this case, the incident ray will be NO and the reflected ray will be ON . In other words, the same line represents incident ray, normal and reflected ray. We should, however, put two arrow heads on the same 'normal' line pointing in opposite directions – one arrow to represent incident ray and the other arrow to represent reflected ray (as shown in Figure 8).

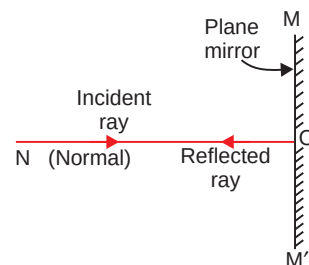


Figure 8. A ray of light incident normally (at 90°) to the mirror surface is reflected back along the same path.

Please note that **the laws of reflection of light apply to all kinds of mirrors, plane mirrors as well as spherical mirrors (like concave mirrors and convex mirrors).** By using the laws of reflection of light, we can find out the nature and position of the images (of objects) formed by the various types of mirrors.

Regular Reflection and Diffuse Reflection of Light

In regular reflection, a parallel beam of incident light is reflected as a parallel beam in one direction. In this case, parallel incident rays remain parallel even after reflection and go only in one direction (see Figure 9). **Regular reflection of light occurs from smooth surfaces like that of a plane mirror (or highly polished metal surfaces).** For example, when a parallel beam of light falls on the smooth surface of a plane mirror, it is reflected as a parallel beam in only one direction as shown in Figure 9. Thus, **a plane mirror produces regular reflection of light.** Images are formed by regular reflection of light. For example, a

smooth surface (like that of a plane mirror) produces a clear image of an object due to regular reflection of light. A highly polished metal surface and a still water surface also produce regular reflection of light and form images. This is why we can see our face in a polished metal object as well as in the still water surface of a pond or lake. A polished wooden table is very smooth and hence produces regular reflection of light.

The regular reflection of light from a smooth surface can be explained as follows : All the particles of a smooth surface (like a plane mirror) are facing in *one direction*. Due to this the angle of incidence for all the parallel rays of light falling on a smooth surface is the same and hence the angle of reflection for all the rays of light is also the same. *Since the angle of incidence and the angle of reflection are the same (or equal), a beam of parallel rays of light falling on a smooth surface is reflected as a beam of parallel light rays in one direction only* (see Figure 9).

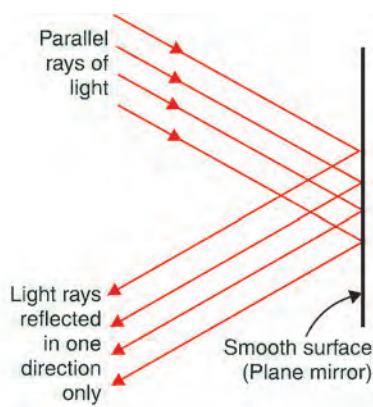


Figure 9. Regular reflection : Incident light is reflected in only one direction.

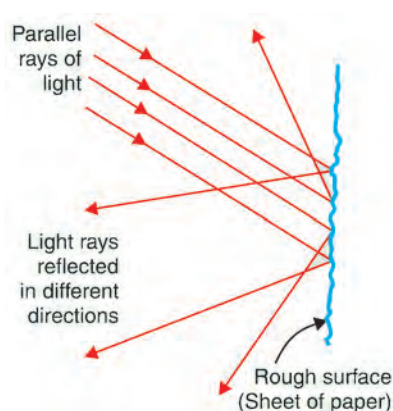


Figure 10. Diffuse reflection : Incident light is reflected in different directions.

In diffuse reflection, a parallel beam of incident light is reflected in different directions. In this case, the parallel incident rays do not remain parallel after reflection, they are scattered in different directions (see Figure 10). The diffuse reflection is also known as *irregular reflection* or scattering. **The diffuse reflection of light takes place from rough surfaces like that of paper, cardboard, chalk, table, chair, walls and unpolished metal objects.** For example, when a parallel beam of light rays falls on the rough surface of a sheet of paper, the light is scattered by making reflected rays in different directions (see Figure 10). Thus, **a sheet of paper produces diffuse reflection of light.** No image is formed in diffuse reflection of light. For example, a rough surface (like that of paper) does not produce an image of the object due to diffuse reflection of light. Actually, the light rays falling on the rough surface of paper are scattered in all directions and hence no image is formed.

The diffuse reflection of light from a rough surface can be explained as follows : The particles of a rough surface (like that of paper) are all facing in *different directions*. Due to this, the angles of incidence for all the parallel rays of light falling on a rough surface are different and hence the angles of reflection for all the rays of light are also different. *Since the angles of incidence and the angles of reflection are different, the parallel rays of light falling on a rough surface go in different directions* (see Figure 10). **Please note that the diffuse reflection of light is not due to the failure of the laws of reflection.** Diffuse reflection is caused by the roughness (or irregularities) in the reflecting surface of an object (like paper or cardboard, etc.). The laws of reflection are valid at each point even on the rough surface of an object.

The surfaces of most of the objects are rough (or uneven) to some extent. So, **most of the objects around us cause diffuse reflection of light and scatter the light falling on them in all directions.** In fact, we can see these objects only because they scatter light rays falling on them in all directions. For example, a book lying on a table can be seen from all parts of the room due to diffuse reflection of light from its surface. The

surface of book, being rough, scatters the incident light in all parts of a room. Hence the book can be seen from all parts of the room. A cinema screen has a rough surface and causes diffuse reflection of light falling on it. The cinema screen receives light from a film projector and scatters it in all directions in the cinema hall so that people sitting anywhere in the hall can see the picture focused on the screen.

Objects and Images

In the study of light, the term 'object' has a special meaning. **Anything which gives out light rays (either its own or reflected by it) is called an object.** A bulb, a candle, a pin-head, an arrow, our face, or a tree, are all examples of objects from the point of view of study of light. The objects can be of two types : very small objects (called point objects) or large objects (called extended objects). In drawing the ray-diagrams for the formation of images, we will be using both type of objects according to our convenience. In ray-diagrams, the point objects (like point sources of light or a pin-head) are represented by a 'dot', and the extended objects are represented by drawing 'an arrow pointing upwards'.

In physics, **image is an optical appearance produced when light rays coming from an object are reflected from a mirror (or refracted through a lens).** This will become clear from the following example. When we look into a mirror, we see our face. What we see in the mirror is actually a 'reflection' of our face and it is called 'image' of our face. Thus, **when we look into a mirror, we see the image of our face in it.** In this case, 'our face' is the 'object' and what we see in the mirror is the 'image'. The image of our face appears to be situated behind the mirror. **While watching a movie in the cinema hall, we see the images of actors and actresses on the cinema screen.** Please note that *an image is formed when the light rays coming from an object meet (or appear to meet) at a point, after reflection from a mirror (or refraction through a lens).* The images are of two types : real images and virtual images. These are discussed below.

Real Images and Virtual Images

The image which *can* be obtained on a screen is called a **real image**. In a cinema hall, we see the images of actors and actresses on the screen. So, **the images formed on a cinema screen is an example of real images** (see Figure 11). A real image is formed when light rays coming from an object actually meet at a point after reflection from a mirror (or refraction through a lens). A real image can be formed on a screen because light rays actually pass through a real image. Real images can be formed by a concave mirror. A convex lens can also form real images. We will study the formation of real images with the help of ray-diagrams after a while.



Figure 11. The image of actors and actresses formed on a cinema screen is a *real* image.



Figure 12. When we look into a mirror, we see our image in it. The image formed in a plane mirror is a *virtual* image.

The image which *cannot* be obtained on a screen is called a **virtual image**. A virtual image can be seen only by looking into a mirror (or a lens). The image of our face formed by a plane mirror cannot be obtained on a screen, it can be seen only by looking into the mirror. So, **the image of our face in a plane mirror is an example of virtual image** (see Figure 12). Virtual images are also called unreal images. A

virtual image is just an illusion. A virtual image is formed when light rays coming from an object only appear to meet at a point when produced backwards (but do not actually meet) after reflection from a mirror (or refraction through a lens). A virtual image cannot be formed on a screen because light rays do not actually pass through a virtual image. A plane mirror always forms virtual images. Similarly, a convex mirror also forms only virtual images. A concave mirror can form a virtual image only when the object is kept within its focus. As we will study in the topic on lenses, a concave lens always forms virtual images but a convex lens forms a virtual image only when the object is within its focus. We will study the formation of virtual images with the help of ray-diagrams after a while.

We will now describe the formation of image in a plane mirror. Before we do that please remember that an object gives out a large number of light rays in all the directions but we will use only two light rays coming from the object to show the formation of image. This is done just to keep the ray-diagram simple. Another point to be noted is that the real rays of light are represented by full lines (solid lines) whereas virtual rays of light are represented by dotted lines (broken lines). The real rays of light can exist only in front of a mirror. The virtual rays of light are those which we show behind a mirror. The virtual light rays do not exist at all (because light cannot reach behind the mirror by passing through it). They just appear to be coming from behind the mirror. We will now describe how the image of an object is formed in a plane mirror by using the laws of reflection of light.

Formation of Image in a Plane Mirror

Consider a small object O (say, a point source of light) placed in front of a plane mirror MM' (see Figure 13). The mirror will form an image I of the object O . Now, we want to know how this image has been formed. This happens as follows : The object O gives out light rays in all directions but we need only two rays of light to locate the image. Now, a ray of light OA coming from the object O is incident on the plane mirror at point A and it gets reflected in the direction AX according to the laws of reflection of light making the angle of reflection r_1 equal to the angle of incidence i_1 (see Figure 13). Another ray of light OB coming from the object O strikes the mirror at point B and gets reflected in the direction BY , again making the angle of reflection r_2 equal to the angle of incidence i_2 .

The two reflected rays AX and BY are diverging (moving away from one another), so they cannot meet on the left side. Let us produce the reflected rays AX and BY backwards (as shown by dotted lines in Figure 13). They meet at point I behind the mirror. Now, when the reflected rays AX and BY enter the eye of a person at position E , the eye sees the rays of light in the straight line direction in which the reflected rays enter it. So, the person looking into the mirror from position E sees the reflected rays as if they are coming from the point I behind the mirror (because I is the point of intersection of the reflected rays when produced backwards to the right side). Thus, point I is the image of the object O formed by the plane mirror. For example, if our face is at position O in front of the plane mirror, then we will see the image of our face in the mirror at point I .

Please note that **the image formed by a plane mirror can be seen only by looking into the mirror.** So, if a screen is placed at position I , no image would be formed on it because the light rays do not actually pass through the point I , they only appear to do so. **An image of this type, which cannot be received on a screen, is known as a virtual image.** Another point to be noted is that the light rays shown by dotted lines behind the mirror are only imaginary light rays. There can be no real light rays behind a mirror because it has a silvered reflecting surface at its back (covered with a paint) which does not allow light rays to pass

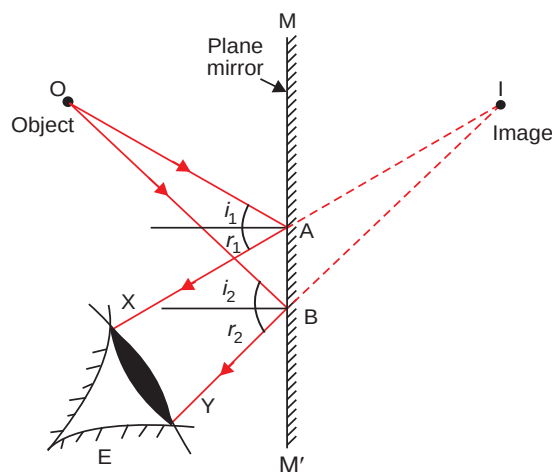


Figure 13. Formation of image in a plane mirror.

through it and go behind the mirror. So, there is no light coming to the person directly from the image point I . It is only the light reflected from the mirror which appears to be coming from the image point I .

Image of an Extended Object (or Finite Object)

We have just studied that a plane mirror forms a point image of a point object. But most of the actual objects are much bigger than a mere 'point' and they are called 'extended objects' or 'finite objects'. An extended object (or bigger object) can be considered to be made up of a very, very large number of point objects. So, the image of an extended object is a collection of the image points corresponding to the various points of the object. We will now describe how a plane mirror forms the image of an extended object. In our ray-diagrams, we will use an 'arrow pointing upwards' to represent an extended object (or finite object).

In Figure 14, an extended object AB (an arrow pointing upwards) is placed in front of a plane mirror MM' . In order to locate the image of arrow AB in the plane mirror, we will first find out the positions of images of its top point A and bottom point B . This can be done as follows :

(i) From point A , we draw an incident ray AC perpendicular to the mirror (see Figure 14). This will be reflected back along the same path. So, CA is the first reflected ray. We now draw another incident ray AD which strikes the mirror at point D . The ray AD is reflected along DE , making an angle of reflection (r_1) equal to the angle of incidence (i_1). Thus, DE is the second reflected ray here. We produce the two reflected rays CA and DE backwards by dotted lines. They meet at point A' (A-dash). So, A' is the virtual image of the top point A of the object.

(ii) From point B , we draw an incident ray BF perpendicular to the mirror. This will be reflected back along the same path, giving the reflected ray FB . Another incident ray BG is reflected along GH making the angle of reflection (r_2) equal to the angle of incidence (i_2). We produce the two reflected rays FB and GH backwards by dotted lines. They meet at point B' (B-dash). So, B' is the virtual image of the bottom point B of the object.

In this way we have located the images of the top point A and bottom point B of the object. Now, each point of the object (or arrow) between A and B will give a point image which will lie between the points A' and B' . So, to get the complete image of object AB , we join the points A' and B' by a dotted line. Thus, $A'B'$ (A-dash B-dash) is the complete image of the object AB which has been formed by the plane mirror (see Figure 14).

The image is virtual, erect (same side up as the object, because both the object and image have arrow-head at the top), and of the same size as the object. Please note that in Figure 14, the image $A'B'$ has been drawn by dotted line just to show that it is a virtual image. Thus, **the nature of image formed by a plane mirror is virtual and erect. And the size of image formed by a plane mirror is equal to that of the object.** The image is at the same distance behind the plane mirror as the object (arrow) is in front of the mirror.

The Position of Image Formed in a Plane Mirror

The image formed in a plane mirror is at the same distance behind the mirror as the object is in front of the mirror (see Figure 15). In other words, the image and object are at equal

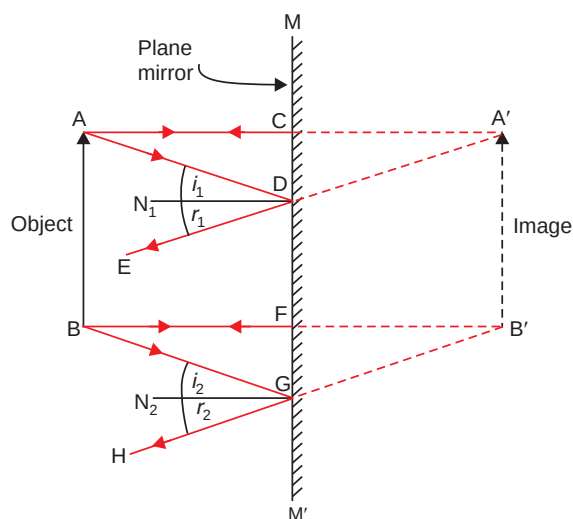


Figure 14. The formation of image of an extended object (here an arrow AB) in a plane mirror.

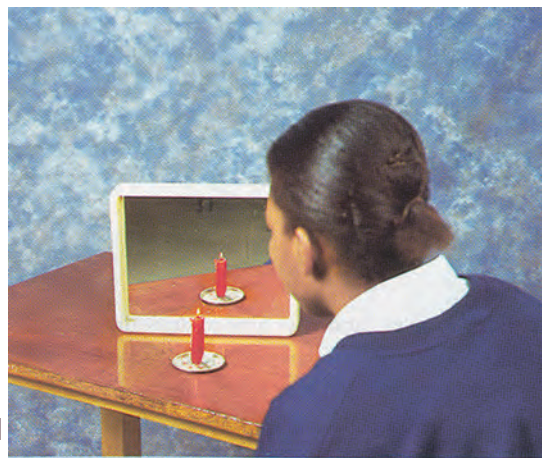


Figure 15. The image of candle in the plane mirror is the same distance behind the mirror as the candle is in front of the mirror.

distances from a plane mirror but they are on the opposite sides of the plane mirror. For example, if an object is placed at a distance of 5 cm in front of a plane mirror, then its image will also be formed at a distance of 5 cm behind the mirror. And the total distance between the object and its image will be $5 + 5 = 10$ cm. That is, the object will be at a distance of 10 cm from its image.

Lateral Inversion

If we stand in front of a big plane mirror, we see the image of our body in it. Though our image appears to be just as we are, but there is a difference. This is because if we lift our right hand, then our image lifts its left hand. And if we lift our left hand, then the image appears to lift its right hand (see Figure 16). This means that **the right side of our body becomes left side in the image; whereas the left side of our body becomes right side in the image**. It appears as if our image has been 'reversed sideways' with respect to our body. This effect of reversing the sides of an object and its image is called *lateral inversion*. And we say that the image formed in a plane mirror is laterally inverted. In other words, the image formed in a plane mirror is 'sideways reversed' with respect to the object. We can now define lateral inversion as follows :

When an object is placed in front of a plane mirror, then the right side of object appears to become the left side of image; and the left side of object appears to become the right side of image. This change of sides of an 'object' and its 'mirror image' is called lateral inversion. The phenomenon of lateral inversion will become clear from the following example. Suppose we have a placard having the word RED written on it [see Figure 17(a)]. When we hold this placard in front of a plane mirror, the image of word RED appears to be like ƆƎЯ [see Figure 17(b)]. Please note that the object (placard) has the letter R on its left side but the image has this letter in reversed form Я on its right side. Similarly, the image of letter E appears to be reversed like Ǝ. And the letter D on the right side of the object (placard) is on the left side of the image in the reversed form as Ɔ. Thus, all the letters written on the placard are reversed from left to right. We say that the image is laterally inverted. This is an example of lateral inversion. **The phenomenon of lateral inversion is due to the reflection of light.**

From the above discussion we conclude that **the image formed in a plane mirror is laterally inverted (or sideways reversed) with respect to the object**. It is due to lateral inversion that the image of our right hand appears to be our left hand. So, when we sit in front of a plane mirror and write with our right hand, it appears in the mirror that we are writing with the left hand. And it is also due to lateral inversion that the parting in our hair on the right appears to be on the left when seen in a mirror. The word AMBULANCE on the hospital vans is written in the form of its mirror image as ƎHƆAƆU8MA (see Figure 18). This is because when we are driving our car and see the hospital van coming from behind in our rear-view mirror, then we will get the laterally inverted image of ƎHƆAƆU8MA and read it as AMBULANCE. Since an ambulance carries seriously ill patients, we can make way for it to pass through and reach the hospital quickly. We are now in a position to give all the **characteristics of an image formed by a plane mirror**.

1. The image formed in a plane mirror is virtual. It cannot be received on a screen.



Figure 16. Our left hand appears to be right hand in the mirror image.

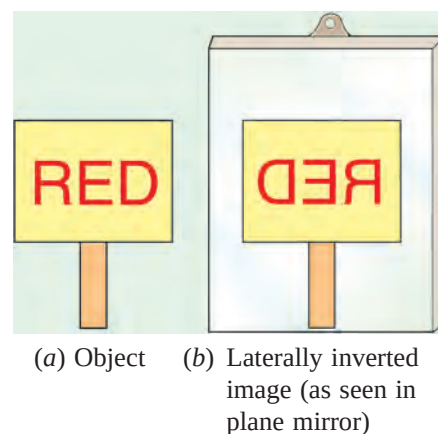


Figure 17. Diagram to show lateral inversion.



Figure 18. An ambulance. Please note the word AMBULANCE written in the form of its mirror image.

- The image formed in a plane mirror is erect. It is the same side up as the object.
- The image in a plane mirror is of the same size as the object.
- The image formed by a plane mirror is at the same distance behind the mirror as the object is in front of the mirror.
- The image formed in a plane mirror is laterally inverted (or sideways reversed).

Uses of Plane Mirrors

- Plane mirrors are used to see ourselves. The mirrors on our dressing table and in bathrooms are plane mirrors.
- Plane mirrors are fixed on the inside walls of certain shops (like jewellery shops) to make them look bigger.
- Plane mirrors are fitted at blind turns of some busy roads so that drivers can see the vehicles coming from the other side and prevent accidents.
- Plane mirrors are used in making periscopes.

We will now solve some problems based on plane mirrors.

Sample Problem 1. An incident ray makes an angle of 35° with the surface of a plane mirror. What is the angle of reflection?

Solution. In order to find out the angle of reflection, we should first know the angle of incidence. In this case, the incident ray makes an angle of 35° with the surface of the mirror (see Figure 19), so the angle of incidence is not 35° . The angle of incidence is the angle between incident ray and normal. So, in this case, the angle of incidence will be $90^\circ - 35^\circ = 55^\circ$. Since the angle of incidence is 55 degrees, therefore, the angle of reflection is also 55 degrees. This is shown clearly in Figure 19.

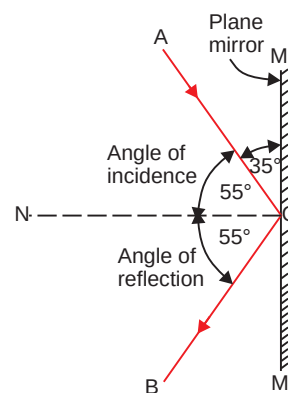


Figure 19.

Sample Problem 2. David is observing his image in a plane mirror. The distance between the mirror and his image is 4 m. If he moves 1 m towards the mirror, then the distance between David and his image will be :

- (a) 3 m (b) 5 m (c) 6 m (d) 8 m

Solution. (i) In this problem, David is the object. Initially, the distance between the mirror and David's image is 4 m. Since the distance of object from plane mirror is equal to distance of image from plane mirror, therefore, the distance of David (which is object), from the plane mirror is also 4 m (see Figure 20).

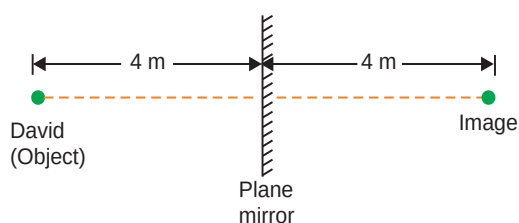


Figure 20.

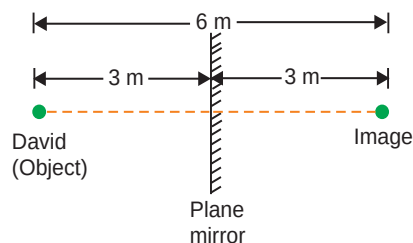


Figure 21.

(ii) David moves 1 m towards the mirror. This means that the distance of David from mirror will be $4 - 1 = 3$ m. Since the image is formed at the same distance behind the mirror as the object is in front of it, therefore, the distance of image from mirror in this case will also be 3 m (see Figure 21)

Now, Distance of David from mirror = 3 m (in front)

And, Distance of image from mirror = 3 m (at the back)

So, Distance between David and his image = $3 \text{ m} + 3 \text{ m}$
 $= 6 \text{ m}$

Thus, the correct option is : (c) 6 m.

Sample Problem 3. The rear view mirror of a car is a plane mirror. A driver is reversing his car at a speed of 2 m/s. The driver sees in his rear view mirror, the image of a truck parked behind his car. The speed at which the image of the truck appears to approach the driver will be :

- (a) 1 m/s (b) 2 m/s (c) 4 m/s (d) 8 m/s

Solution. In a plane mirror, the object and its image always remain at the same distance from the mirror. So, when the car reverses at a speed of 2 m/s, then the image will also appear to move towards the mirror at the same speed of 2 m/s. So, the speed at which the image of truck appears to approach the car driver will be $2 \text{ m/s} + 2 \text{ m/s} = 4 \text{ m/s}$. Thus, the correct option will be : (c) 4 m/s.

Before we go further and discuss spherical mirrors, **please answer the following questions :**

Very Short Answer Type Questions

1. What happens when a ray of light falls normally (or perpendicularly) on the surface of a plane mirror ?
2. A ray of light is incident on a plane mirror at an angle of 30° . What is the angle of reflection ?
3. A ray of light strikes a plane mirror at an angle of 40° to the mirror surface. What will be the angle of reflection ?
4. A ray of light is incident normally on a plane mirror. What will be the :
(a) angle of incidence ?
(b) angle of reflection ?
5. What type of image is formed :
(a) in a plane mirror ?
(b) on a cinema screen ?
6. What kind of mirror is required for obtaining a virtual image of the same size as the object ?
7. What is the name of the phenomenon in which the right side of an object appears to be the left side of the image in a plane mirror ?
8. Name the phenomenon responsible for the following effect :
When we sit in front of a plane mirror and write with our right hand, it appears in the mirror that we are writing with the left hand.
9. If an object is placed at a distance of 10 cm in front of a plane mirror, how far would it be from its image ?
10. Which property of light makes a pencil cast a shadow when it is held in front of a light source ?
11. The image seen in a plane mirror cannot be formed on a screen. What name is given to this type of image ?
12. Fill in the following blank with a suitable word :
When light is reflected, the angles of incidence and reflection are..... .
13. State whether the following statement is true or false :
A student says that we can see an object because light from our eyes is reflected back by the object.
14. Where is the image when you look at something in a mirror ?
15. A ray of light strikes a plane mirror such that its angle of incidence is 30° . What angle does the reflected ray make with the mirror surface ?

Short Answer Type Questions

16. What is the difference between a real image and a virtual image ? Give one example of each type of image.
17. The letter F is placed in front of a plane mirror :
(a) How would its image look like when seen in a plane mirror ?
(b) What is the name of the phenomenon involved ?
18. What is lateral inversion ? Explain by giving a suitable example.
19. Write the word AMBULANCE as it would appear when reflected in a plane mirror. Why is it sometimes written in this way (as its mirror image) on the front of an ambulance ?
20. What are the important differences between looking at a photograph of your face and looking at yourself in a plane mirror ?

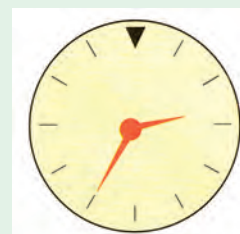
21. (a) A wall reflects light and a mirror also reflects light. What difference is there in the way they reflect light ?
(b) Which type of reflection of light leads to the formation of images ?
22. What is the difference between regular reflection of light and diffuse reflection of light ? What type of reflection of light takes place from :
(a) a cinema screen (b) a plane mirror (c) a cardboard (d) still water surface of a lake
23. What can you see in a completely dark room ? If you switch on an electric bulb in this dark room as a light source, explain how you could now see :
(a) the electric bulb.
(b) a piece of white paper.
24. (a) A boy with a mouth 5 cm wide stands 2 m away from a plane mirror. Where is his image and how wide is the image of his mouth ?
(b) The boy walks towards the mirror at a speed of 1 m/s. At what speed does his image approach him ?
25. (a) An extended object in the form of an arrow pointing upward has been placed in front of a plane mirror. Draw a labelled ray-diagram to show the formation of its image.
(b) State the uses of plane mirrors.

Long Answer Type Questions

26. What is meant by 'reflection of light' ? Define the following terms used in the study of reflection of light by drawing a labelled ray-diagram :
(a) Incident ray (b) Point of incidence (c) Normal
(d) Reflected ray (e) Angle of incidence (f) Angle of reflection
27. State and explain the laws of reflection of light at a plane surface (like a plane mirror), with the help of a labelled ray-diagram. Mark the angles of 'incidence' and 'reflection' clearly on the diagram. If the angle of reflection is 47.5° , what will be the angle of incidence ?
28. With the help of a labelled ray-diagram, describe how a plane mirror forms an image of a point source of light placed in front of it. State the characteristics of the image formed in a plane mirror.
29. (a) Explain why, though both a plane mirror and a sheet of paper reflect light but we can see the image of our face in a plane mirror but not in a sheet of paper.
(b) The image in a plane mirror is virtual and laterally inverted. What does this statement mean ?
(c) Write all the capital letters of the alphabet which look the same in a plane mirror.

Multiple Choice Questions (MCQs)

30. The angle of reflection is equal to the angle of incidence :
(a) always (b) sometimes (c) under special conditions (d) never
31. The angle between an incident ray and the plane mirror is 30° . The total angle between the incident ray and reflected ray will be :
(a) 30° (b) 60° (c) 90° (d) 120°
32. A ray of light is incident on a plane mirror making an angle of 90° with the mirror surface. The angle of reflection for this ray of light will be :
(a) 45° (b) 90° (c) 0° (d) 60°
33. The image of an object formed by a plane mirror is :
(a) virtual (b) real (c) diminished (d) upside-down
34. The image formed by a plane mirror is :
(a) virtual, behind the mirror and enlarged.
(b) virtual, behind the mirror and of the same size as the object.
(c) real, at the surface of the mirror and enlarged.
(d) real, behind the mirror and of the same size as the object.
35. The figure given alongside shows the image of a clock as seen in a plane mirror. The correct time is :
(a) 2.25 (b) 2.35 (c) 6.45 (d) 9.25

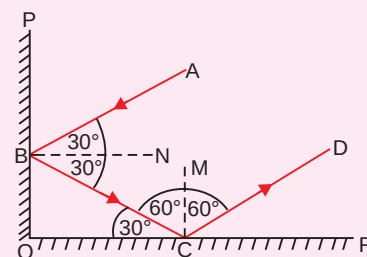


Questions Based on High Order Thinking Skills (HOTS)

36. A man stands 10 m in front of a large plane mirror. How far must he walk before he is 5 m away from his image ?
37. An object is placed 20 cm in front of a plane mirror. The mirror is moved 2 cm towards the object. The distance between the positions of the original and final images seen in the mirror is :
 (a) 2 cm (b) 4 cm (c) 10 cm (d) 22 cm
38. A man sits in an optician's chair, looking into a plane mirror which is 2 m away from him and views the image of a chart which faces the mirror and is 50 cm behind his head. How far away from his eyes does the chart appear to be ?
39. A ray of light strikes a plane mirror PQ at an angle of incidence of 30° , is reflected from the plane mirror and then strikes a second plane mirror QR placed at right angles to the first mirror. The angle of reflection at the second mirror is :
 (a) 30° (b) 45° (c) 60° (d) 90°
- Draw a ray-diagram to illustrate your answer.
40. Explain how to read the following message which was found on some blotting paper :

**ANSWERS**

2. 30° 3. 50° 4. (a) 0° (b) 0° 6. Plane mirror 7. Lateral inversion 8. Lateral inversion
 9. 20 cm 10. Light travels in straight lines 11. Virtual image 12. equal 13. False 14. Behind
 the mirror 15. 60° 17. (a) 1 (b) Lateral inversion 24. (a) 2 m behind the
 plane mirror ; 5 cm wide (b) 2 m/s 27. 47.5° 29. (c) A, H, I, M, O,
 T, U, V, W, X, Y 30. (a) 31. (d) 32. (c) 33. (a) 34. (b)
 35. (d) 36. 7.5 m 37. (b) 38. 4.5 m 39. (c) 60°
40. The impression on blotting paper is the mirror image of the written message ; Hold the written message in front of a mirror to read it.



REFLECTION OF LIGHT FROM CURVED SURFACES : SPHERICAL MIRRORS

So far we have discussed the reflection of light from plane surfaces like that of a plane mirror. **When a parallel beam of light rays falls on a plane mirror, it is reflected as a parallel beam.** So, a plane mirror changes only the direction of incident light rays, it does not 'converge' or 'diverge' the parallel rays of light (To bring the parallel rays of light 'closer together' is called 'to converge' the light rays whereas 'to spread out' the parallel rays of light is called 'to diverge' the light rays). **We will now describe the spherical mirrors which can converge or diverge the parallel rays of light which fall on them.** Please note that the spherical mirrors have a curve-like surface, so they are also known as curved mirrors. We will first define the spherical mirrors and then study the reflection of light from these spherical mirrors.

A spherical mirror is that mirror whose reflecting surface is the part of a hollow sphere of glass. The spherical mirrors are of two types : Concave mirrors, and Convex mirrors.

(i) **A concave mirror is that spherical mirror in which the reflection of light takes place at the concave surface (or bent-in surface).** A concave mirror is shown in Figure 22(a), in which the concave reflecting surface has been marked A. The other surface B in Figure 22(a), having short, oblique lines is the non-reflecting surface. The inner shining surface of a steel spoon is an example of concave mirror (see Figure 23). In our ray-diagrams, we use only the side-view of a concave mirror as shown in Figure 22(a). If, however,

we look at a concave mirror from the front, it appears to be like a piece of thin round glass whose front surface is shining and bent inward whereas back surface is covered with a paint and bulging outward.

(ii) **A convex mirror is that spherical mirror in which the reflection of light takes place at the convex surface (or bulging-out surface).**

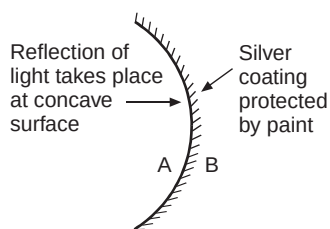
A convex mirror is shown in Figure 22(b), in which the convex reflecting surface has been marked B. The other surface A in Figure 22(b), having short, oblique lines is the non-reflecting surface. The back-side of a shining steel spoon is an example of convex mirror (see Figure 23). In our ray diagrams, we use only the side-view of a convex mirror as shown in Figure 22(b). If, however, we look at a convex mirror from the front, it appears like a piece of thin, round glass whose front shining surface is bulging outward but the back surface covered with paint is bent inward. **A spherical mirror (concave mirror or convex mirror) works on the reflection of light.** We will now understand the meaning of some new terms such as centre of curvature, radius of curvature, pole, and principal axis, which are used in the study of spherical mirrors.

Centre of Curvature, Radius of Curvature, Pole and Principal Axis of a Spherical Mirror

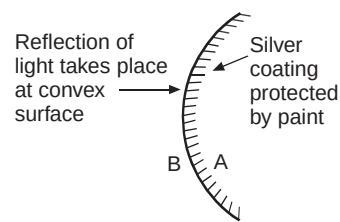
The centre of curvature of a spherical mirror is the centre of the hollow sphere of glass of which the mirror is a part. The centre of curvature of a mirror is represented by the letter C. In Figure 24(a), C is the centre of curvature of the concave mirror and in Figure 24(b), C is the centre of curvature of the convex mirror. The centre of curvature is not a part of the mirror. It lies outside the reflecting surface of the mirror. It should be noted that **the centre of curvature of a concave mirror is in front of it but the centre of curvature of a convex mirror is behind it.**

The radius of curvature of a spherical mirror is the radius of the hollow sphere of glass of which the mirror is a part. In Figure 24(a), the distance CP is the radius of curvature of the concave mirror and in Figure 24(b), the distance CP is the radius of curvature of the convex mirror. The radius of curvature of a mirror is represented by the letter R.

The centre of a spherical mirror is called its pole. In other words, the middle point of a spherical mirror is called its pole. In Figure 24(a), P is the pole of the concave mirror and in Figure 24(b), P is the pole of the convex mirror. The pole of a spherical mirror lies on the surface of the mirror.



(a) A concave mirror



(b) A convex mirror

Figure 22.

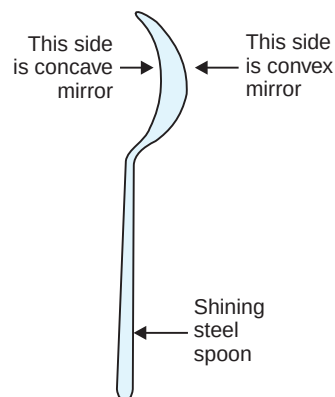
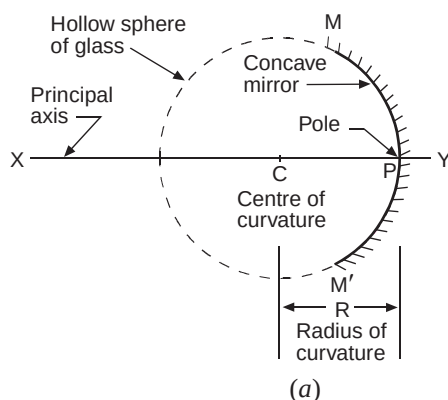
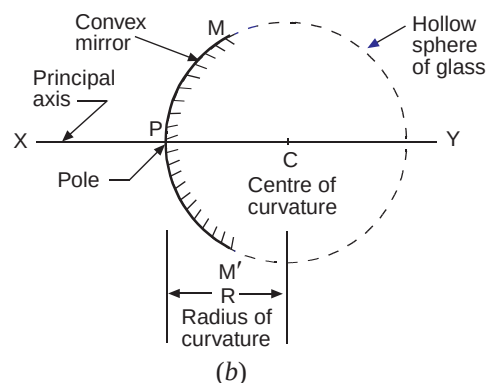


Figure 23. The front side of a shining steel spoon is a concave mirror whereas its back side is a convex mirror.



(a)



(b)

Figure 24. Diagrams to show how a concave mirror and a convex mirror can be considered to be the part of a hollow sphere of glass.



The straight line passing through the centre of curvature and pole of a spherical mirror is called its **principal axis**. In Figure 24(a), C is the centre of curvature of the concave mirror and P is the pole of the concave mirror, so the line XY , passing through C and P is the principal axis of the concave mirror. Similarly, in Figure 24(b), XY is the principal axis of the convex mirror. The principal axis is normal (or perpendicular) to the mirror at its pole.

That portion of a mirror from which the reflection of light actually takes place is called the **aperture of the mirror**. The aperture of a spherical mirror is represented by the diameter of its reflecting surface. In Figure 24, the distance MM' is the aperture of the mirror. In fact, *the aperture of a mirror represents the size of the mirror*.

Principal Focus and Focal Length of a Concave Mirror

The principal focus of a concave mirror is a point on its principal axis to which all the light rays which are parallel and close to the axis, converge after reflection from the concave mirror. Look at Figure 25 in which a parallel beam of light rays is falling on a concave mirror MM' . In Figure 25, point F is the principal focus of the concave mirror because all the parallel rays of light converge at this point after getting reflected from the concave mirror. Since all the reflected light rays actually pass through the focus (F) of a concave mirror, therefore, **a concave mirror has a real focus**. The focus of a concave mirror is in front of the mirror. Since a concave mirror converges a parallel beam of light rays, it is also called a converging mirror. **The focal length of a concave mirror is the distance between its pole and principal focus**. In Figure 25, P is the pole of the concave mirror and F is the principal focus, so the distance PF is the focal length of this concave mirror. The focal length of a mirror is denoted by the letter f .

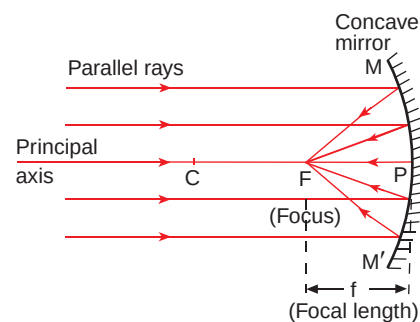


Figure 25. A concave mirror converges (brings closer) a beam of parallel rays of light.

We will now describe how an image of the sun can be formed by a concave mirror. The sun is at a far off distance from us, so the sunlight rays reaching us are parallel rays. **An image of the sun can be produced by a concave mirror at its focus** (see Figure 26). This can be done by performing a simple experiment as follows. We take a concave mirror and point it towards the sun. Hold a piece of paper in front of the concave mirror in such a way that the sunlight reflected by concave mirror falls on the paper (see Figure 26). A small patch of bright reflected light will appear on the paper. Adjust the distance of paper from the concave mirror in such a way that the sharpest point of bright light is obtained. This sharp point of light on paper is the image of sun formed by the concave mirror (see Figure 26). This image of the sun is formed at the focus of concave mirror (where the paper is held by us). If we keep this piece of paper in this position for a few minutes, the paper would start burning at the point of sun's image and a hole will be formed in it. This is because the concave mirror converges (or concentrates) a lot of sun's rays to a small point on paper. The heat energy of these concentrated sun rays burns the paper. Please note that the image of the sun formed by the concave mirror is real because it can be received on screen (such as a sheet of paper).

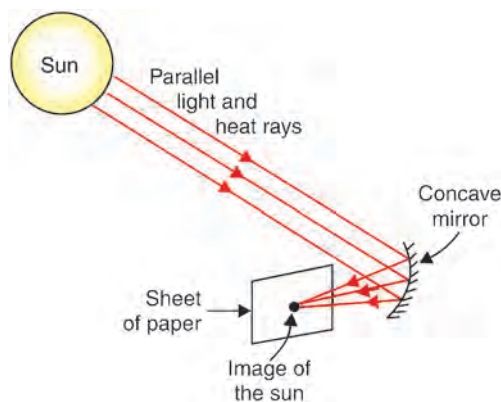


Figure 26. A concave mirror forms a real image of the sun.

Since the sun's image is formed at the focus of the concave mirror, therefore, the distance of sun's image (or paper) from the concave mirror gives us an approximate value of the focal length of the concave mirror.

Principal Focus and Focal Length of a Convex Mirror

The principal focus of a convex mirror is a point on its principal axis from which a beam of light rays, initially parallel to the axis, appears to diverge after being reflected from the convex mirror. In Figure 27, a parallel beam of light rays is incident on a convex mirror MM' . Each ray of light is reflected by the convex mirror, and the reflected rays diverge (spread out) from the mirror surface. Let us produce all the reflected rays backwards (as shown by dotted lines) so that they appear to meet at a point F behind the convex mirror. Now, to a person looking into the mirror from the left side, all the reflected rays appear to be coming (or diverging) from the same point F behind the convex mirror. This point F is the principal focus of the convex mirror. It should be noted that the reflected rays do not actually pass through the focus (F) of a convex mirror, therefore, **a convex mirror has virtual focus**. Another point to be noted is that **the focus of a convex mirror is situated behind the mirror**. The focal length of a convex mirror is the distance from the pole P to its principal focus F . Thus, in Figure 27, the distance PF is the focal length of the convex mirror.

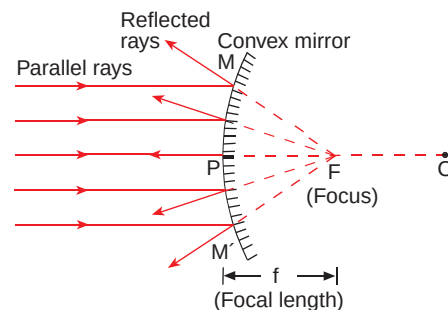


Figure 27. A convex mirror diverges (spreads out) a beam of parallel rays of light.



A plane mirror neither converges parallel rays of light nor diverges them. The focal length of a plane mirror can be considered to be 'infinite' or 'infinity' (which means very, very great or limitless).

Relation between Radius of Curvature and Focal Length of a Spherical Mirror

For a spherical mirror having small aperture, the principal focus (F) lies exactly mid-way between the pole (P) and centre of curvature (C) (see Figures 26 and 27). So, **the focal length of a spherical mirror (a concave mirror or a convex mirror) is equal to half of its radius of curvature**. If f is the focal length of a spherical mirror and R is its radius of curvature, then :

$$f = \frac{R}{2}$$

Let us solve one problem now.

Sample Problem. If the radius of curvature of a spherical mirror is 20 cm, what is its focal length ?
(NCERT Book Question)

Solution. We know that : $f = \frac{R}{2}$

Here, Focal length, $f = ?$ (To be calculated)

And, Radius of curvature, $R = 20$ cm

So, $f = \frac{20}{2}$ cm
 $= 10$ cm

Thus, the focal length of this spherical mirror is 10 cm.

We are now in a position to **answer the following questions** :

Very Short Answer Type Questions

- Name the spherical mirror which has :
(a) virtual principal focus.
(b) real principal focus.
- Out of convex mirror and concave mirror, whose focus is situated behind the mirror ?
- Find the focal length of a concave mirror whose radius of curvature is 32 cm.
- If the focal length of a convex mirror is 25 cm, what is its radius of curvature ?

5. Fill in the following blanks with suitable words :
- Parallel rays of light are reflected by a concave mirror to a point called the
 - The focal length of a concave mirror is the distance from theto the mirror.
 - A concave mirror.....rays of light whereas a convex mirror.....rays of light.
 - For a convex mirror, parallel rays of light appear to diverge from a point called the..... .

Short Answer Type Questions

- What is a spherical mirror ? Distinguish between a concave mirror and a convex mirror.
- Name the two types of spherical mirrors. What type of mirror is represented by the :
 - back side of a shining steel spoon ?
 - front side of a shining steel spoon ?
- What is the relation between the focal length and radius of curvature of a spherical mirror (concave mirror or convex mirror) ? Calculate the focal length of a spherical mirror whose radius of curvature is 25 cm.
- Explain with a suitable diagram, how a concave mirror converges a parallel beam of light rays. Mark clearly the pole, focus and centre of curvature of concave mirror in this diagram.
- Describe with a suitable diagram, how a convex mirror diverges a parallel beam of light rays. Mark clearly the pole, focus and centre of curvature of convex mirror in this diagram.

Long Answer type Questions

- Define (a) centre of curvature (b) radius of curvature (c) pole (d) principal axis, and (e) aperture, of a spherical mirror with the help of a labelled diagram.
- (a) Define (i) principal focus of a concave mirror, and (ii) focal length of a concave mirror.
(b) Draw diagram to represent the action of a concave mirror on a beam of parallel light rays. Mark on this diagram principal axis, focus F, centre of curvature C, pole P and focal length f , of the concave mirror.
- (a) What is meant by (i) principal focus of a convex mirror, and (ii) focal length of a convex mirror ?
(b) Draw diagram to show the action of convex mirror on a beam of parallel light rays. Mark on this diagram principal axis, focus F, centre of curvature C, pole P and focal length f , of the convex mirror.

Multiple Choice Questions (MCQs)

- In a convex spherical mirror, reflection of light takes place at :
 - a flat surface
 - a bent-in surface
 - a bulging-out surface
 - an uneven surface
- A diverging mirror is :
 - a plane mirror
 - a convex mirror
 - a concave mirror
 - a shaving mirror
- If R is the radius of curvature of a spherical mirror and f is its focal length, then :
 - $R = f$
 - $R = 2f$
 - $R = \frac{f}{2}$
 - $R = 3f$
- The focal length of a spherical mirror of radius of curvature 30 cm is :
 - 10 cm
 - 15 cm
 - 20 cm
 - 30 cm
- If the focal length of a spherical mirror is 12.5 less cm, its radius of curvature will be :
 - 25 cm
 - 15 cm
 - 20 cm
 - 35 cm

Questions Based on High Order Thinking Skills (HOTS)

- A communications satellite in orbit sends a parallel beam of signals down to earth. If these signals obey the same laws of reflection as light and are to be focussed onto a small receiving aerial, what should be the best shape of the metal 'dish' used to collect them ?
- When a spherical mirror is held towards the sun and its sharp image is formed on a piece of carbon paper for some time, a hole is burnt in the carbon paper.
 - What is the nature of spherical mirror ?
 - Why is a hole burnt in the carbon paper ?
 - At which point of the spherical mirror the carbon paper is placed ?
 - What name is given to the distance between spherical mirror and carbon paper ?
 - What is the advantage of using a carbon paper rather than a white paper ?

ANSWERS

1. (a) Convex mirror (b) Concave mirror 2. Convex mirror 3. 16 cm 4. 50 cm 5. (a) principal focus (b) principal focus (c) converges ; diverges (d) principal focus 8. 12.5 cm 14. (c) 15. (b) 16. (b) 17. (b) 18. (a) 19. Concave metal dish : It will collect the parallel beam of satellite signals at its focus where receiving aerial is fixed. 20. (a) Concave mirror (b) A lot of sun's heat rays are concentrated at the point of sun's image which burn the hole in carbon paper (c) At the focus (d) Focal length (e) A black carbon paper absorbs more heat rays and hence burns a hole more easily (than a white paper).

RULES FOR OBTAINING IMAGES FORMED BY CONCAVE MIRRORS

When an object is placed in front of a concave mirror, an image is formed. **The image is formed at that point where at least two reflected rays intersect (or appear to intersect).** Now, to find out the position of an image formed by a concave mirror, we will use only those two rays of light (starting from the object), whose paths, after reflection from the mirror, are known to us and easy to draw. The following rays of light are usually used to locate the images formed by concave mirrors. We can call them rules for obtaining images in concave mirrors.

Rule 1. A ray of light which is parallel to the principal axis of a concave mirror, passes through its focus after reflection from the mirror.

This is shown in Figure 28. Here, we have a concave mirror M and its principal axis is XP . The centre of curvature is C and its focus is F . Now, a ray of light AB is parallel to the principal axis of the mirror. It strikes the mirror at point B and gets reflected. After reflection its path changes and it passes through the focus F and goes in the direction BY (see Figure 28).

my study gear

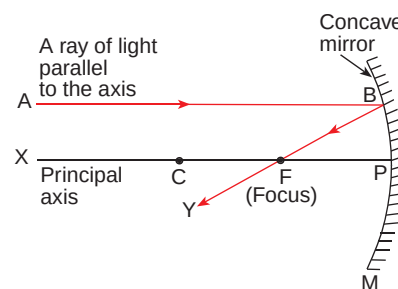


Figure 28.

Rule 2. A ray of light passing through the centre of curvature of a concave mirror is reflected back along the same path (because it strikes the mirror normally or perpendicularly). This is shown in Figure 29. A ray of light AD is passing through the centre of curvature C of the concave mirror. It falls normally (or perpendicularly) on the mirror at point D and gets reflected back along the same path DA . It should be noted that we have put two arrow-heads on the line AD which point in the opposite directions. The arrow pointing from left to right indicates the direction of incident ray and the arrow pointing from right to left indicates the direction of reflected ray (see Figure 29). The ray of light passing through the centre of curvature of a concave mirror is reflected back along the same path because it strikes the concave mirror at right angles (90°) to its surface due to which the angle of incidence and angle of reflection both are 0° .

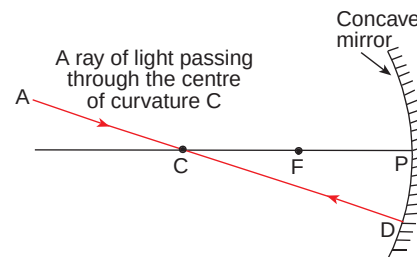


Figure 29.

Rule 3. A ray of light passing through the focus of a concave mirror becomes parallel to the principal axis after reflection. This rule is just the reverse case of the first rule and it is shown in Figure 30. Here, the incident ray AE is passing through the focus F of the concave mirror. It strikes the mirror surface at point E and gets reflected. After reflection, it becomes parallel to the axis and goes in the direction EG .

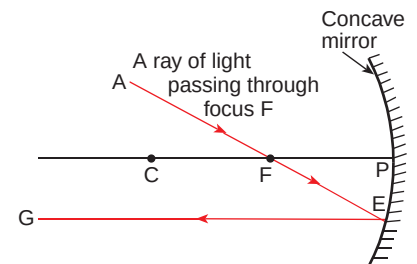


Figure 30.

Rule 4. A ray of light which is incident at the pole of a concave mirror is reflected back making the same angle with the principal axis. This is shown in Figure 31. Here a ray of light AP is incident at the pole P of the concave mirror making an angle of incidence i with the principal axis. It gets reflected along the direction PH such that the angle of reflection (r) is equal to the angle of incidence (i).

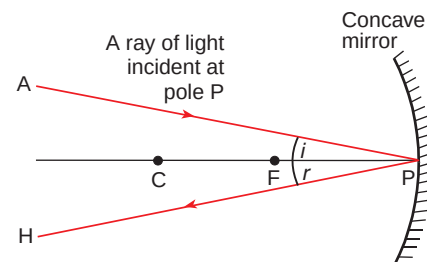


Figure 31.

Please note that if a ray of light is incident on a concave mirror along its principal axis, then it is reflected back along the same path (because it will be normal to the mirror surface). The angle of incidence as well as the angle of reflection for such a ray of light will be zero.

FORMATION OF DIFFERENT TYPES OF IMAGES BY A CONCAVE MIRROR

The type of image formed by a concave mirror depends on the position of object in front of the mirror. We can place the object at different positions (or different distances) from a concave mirror to get different types of images. For example, we can place the object :

- (i) between the pole (P) and focus (F) (see Figure 32),
- (ii) at the focus (F),
- (iii) between focus (F) and centre of curvature (C),
- (iv) at the centre of curvature (C),
- (v) beyond the centre of curvature (C), and
- (vi) at far-off distance called infinity (This distance cannot be shown in the Figure).

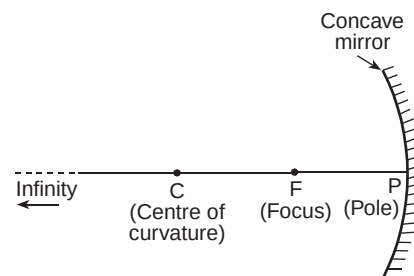


Figure 32.

Please note that in all our ray-diagrams the object, an arrow AB pointing upwards, will be placed on the principal axis. So, a ray of light coming from the bottom B of the object always travels along the principal axis and gets reflected back along the same path. We, however, do not put arrows on the principal axis to show this ray of light.

We will now draw the ray-diagrams to show the images formed by a concave mirror for the different positions of an object. We will consider all the six positions of the object, one by one.

Case 1. Image formed by a concave mirror when the object is placed between pole and focus of the mirror (Object between P and F)

In Figure 33, we have an object AB placed between the pole (P) and focus (F) of a concave mirror, that is, the object is within the focus of the concave mirror. To find out the position and nature of the image, starting from A , we draw a ray AD parallel to the axis (see Figure 33). This ray gets reflected at D and then passes through the focus F . A second ray of light AE passing through the centre of curvature C strikes the mirror normally (or perpendicularly) at point E and gets reflected back along the same path.

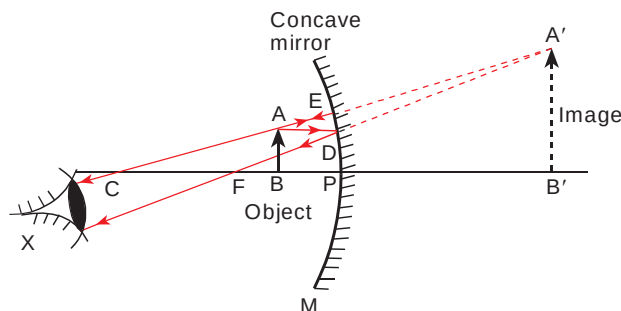


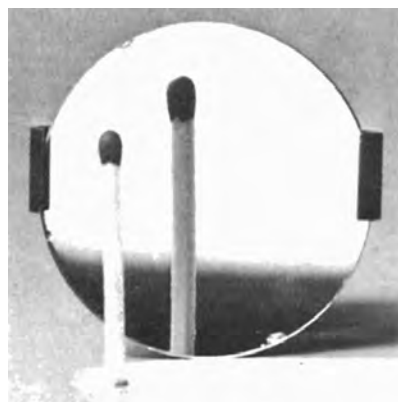
Figure 33. Formation of image by a concave mirror when the object is placed between its pole (P) and focus (F).

Now, the two reflected rays DF and EC are diverging rays and, therefore, do not intersect each other on the left side. The reflected rays DF and EC are produced backwards (as shown by dotted lines). On producing backwards, they appear to intersect at point A' behind the concave mirror. Thus, A' is the virtual image of point A of the object. To get the complete image of the object we draw $A'B'$ perpendicular to the axis from point A' . Thus $A'B'$ is the image of the object AB formed by the concave mirror. Now, an eye at position X , sees the reflected rays as if they have come from points $A'B'$ behind the mirror. Since there is no actual intersection of the reflected rays, so the image $A'B'$ is a virtual image. This image can be seen only by looking into the concave mirror, it cannot be obtained on a screen. We can see from Figure 33 that the image is same side up as object (both of them have arrow-head at the top), so it is an erect image. And if we compare the size of the object AB and its image $A'B'$, the image appears to be bigger in size than the object. Thus, it is an enlarged image or magnified image. From the above discussion we conclude that : **When an object is placed between the pole (P) and focus (F) of a concave mirror, the image formed is :**

- (i) behind the mirror,
- (ii) virtual and erect, and
- (iii) larger than the object (or magnified).

It is clear from the above discussion that to obtain a magnified and erect image with a concave mirror, the object should be placed between the pole and focus of the concave mirror, that is, the object should be placed within the focus (or focal length) of the concave mirror. For example, if the focal length of a concave mirror is 10 cm, then its focus (F) will be at a distance of 10 cm from the pole of the mirror. So, to obtain a magnified and erect image with this concave mirror, we will have to place the object at a distance of less than 10 cm so that it remains between the pole and focus. Thus, if we place the object at a distance of, say 8 cm, from this concave mirror, the object will be within the focus and a magnified, virtual and erect image will be formed behind the concave mirror (which can be seen by looking into the concave mirror).

If we hold a matchstick (as object) within the focus of a concave mirror, then a magnified, virtual and erect image of the matchstick is seen on looking into the concave mirror [as shown in Figure 34(a)]. This



(a) A concave mirror can be used to magnify objects. Here it is magnifying a matchstick kept close to it



(b) A concave mirror being used as a make-up mirror. It is magnifying a part of the face here.



(c) Dentist's mirror is a small concave mirror fitted in a frame with a long handle. It gives a magnified image of tooth.

Figure 34.

also explains the use of a concave mirror as a shaving mirror. When the face is placed close to a concave mirror (so that the face is within its focus) the concave mirror produces a magnified (large) and erect image of the face. Since a large image of the face is seen in the concave mirror, it becomes easier to make a smooth shave. Thus, **while using a concave mirror as a shaving mirror, the face should be close to the concave mirror.** For the same reason, concave mirrors are also used as make-up mirrors by women for putting on make-up (such as painting eye-lashes) [see Figure 34(b)]. In order to use a concave mirror as a make-up mirror, the face is held close to the concave mirror so that it lies within the focus of concave mirror. Dentists use concave mirrors to see the large image of the teeth for examining the teeth of a person (see Figure 34(c)). For this purpose, the dentist holds a small concave mirror in such a way that the tooth lies within its focus. A magnified image of the tooth is then seen by the dentist in the concave mirror. Since the tooth looks much bigger, it becomes easy to examine the defect in the tooth.

Case 2. When the object is placed at the focus of a concave mirror (Object at F)

In Figure 35, the object AB has been placed at the focus (F) of the concave mirror. Now, the parallel ray of light AD (coming from the top of the object) gets reflected at D and passes through the focus F , giving us the reflected ray DX . A second ray of light AE passing through the centre of curvature C , is reflected back along the same path giving us another reflected ray EY . We find that the reflected rays DX and EY are parallel to one another. These parallel rays will intersect (or meet) at a far off distance to

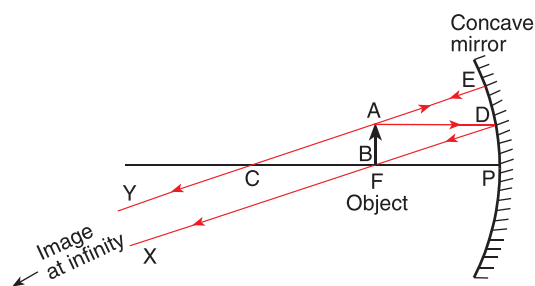


Figure 35. Formation of image by a concave mirror when the object is placed at its focus (F).

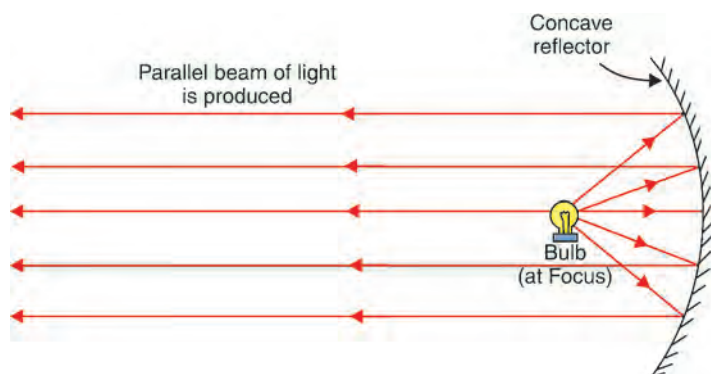
form an image 'at infinity'. And since the image is formed at infinity, it is not possible to show it in our diagram. From this discussion we conclude that : **When an object is placed at the focus of a concave mirror, the image formed is :**



- (i) at infinity,
- (ii) real and inverted, and
- (iii) highly magnified (or highly enlarged).

Suppose we have a concave mirror of focal length 10 cm, then its focus (F) will be at a distance of 10 cm from it. So, by saying that an object is placed at the focus of this concave mirror, we will mean that the object is placed at a distance of 10 cm from the concave mirror. In this case, the concave mirror converts the diverging rays of light coming from the object into a parallel beam of light rays (which form image at infinity).

When a lighted bulb is placed at the focus of a concave mirror reflector, the diverging light rays of the bulb are collected by the concave reflector and then reflected to produce a strong, parallel-sided beam of light [see Figure 36(a)]. This explains the use of concave reflectors in torches, car head-lights, and



(a) When a lighted bulb is placed at the focus of a concave reflector, a parallel beam of light is produced.



(b) This picture shows the concave reflector of a torch.

Figure 36.

search lights to produce a strong beam of light (which travels a considerable distance in the darkness of night).

Case 3. When the object is placed between focus and centre of curvature (Object between F and C)

In Figure 37, the object AB has been placed between the focus F and centre of curvature C of a concave mirror. Now, a ray of light AD parallel to the principal axis gets reflected at point D and then passes through the focus F . A second ray of light AE passing through the centre of curvature C falls normally on the mirror surface at E and returns along the same path. Thus, we have two reflected rays DF and EC which are converging in the downward direction. If we extend these rays further in the downward direction, they actually intersect at point A' . Thus A' is the real image of point A of the object. To get the complete image we draw $A'B'$ perpendicular to the axis from point A' . Thus, $A'B'$ is the real image of the object AB and it can be received on a screen. For example, if we take a lighted candle as the object then the image of its wick will be formed on a white screen placed at the image position. If we look at the image $A'B'$, we find that it is wrong side up having arrow-head at its bottom. So, we say that the image is inverted. The size of image is larger than the object, so the image is magnified, and

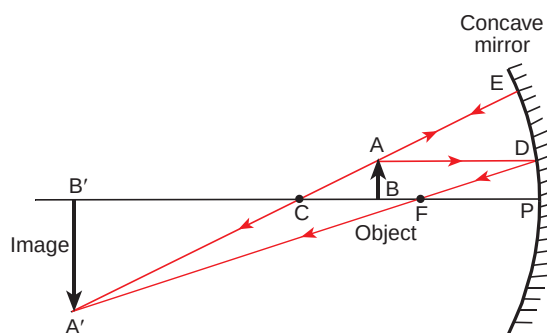


Figure 37. Formation of image by a concave mirror when the object is placed between focus (F) and centre of curvature (C).

it has been formed beyond the centre of curvature of the mirror. From the above discussion we conclude that : **When an object is placed between the focus (F) and centre of curvature (C) of a concave mirror, the image formed is :**

- (i) **beyond the centre of curvature,**
- (ii) **real and inverted,** and
- (iii) **larger than the object (or magnified).**

Please remember that a **real image is always inverted**, and a **virtual image is always erect**. Suppose we have a concave mirror of focal length 10 cm. Then its focus (F) will be at a distance of 10 cm and centre of curvature (C) will be at a distance of $2 \times 10 = 20$ cm from it. Now, by saying that the object is placed between the focus and centre of curvature of this concave mirror, we mean that the object is at any distance between 10 cm and 20 cm from the concave mirror. For example, it may be at a distance of say, 15 cm from the concave mirror. And by saying that the image is formed beyond centre of curvature, we mean that it is at a distance greater than 20 cm from this concave mirror.

Case 4. When the object is placed at the centre of curvature of a concave mirror (Object at C)

In Figure 38, the object AB has been placed at the centre of curvature C of the concave mirror. A ray of light AD which is parallel to the principal axis passes through the focus F after reflection. Now, the second ray of light that we usually use is the one passing through the centre of curvature C . But in this case the object itself is placed at the centre of curvature, so we cannot use this ray of light to locate the image. Here we will use rule No. 3 of image formation which says that "A ray of light passing through the focus of a concave mirror becomes parallel to the principal axis after reflection". So, we now take the ray AE passing through the focus F . It strikes the mirror at point E and gets reflected in the direction EA' parallel to the principal axis. The reflected rays DA' and EA' meet at point A' , so A' is the real image of point A of the object. To get the complete image, we draw $A'B'$ perpendicular to the principal axis. Thus, $A'B'$ is the real image of the object AB (Note that B and B' is just the same point). The image is of the same size as the object, it is real, inverted and formed at the centre of curvature, where the object itself is placed. From this discussion we conclude that : **When an object is placed at the centre of curvature (C) of a concave mirror, the image formed is :**

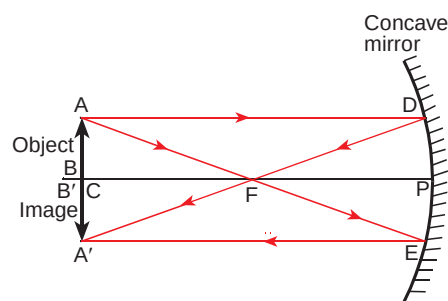


Figure 38. Formation of image by a concave mirror when the object is placed at the centre of curvature (C).

The reflected rays DA' and EA' meet at point A' , so A' is the real image of point A of the object. To get the complete image, we draw $A'B'$ perpendicular to the principal axis. Thus, $A'B'$ is the real image of the object AB (Note that B and B' is just the same point). The image is of the same size as the object, it is real, inverted and formed at the centre of curvature, where the object itself is placed. From this discussion we conclude that : **When an object is placed at the centre of curvature (C) of a concave mirror, the image formed is :**

- (i) **at the centre of curvature (C),**
- (ii) **real and inverted,** and
- (iii) **same size as the object.**

Suppose we have a concave mirror of focal length 10 cm, then its focus (F) will be at a distance of 10 cm and its centre of curvature (C) will be at a distance of $2 \times 10 = 20$ cm. So, by saying that the object is at the centre of curvature, we mean that the object is at a distance of 20 cm from this concave mirror. And by saying that the image is formed at the centre of curvature, we mean that the image is also formed at a distance of 20 cm from the concave mirror.

Case 5. When the object is beyond the centre of curvature of the concave mirror (Object beyond C)

In Figure 39, the object AB has been placed beyond the centre of curvature C of the concave mirror. A ray of light AD which is parallel to the principal axis, passes through the focus F after reflection. A second ray of light AE passing through the centre of curvature falls normally on the mirror surface at E and returns

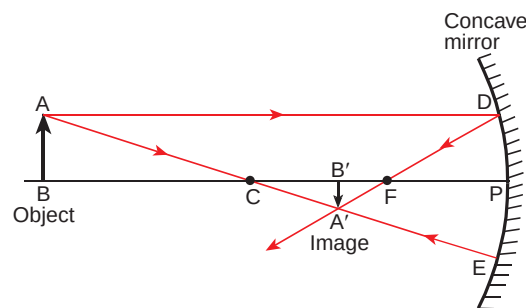


Figure 39. Formation of image by a concave mirror when the object is placed beyond the centre of curvature (C).



along the same path. The two reflected rays intersect at A' . Thus, A' is the real image of point A of the object. To get the complete image, we draw $A'B'$ perpendicular to the axis. Thus, $A'B'$ is the complete image of the object AB . It is clear from Figure 39 that the image $A'B'$ is formed between the focus (F) and centre of curvature (C). The image is real and inverted. It is smaller in size than the object AB , so we call it a diminished image. From the above discussion we conclude that : **When an object is placed beyond the centre of curvature (C) of a concave mirror, the image formed is :**

- (i) **between the focus and centre of curvature,**
- (ii) **real and inverted, and**
- (iii) **smaller than the object (or diminished).**

Suppose we have a concave mirror of focal length 10 cm, then its focus (F) will be at a distance of 10 cm and its centre of curvature (C) will be at a distance of $2 \times 10 = 20$ cm from it. Now, by saying that an object is placed beyond the centre of curvature of this concave mirror, we mean that the object is placed at a distance greater than 20 cm (say, 25 cm) from the concave mirror. And by saying that the image is formed between focus and centre of curvature, we mean that the image is formed at a distance between 10 cm and 20 cm.

We know that the front side of a shining steel spoon is a kind of concave mirror. So, if we keep our face at a fairly good distance from the front side of a shining steel spoon (so that it is beyond its centre of curvature), we will see a real, inverted and smaller image (diminished image) of our face in the spoon (as shown in Figure 40).

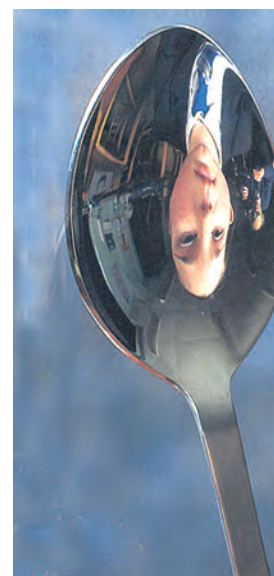


Figure 40. The formation of a real, inverted and diminished image of face in the front side (concave surface) of a shining steel spoon.

Case 6. When the Object is at Infinity

When the object is at a very large distance, we say that the object is at infinity. In Figure 41 we have a concave mirror M . Suppose an object (an arrow pointing upwards) has been placed at infinity in front of the concave mirror (Since the object is very far off, it cannot be shown in the diagram). Because the object AB is very far off, the two rays AD and AP coming from its top point A are parallel to one-another but at an angle to the principal axis as shown in Figure 41. These parallel rays get reflected at points D and P and then intersect at point A' in the focal plane of the mirror. Thus, A' is the real image of the top point A of the object. To get the full image of the object, we draw $A'B'$ perpendicular to the principal axis from A' . Thus, $A'B'$ is the image of the object AB placed at infinity. We find that image $A'B'$ is formed at the focus (F) of the concave mirror.

It is real, inverted and much smaller than the object or highly diminished. From the above discussion we conclude that : **When an object is at infinity from a concave mirror, the image formed is :**

- (i) **at the focus (F),**
- (ii) **real and inverted, and**
- (iii) **much smaller than the object (or highly diminished).**

It should be noted that this case is just opposite of case No. 2 which we have already discussed. There we studied that “**when the object is at focus of concave mirror, the image is formed at infinity**” and here we have seen that “**when the object is at infinity, the image is formed at the focus of concave mirror**”. This means that a concave mirror can concentrate all the parallel rays of light to its focus.

Suppose we have a concave mirror of focal length 10 cm, then its focus (F) will be at a distance of 10 cm

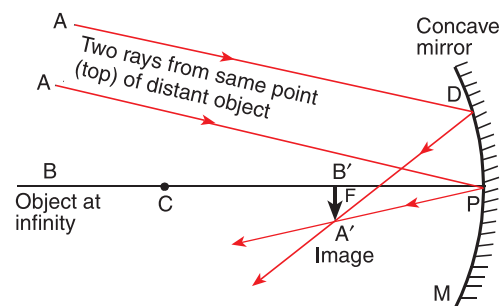


Figure 41. Formation of image by a concave mirror when the object is at a considerable distance (or infinity).



from it. Now, by saying that the object is at infinity, we mean that the object is at a distance which is many, many times its focal length : it may be 1000 cm, 10,000 cm or even more. And by saying that the image is formed at the focus of this concave mirror, we mean that the image is formed at a distance of 10 cm from the pole of this concave mirror.

Please note that when the object kept at infinity in front of a concave mirror is assumed to be a big arrow pointing upwards, then its image is formed at focus according to the ray-diagram shown in Figure 41. If, however, the object kept at infinity in front of a concave mirror is round in shape (like the sun), then its image is formed at the focus according to the ray-diagram shown in Figure 25 on page 177.

The fact that when a parallel beam of light rays falls on a concave mirror, the concave mirror concentrates all the light at its focus, explains the use of concave mirror as a doctor's head-mirror. **A concave mirror is used as a "head-mirror" by the doctors to concentrate light coming from a lamp on to the body part of a patient (like ear, nose, throat, etc.) to be examined.** This is done as follows : A concave mirror is fixed to the front part of a doctor's head with the help of a strap [as shown in Figure 42(a)]. When the beam of light coming from a lamp falls on the doctor's concave head-mirror, the mirror focusses this light on to the body part of the patient. The body part gets illuminated brightly with this light due to which the doctor can look into the body part (like ear, nose, throat, etc.) very clearly.



(a) The doctor's head-mirror is a concave mirror



(b) The metal 'dish' of dish antenna of television is concave.

Figure 42.

The concave 'metal dishes' are used in dish antennas (or dish aerials) of televisions to receive TV signals from the very distant communications satellites which are high up in the sky [see Figure 42(b)]. The dish is a concave reflector. The dish collects a large amount of parallel beams of TV signals coming from the far off satellite and converges them to its focus. The antenna (or signal detector) is fixed in front of the concave dish at its focus. Since the antenna is fixed at the focus of dish, it receives the strongest possible TV signals from the satellite which make our television work. The TV signals coming from satellite can be made even more stronger by using a bigger dish in the dish antenna. Please note that though the TV signals coming from the satellite are not light rays, they obey the laws of reflection of light.

How to find out the Focal Length of a Concave Mirror Quickly but Approximately

The fact that "when the object is at a considerable distance (or at infinity) from a concave mirror, then its image is formed at the focus" can be used to find out the focal length of a concave mirror quickly but approximately. We focus a distant object (several metres away) like a window or tree on a screen by using a concave mirror whose focal length is to be determined (see Figure 43). The sharp image of window or tree will be formed at the focus of the concave mirror. That is, the distance of image (or screen) from the

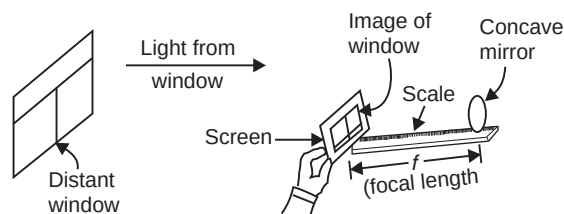


Figure 43. Arrangement for measuring the focal length of a concave mirror quickly but approximately.

concave mirror will be equal to the focal length of concave mirror. This distance can be measured with a scale. It will give us the approximate focal length of the concave mirror. And before we conclude this discussion, here is a summary of the images formed by a concave mirror.

Summary of the Images Formed by a Concave Mirror

<i>Position of object</i>	<i>Position of image</i>	<i>Size of image</i>	<i>Nature of image</i>
1. Within focus (between pole P and focus F)	Behind the mirror	Enlarged	Virtual and erect
2. At focus (F)	At infinity	Highly enlarged	Real and inverted
3. Between F and C	Beyond C	Enlarged	Real and inverted
4. At C	At C	Equal to object	Real and inverted
5. Beyond C	Between F and C	Diminished	Real and inverted
6. At infinity	At focus (F)	Highly diminished	Real and inverted

Uses of Concave Mirrors

(i) **Concave mirrors are used as shaving mirrors to see a large image of the face.** This is because when the face is held within the focus of a concave mirror, then an enlarged image of the face is seen in the concave mirror. This helps in making a smooth shave. For the same reason, **concave mirrors are used as make-up mirrors.**

(ii) **Concave mirrors are used by dentists to see the large images of the teeth of patients.** This is because when a tooth is within the focus of a concave mirror, then an enlarged image of the tooth is seen in the concave mirror. Due to this, it becomes easier to locate the defect in the tooth. The concave mirrors used by dentists are very small in size. They are fitted in a frame with a long handle.

(iii) **Concave mirrors are used as reflectors in torches, vehicle head-lights and search lights to get powerful beams of light.** This is because when a lighted bulb is placed at the focus of a concave reflector, then the concave reflector produces a powerful beam of parallel light rays. This helps us see things up to a considerable distance in the darkness of night. **Concave reflectors are also used in room heaters.** The concave reflectors of room heaters direct heat rays into the whole room.

(iv) **Concave mirrors are used as doctor's head-mirrors to focus light coming from a lamp on to the body parts of a patient (such as eye, ear, nose, throat, etc.) to be examined by the doctor.**

(v) **Concave dishes are used in TV dish antennas to receive TV signals from the distant communications satellites.** The concave dish collects a lot of TV signals and focusses them on to an antenna (or aerial) fixed at its focus, so as to produce strong signals to run the television.

(vi) **Large concave mirrors are used in the field of solar energy to focus sun's rays for heating solar furnaces.** This can be explained as follows : The solar furnace is placed at the focus of a large concave reflector. The concave reflector focuses the sun's heat rays on the furnace due to which the solar furnace gets very hot. Even steel can be melted in this solar furnace.

Let us solve some problems now.

Sample Problem 1. We wish to obtain an erect image of an object using a concave mirror of focal length 15 cm. What should be the range of distance of the object from the mirror ? What is the nature of the image ? Is the image larger or smaller than the object ? Draw a ray-diagram to show the image formation in this case. **(NCERT Book Question)**

Solution. In order to obtain an erect image of an object with a concave mirror, the object should be at a distance less than its focal length. Here the focal length of concave mirror is 15 cm. So to obtain an erect image of the object by using this concave mirror, the object should be placed at any distance which is less than 15 cm from the mirror. The nature of image will be virtual. The image will be larger than the object (For ray-diagram, see Figure 33 on page 181).

Sample Problem 2. The image formed by a concave mirror is seen to be virtual, erect and larger than the object. The position of the object must then be :

- (i) between the focus and centre of curvature.
- (ii) at the centre of curvature.
- (iii) beyond the centre of curvature.
- (iv) between the pole of the mirror and its focus.

Choose the correct alternative.

(NCERT Book Question)

Solution. The correct alternative is (iv) : between the pole of the mirror and its focus.

Sample Problem 3. A concave mirror has a focal length of 10 cm. Where should an object be placed in front of this concave mirror so as to obtain an image which is real, inverted and same size as the object ?

Solution. When the image formed by a concave mirror is real, inverted and of the same size as the object, then the object must be placed at its centre of curvature (C). Now, the centre of curvature of a concave mirror is at a distance of 'twice the focal length' or ' $2f$ '. Here,

$$\begin{aligned}\text{Focal length, } f &= 10 \text{ cm} \\ \text{So, } 2f &= 2 \times 10 \text{ cm} \\ &= 20 \text{ cm}\end{aligned}$$

Thus, the object should be placed at a distance of 20 cm in front of this concave mirror.

Sample Problem 4. An object is placed at the following distances from a concave mirror of focal length 10 cm :

- (a) 8 cm (b) 15 cm (c) 20 cm (d) 25 cm

Which position of the object will produce :

- (i) a diminished real image ?
- (ii) a magnified real image ?
- (iii) a magnified virtual image ?
- (iv) an image of the same size as the object ?

Solution. In this case the focal length of concave mirror is 10 cm. This means that the focus (F) of this concave mirror is at a distance of 10 cm from the mirror and its centre of curvature (C) is at a distance of $2 \times 10 = 20$ cm from the mirror.

(i) A diminished real image is formed by a concave mirror when the object is beyond C. Here C is at 20 cm. So, the diminished real image will be formed when the object is at a distance greater than 20 cm, which in this problem is 25 cm. So, the position of object for a diminished real image is **25 cm**.

(ii) A magnified real image is formed by a concave mirror when the object is between F and C. Here F is at 10 cm and C is at 20 cm. So, the magnified real image will be formed when the object is at a distance between 10 cm and 20 cm, which in this problem is 15 cm. So, the position of object for a magnified real image is **15 cm**.

(iii) A magnified virtual image is formed by a concave mirror when the object is within focus (F) at a distance less than focal length or less than 10 cm, which in this problem is 8 cm. So, the position of object for a magnified virtual image is **8 cm**.

(iv) An image of the same size as object is formed by a concave mirror when the object is at centre of curvature (C). Here C is at 20 cm. So, the image of same size as the object will be formed when the object is at 20 cm from the concave mirror. Thus, the position of object for an image of same size as the object is **20 cm**.

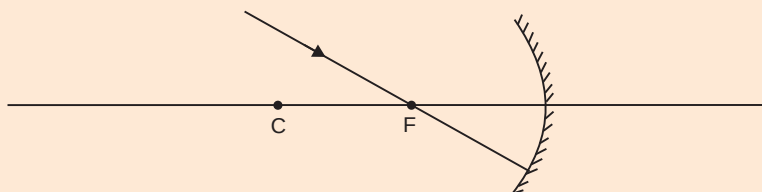
Please note that we have given so many details in answering the above question just to make you understand clearly. There is, however, no need to give so many details in the examination. In the examination, the answer to the above question can just be written as :

- (i) 25 cm (ii) 15 cm (iii) 8 cm (iv) 20 cm

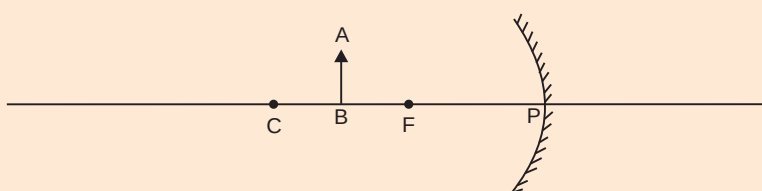
Before we go further and discuss the sign convention for spherical mirrors, **please answer the following questions :**

Very Short Answer Type Questions

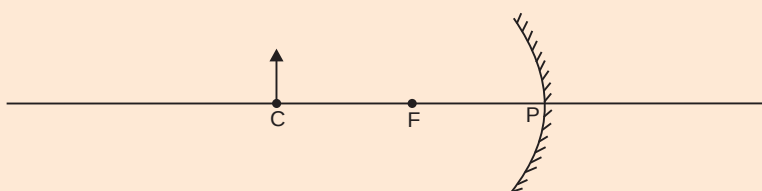
1. For what position of an object, a concave mirror forms a real image equal in size to the object ?
2. Where should an object be placed in front of the concave mirror so as to obtain its virtual, erect and magnified image ?
3. For which positions of the object does a concave mirror produce an inverted, magnified and real image ?
4. If an object is placed at the focus of a concave mirror, where is the image formed ?
5. If an object is at infinity (very large distance) in front of a concave mirror, where is the image formed ?
6. For what position of an object, a real and diminished image is formed by a concave mirror ?
7. Copy this figure in your answer book and show the direction of the light ray after reflection :



8. Draw the following diagram in your answer book and show the formation of image of the object AB with the help of suitable rays :



9. Draw the following diagram in your answer book and show the formation of image with the help of suitable rays :



10. Which type of mirror could be used as a dentist's mirror ?

Short Answer Type Questions

11. Which kind of mirror is used in the headlights of a car ? Why is it used for this purpose ?
12. Explain why, a ray of light passing through the centre of curvature of a concave mirror gets reflected back along the same path.
13. What is the minimum number of rays required for locating the image formed by a concave mirror for an object ? Draw a ray diagram to show the formation of a virtual image by a concave mirror.
14. With the help of a ray diagram, determine the position, nature and size of the image formed of an object placed at the centre of curvature of a concave mirror.
15. Describe with the help of a diagram, the nature, size and position of the image formed when an object is placed beyond the centre of curvature of a concave mirror.
16. If an object is placed at a distance of 8 cm from a concave mirror of focal length 10 cm, discuss the nature of the image formed by drawing the ray diagram.
17. Draw a ray diagram showing how a concave mirror can be used to produce a real, inverted and diminished image of an object.

18. Which mirror is used as a torch reflector ? Draw a labelled diagram to show how a torch reflector can be used to produce a parallel beam of light. Where is the bulb placed in relation to the torch reflector ?
19. State where an object must be placed so that the image formed by a concave mirror is :
 - (a) erect and virtual.
 - (b) at infinity.
 - (c) the same size as the object.
20. With the help of a labelled ray diagram, describe how a converging mirror can be used to give an enlarged upright image of an object.
21. Make labelled ray diagrams to illustrate the formation of :
 - (a) a real image by a converging mirror.
 - (b) a virtual image by a converging mirror.
 Mark clearly the pole, focus, centre of curvature and position of object in each case.
22. Briefly describe how you would find the focal length of a concave mirror quickly but approximately.
23. Which type of mirror is used in a solar furnace ? Support your answer with reason.
24. Name the type of mirror used by dentists. How does it help ?
25. Explain why, concave mirrors are used as shaving mirrors.
26. Give two uses of concave mirrors. Explain why you would choose concave mirrors for these uses.

Long Answer Type Questions

27. (a) Draw ray-diagrams to show the formation of images when the object is placed in front of a concave mirror (converging mirror) :
 - (i) between its pole and focus
 - (ii) between its centre of curvature and focus
 Describe the nature, size and position of the image formed in each case.
- (b) State one use of concave mirror based on the formation of image as in case (i) above.
28. (a) Give two circumstances in which a concave mirror can form a magnified image of an object placed in front of it. Illustrate your answer by drawing labelled ray diagrams for both.
- (b) Which one of these circumstances enables a concave mirror to be used as a shaving mirror ?

Multiple Choice Questions (MCQs)

29. The real image formed by a concave mirror is larger than the object when the object is :

(a) at a distance equal to radius of curvature	(b) at a distance less than the focal length
(c) between focus and centre of curvature	(d) at a distance greater than radius of curvature
30. The real image formed by a concave mirror is smaller than the object if the object is :

(a) between centre of curvature and focus	(b) at a distance greater than radius of curvature
(c) at a distance equal to radius of curvature	(d) at a distance equal to focal length
31. The image formed by a concave mirror is virtual, erect and magnified. The position of object is :

(a) at focus	(b) between focus and centre of curvature
(c) at pole	(d) between pole and focus
32. The image formed by a concave mirror is real, inverted and of the same size as the object. The position of the object must then be :

(a) at the focus	(b) between the centre of curvature and focus
(c) at the centre of curvature	(d) beyond the centre of curvature
33. The image formed by a concave mirror is real, inverted and highly diminished (much smaller than the object). The object must be :

(a) between pole and focus	(b) at focus
(c) at the centre of curvature	(d) at infinity
34. The angle of incidence for a ray of light passing through the centre of curvature of a concave mirror is :

(a) 45°	(b) 90°	(c) 0°	(d) 180°
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35. In the concave reflector of a torch, the bulb is placed :
 (a) between the pole and focus of reflector (b) at the focus of reflector
 (c) between focus and centre of curvature of reflector (d) at the centre of curvature of reflector
36. The focal length of a small concave mirror is 2.5 cm. In order to use this concave mirror as a dentist's mirror, the distance of tooth from the mirror should be :
 (a) 2.5 cm (b) 1.5 cm (c) 4.5 cm (d) 3.5 cm

Questions Based on High Order Thinking Skills (HOTS)

37. An object is 100 mm in front of a concave mirror which produces an upright image (erect image). The radius of curvature of the mirror is :
 (a) less than 100 mm (b) between 100 mm and 200 mm
 (c) exactly 200 mm (d) more than 200 mm
38. A virtual, erect and magnified image of an object is to be produced with a concave mirror of focal length 12 cm. Which of the following object distance should be chosen for this purpose ?
 (i) 10 cm (ii) 15 cm (iii) 20 cm
 Give reasons for your choice.
39. A concave mirror has a focal length of 25 cm. At which of the following distance should a person hold his face from this concave mirror so that it may act as a shaving mirror ?
 (a) 45 cm (b) 20 cm (c) 25 cm (d) 30 cm
 Give reason for your choice.
40. An object is placed at the following distances from a concave mirror of focal length 15 cm, turn by turn :
 (a) 35 cm (b) 30 cm (c) 20 cm (d) 10 cm
 Which position of the object will produce :
 (i) a magnified real image ?
 (ii) a magnified virtual image ?
 (iii) a diminished real image ?
 (iv) an image of same size as the object ?

ANSWERS

1. At the centre of curvature 2. Between pole and focus 3. Between focus and centre of curvature
 4. At infinity 5. At focus 6. Beyond centre of curvature 12. It falls normally (perpendicularly) on the mirror surface; The angle of incidence is 0° and the angle of reflection is also 0° 13. Two rays 16. Virtual, erect and magnified 29. (c) 30. (b) 31. (d) 32. (c) 33. (d) 34. (c) 35. (b) 36. (b) 37. (d) 38. 10 cm ; Because it is less than focal length 39. 20 cm 40. (i) 20 cm (ii) 10 cm (iii) 35 cm (iv) 30 cm.

SIGN CONVENTION FOR SPHERICAL MIRRORS

These days New Cartesian Sign Convention is used for measuring the various distances in the ray-diagrams of spherical mirrors (concave mirrors and convex mirrors). According to the New Cartesian Sign Convention :

- (i) All the distances are measured from *pole* of the mirror as origin.
- (ii) Distances measured in the *same* direction as that of incident light are taken as *positive*.
- (iii) Distances measured *against* the direction of incident light are taken as *negative*.
- (iv) Distances measured *upward* and perpendicular to the principal axis are taken as *positive*.
- (v) Distances measured *downward* and perpendicular to the principal axis are taken as *negative*.

The New Cartesian Sign Convention for mirrors is shown in Figure 44. **The object is always placed on the left side of the mirror** (as shown in Figure 44) so that the direction of incident light is from

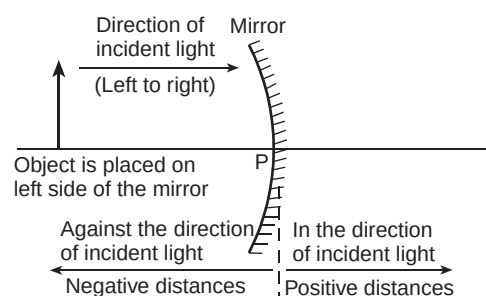


Figure 44. Sign convention for mirrors.

left to right. Since the incident light always goes from left to right, **all the distances measured from the pole (P) of mirror to the right side will be considered positive** (because they will be in the same direction as the incident light). On the other hand, **all the distances measured from pole (P) of mirror to the left side will be negative** (because they are measured against the direction of incident light).

Some Important Conclusions

According to the sign convention, the distances towards the left of the mirror are negative. Since an object is always placed to the left side of a mirror, therefore, **the object distance (u) is always negative**.

The images formed by a concave mirror can be either behind the mirror (virtual) or in front of the mirror (real). So, the image distance (v) for a concave mirror can be either positive or negative depending on the position of the image. **If an image is formed behind a concave mirror (to the right side), the image distance (v) is positive but if the image is formed in front of the mirror (on the left side), then the image distance will be negative.**

In a convex mirror, the image is always formed on the right hand side (behind the mirror), so **the image distance (v) for a convex mirror will be always positive**.

The focus of concave mirror is in front of the mirror on the left side, so **the focal length of a concave mirror is considered negative** (and written with a minus sign, say, -10 cm). On the other hand, the focus of a convex mirror is behind the mirror on the right side, so **the focal length of a convex mirror is positive** (and written with a plus sign, say $+20$ cm or just 20 cm).

We will now discuss the signs for the heights of objects and images. An object is always placed above the principal axis in the upward direction, so **the height of an object is always considered positive**. On the other hand, an image can be formed above the principal axis or below the principal axis. **If an image is formed above the principal axis, its height is taken as positive and if an image is formed below the principal axis, then its height is taken as negative**. Now, we know that all the virtual images are erect and are formed above the principal axis, so **the height of all the virtual and erect images is considered positive**. On the other hand, all the real images are inverted and they are formed below the principal axis, so **the height of all the real and inverted images is taken as negative**. We are now in a position to answer the following questions :

Very Short Answer Type Questions

- According to the "New Cartesian Sign Convention" for mirrors, what sign has been given to the focal length of :
(i) a concave mirror ?
(ii) a convex mirror ?
- Which type of mirror has :
(a) positive focal length ?
(b) negative focal length ?
- What is the nature of a mirror having a focal length of, $+10$ cm ?
- What kind of mirror can have a focal length of, -20 cm ?
- Complete the following sentence :
All the distances are measured from the of a spherical mirror.
- What sign (+ve or -ve) has been given to the following on the basis of Cartesian Sign Convention ?
(a) Height of a real image.
(b) Height of a virtual image.

Short Answer Type Questions

- Describe the New Cartesian Sign Convention used in optics. Draw a labelled diagram to illustrate this sign convention.

8. Giving reasons, state the 'signs' (positive or negative) which can be given to the following :
- object distance (u) for a concave mirror or convex mirror
 - image distances (v) for a concave mirror
 - image distances (v) for a convex mirror

Multiple Choice Questions (MCQs)

9. According to New Cartesian Sign Convention :
- focal length of concave mirror is positive and that of convex mirror is negative
 - focal length of both concave and convex mirrors is positive
 - focal length of both concave and convex mirrors is negative
 - focal length of concave mirror is negative and that of convex mirror is positive
10. One of the following does not apply to a concave mirror. This is :
- focal length is negative
 - image distance can be positive or negative
 - image distance is always positive
 - height of image can be positive or negative

ANSWERS

3. Convex mirror

4. Concave mirror

5. pole

9. (d)

10. (c)

MIRROR FORMULA

The distance of an object from the pole of a mirror is known as object distance. Object distance is denoted by the letter u . The distance of image from the pole of a mirror is known as image distance. Image distance is denoted by the letter v . The distance of focus from the pole of a mirror is known as focal length. Focal length is denoted by the letter f . There is a relationship between the object distance, image distance and focal length of a spherical mirror (concave mirror or convex mirror). This relationship is given by the mirror formula.

A formula which gives the relationship between image distance (v), object distance (u) and focal length (f) of a spherical mirror is known as the mirror formula. The mirror formula can be written as :

$$\frac{1}{\text{Image distance}} + \frac{1}{\text{Object distance}} = \frac{1}{\text{Focal length}}$$

$$\text{or} \quad \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

where v = distance of image from mirror

u = distance of object from mirror

and f = focal length of the mirror

The mirror formula has three values in it. If any two values are known, the third value can be calculated. The known values should be put in this formula with their proper signs but no sign should be given to the unknown value to be calculated. Its proper sign will come by itself by calculations. The proper signs of the known values of u , v or f can be found by using New Cartesian Sign Convention for spherical mirrors. Another point to be noted is that the mirror formula can be applied to both type of spherical mirrors : concave mirrors as well as convex mirrors. By using this mirror formula, we can find out the position and nature of the images formed by concave mirrors and convex mirrors. Before we do that, we will discuss the 'magnification' produced by spherical mirrors.

Linear Magnification Produced by Mirrors

The size of image formed by a spherical mirror depends on the position of the object from the mirror. The image formed by a spherical mirror can be bigger than the object, equal to the object or smaller than the object. The size of the image relative to the object is given by the linear magnification. This is discussed on the next page.

The ratio of the height of image to the height of object is known as linear magnification. That is,

$$\text{Magnification} = \frac{\text{height of image}}{\text{height of object}}$$

$$\text{or } m = \frac{h_2}{h_1}$$

where m = magnification

h_2 = height of image

and h_1 = height of object

In our ray-diagrams, the object is always placed above the principal axis, so **the height (h_1) of the object will always be positive**. We also know that a virtual image is always formed above the principal axis, therefore, **the height (h_2) of a virtual image will be positive**. But a real image is formed below the principal axis, so **the height (h_2) of a real image will be negative** because it is measured in the downward direction. From this discussion we conclude that though the height of object h_1 is always positive, the height h_2 of the image can be either positive or negative.

Now, for a virtual image h_2 is positive and h_1 is also positive, so the magnification $\left(\frac{h_2}{h_1}\right)$ of a virtual (and erect) image is always positive. In other words, **if the magnification has a plus sign, then the image is virtual and erect**. For a real image, h_2 is negative and h_1 is positive, so the magnification $\left(\frac{h_2}{h_1}\right)$ for a real (and inverted) image is always negative. In other words, **if the magnification has a minus sign, then the image is real and inverted**.

Since a concave mirror can produce virtual images as well as real images, the magnification produced by a concave mirror can be either positive or negative. A convex mirror, however, forms only virtual images, so the magnification produced by a convex mirror is always positive. Another point to be noted is that if the magnification m has a value greater than 1 then the image is bigger than the object, that is, the image is magnified or enlarged. And if the magnification m is exactly 1, then the image is of the same size as the object. But if the magnification is less than 1 then the image is smaller than the object (or diminished).

A concave mirror can form images which are smaller than the object, equal to the object or bigger than the object, therefore, the linear magnification (or just magnification) (m) produced by a concave mirror can be less than 1, equal to 1 or more than 1. On the other hand, a convex mirror forms images which are always smaller than the object, so the linear magnification (m) produced by a convex mirror is always less than 1. A plane mirror forms images which are always of the same size as the object, therefore, the magnification (m) produced by a plane mirror is always 1.

If we know the height (or size) of the object and that of the image, then we can calculate the magnification by using the formula given above. Many times, however, we do not know their heights, so we will now write another formula for calculating the magnification produced by a spherical mirror in terms of "object distance" and "image distance".

The linear magnification produced by a mirror is equal to the ratio of the image distance to the object distance, with a minus sign. That is,

$$\text{Magnification} = - \frac{\text{Image distance}}{\text{Object distance}}$$

$$\text{or } m = - \frac{v}{u}$$

where m = magnification

v = image distance

and u = object distance

Thus, if we know the image distance v and the object distance u , then the magnification m can be calculated.

So, now we have two formulae for calculating the magnification :

$$m = \frac{h_2}{h_1} \quad \text{and} \quad m = -\frac{v}{u}$$

We will use these two formulae to solve numerical problems. We can also combine these two formulae to get another formula :

$$\frac{h_2}{h_1} = -\frac{v}{u}$$

This will also be used in solving problems.

Numerical Problems Based on Concave Mirrors

We will now solve some numerical problems based on concave mirrors by using the mirror formula and the magnification formulae. Here are some examples.

Sample Problem 1. Find the size, nature and position of image formed when an object of size 1 cm is placed at a distance of 15 cm from a concave mirror of focal length 10 cm.

Solution. Here we have been given the object distance and focal length, so first of all we will find out the image distance which will give us the position of image.

(i) Position of image

Here, Object distance, $u = -15$ cm (To the left of mirror)
 Image distance, $v = ?$ (To be calculated)
 And, Focal length, $f = -10$ cm (It is concave mirror)

Now, putting these values in the mirror formula :

$$\begin{aligned} \frac{1}{v} + \frac{1}{u} &= \frac{1}{f} \\ \text{we get : } \frac{1}{v} + \frac{1}{-15} &= \frac{1}{-10} \\ \text{or } \frac{1}{v} - \frac{1}{15} &= -\frac{1}{10} \\ \text{or } \frac{1}{v} &= -\frac{1}{10} + \frac{1}{15} \\ \frac{1}{v} &= \frac{-3+2}{30} \\ \frac{1}{v} &= -\frac{1}{30} \\ \text{So, Image distance, } v &= -30 \text{ cm} \end{aligned}$$

Thus, the position of image is 30 cm to the left side of mirror or 30 cm in front of mirror (Minus sign shows the left side of mirror).

(ii) Nature of image. Since the image is formed in front of the concave mirror, its nature will be "Real and Inverted".

(iii) Size of image. To find the size of image, we will have to calculate the magnification first. The magnification produced by a mirror is given by :

$$m = -\frac{v}{u}$$

Here, Image distance, $v = -30$ cm

Object distance, $u = -15$ cm

$$\text{So, } m = -\frac{(-30)}{(-15)}$$

$$m = -\frac{30}{15}$$

$$\text{Magnification, } m = -2$$

We also have another formula for magnification, which is :

$$m = \frac{h_2}{h_1}$$

Here, Magnification, $m = -2$ (Found above)

Height of image, $h_2 = ?$ (To be calculated)

Height of object, $h_1 = 1 \text{ cm}$ (Given)

Now, putting these values in the above magnification formula, we get :

$$-2 = \frac{h_2}{1}$$

$$\text{So, Height of image, } h_2 = -2 \times 1 \\ = -2 \text{ cm}$$

Thus, the size of image is 2 cm long. The minus sign here shows that the image is formed below the principal axis. That is, it is a real and inverted image.

Sample Problem 2. An object 2 cm high is placed at a distance of 16 cm from a concave mirror which produces a real image 3 cm high.

(i) What is the focal length of the mirror ?

(ii) Find the position of the image.

Solution. (i) **Calculation of position of image**

The height of object is 2 cm and that of real image is 3 cm. So, we will first calculate the magnification by using the formula :

$$m = \frac{h_2}{h_1}$$

Please note that an object is always placed above the axis, so the height of object is always taken as positive and written with a plus sign. A real image is formed below the axis. So, the height of a real image is taken as negative and written with a minus sign.

Here, Height of real image, $h_2 = -3 \text{ cm}$

Height of object, $h_1 = +2 \text{ cm}$

So, putting these values in the above magnification formula, we get :

$$\text{Magnification, } m = \frac{-3}{+2}$$

$$\text{or } m = -1.5$$

Now, we have another magnification formula for mirrors, which is :

$$m = -\frac{v}{u}$$

Here, Magnification, $m = -1.5$ (Found above)

Image distance, $v = ?$ (To be calculated)

and Object distance, $u = -16 \text{ cm}$ (To the left of mirror)

So, putting these values in the above formula, we get :

$$-1.5 = -\frac{v}{(-16)}$$

$$\text{or } -1.5 = \frac{v}{16}$$

$$\text{So, } v = -1.5 \times 16$$

$$\text{or } v = -24 \text{ cm}$$

Thus, the position of image is 24 cm in front of the mirror, to the left side of mirror (The minus sign shows that the image is on the left side of the mirror).

(ii) Calculation of focal length

Here, Object distance, $u = -16$ cm
 Image distance, $v = -24$ cm
 Focal length, $f = ?$ (To be calculated)

Now, putting these values in the mirror formula :

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

we get : $\frac{1}{-24} + \frac{1}{-16} = \frac{1}{f}$

or $-\frac{1}{24} - \frac{1}{16} = \frac{1}{f}$

$$\frac{-2-3}{48} = \frac{1}{f}$$

$$-\frac{5}{48} = \frac{1}{f}$$

$$f = -\frac{48}{5}$$

So, Focal length, $f = -9.6$ cm

Sample Problem 3. A concave mirror produces three times magnified (enlarged) real image of an object placed at 10 cm in front of it. Where is the image located ? **(NCERT Book Question)**

Solution. In this problem we have been given magnification (m) and object distance (u). We are to find out the image distance (v). Now,

Magnification, $m = -3$ (Image is real)
 Object distance, $u = -10$ cm (To the left of mirror)

And, Image distance, $v = ?$ (To be calculated)

Putting these values in the magnification formula for a mirror :

$$m = -\frac{v}{u}$$

we get : $-3 = -\frac{v}{-10}$

$$-3 \times -10 = -v$$

$$v = -30 \text{ cm}$$

Thus, the image is located at a distance of 30 cm in front of the mirror (on its left side).

Sample Problem 4. The magnification produced by a plane mirror is +1. What does this mean ? **(NCERT Book Question)**

Answer. The plus sign (+) of the magnification shows that the image is virtual and erect. And the value 1 for magnification shows that the image is exactly of the same size as the object. So, the magnification of +1 produced by a plane mirror means that the image formed in a plane mirror is virtual and erect, and of the same size as the object.

Sample Problem 5. What is the nature of the image formed by a concave mirror if the magnification produced by the mirror is +3 ?

Answer. If the magnification has a plus sign (+), then the image is virtual and erect. In this case, the magnification has a plus sign (it is +3), therefore, the nature of this image is virtual and erect.

Sample Problem 6. What is the nature of the image formed by a concave mirror if the magnification produced by the mirror is, -0.75 ?

Answer. If the magnification has a minus sign ($-$), then the image is real and inverted. In this case, the magnification has a minus sign (it is, -0.75), so the nature of image is real and inverted.

Before we go further and discuss the images formed by convex mirrors, **please answer the following questions and problems yourself :**

Very Short Answer Type Questions

1. If the magnification of a body of size 1 m is 2, what is the size of the image ?
2. What is the position of the image when an object is placed at a distance of 20 cm from a concave mirror of focal length 20 cm ?
3. What is the nature of image formed by a concave mirror if the magnification produced by the mirror is (a) $+4$, and (b) -2 ?
4. State the relation between object distance, image distance and focal length of a spherical mirror (concave mirror or convex mirror).
5. Write the mirror formula. Give the meaning of each symbol which occurs in it.
6. What is the ratio of the height of an image to the height of an object known as ?
7. Define linear magnification produced by a mirror.
8. Write down a formula for the magnification produced by a concave mirror.
(a) in terms of height of object and height of image
(b) in terms of object distance and image distance
9. Describe the nature of image formed when the object is placed at a distance of 20 cm from a concave mirror of focal length 10 cm.
10. Fill in the following blanks with suitable words :
(a) If the magnification has a plus sign, then image is.....and.....
(b) If the magnification has a minus sign, then the image is.....and.....

Short Answer Type Questions

11. An object is placed at a distance of 10 cm from a concave mirror of focal length 20 cm.
(a) Draw a ray diagram for the formation of image.
(b) Calculate the image distance.
(c) State two characteristics of the image formed.
12. If an object of 10 cm height is placed at a distance of 36 cm from a concave mirror of focal length 12 cm, find the position, nature and height of the image.
13. At what distance from a concave mirror of focal length 10 cm should an object 2 cm long be placed in order to get an erect image 6 cm tall ?
14. When an object is placed at a distance of 15 cm from a concave mirror, its image is formed at 10 cm in front of the mirror. Calculate the focal length of the mirror.
15. An object 3 cm high is placed at a distance of 8 cm from a concave mirror which produces a virtual image 4.5 cm high :
(i) What is the focal length of the mirror ?
(ii) What is the position of image ?
(iii) Draw a ray-diagram to show the formation of image.
16. A converging mirror forms a real image of height 4 cm of an object of height 1 cm placed 20 cm away from the mirror :
(i) Calculate the image distance.
(ii) What is the focal length of the mirror ?
17. An object of size 7.0 cm is placed at 27 cm in front of a concave mirror of focal length 18 cm. At what distance from the mirror should a screen be placed so that a sharp focussed image can be obtained ? Find the size and nature of image.

[Hint. Find the value of image distance (v) first. The screen should be placed from the mirror at a distance equal to image distance].

18. An object 3 cm high is placed at a distance of 10 cm in front of a converging mirror of focal length 20 cm. Find the position, nature and size of the image formed.
19. A concave mirror has a focal length of 4 cm and an object 2 cm tall is placed 9 cm away from it. Find the nature, position and size of the image formed.
20. When an object is placed 20 cm from a concave mirror, a real image magnified three times is formed. Find :
(a) the focal length of the mirror.
(b) Where must the object be placed to give a virtual image three times the height of the object ?
21. A dentist's mirror has a radius of curvature of 3 cm. How far must it be placed from a small dental cavity to give a virtual image of the cavity that is magnified five times ?
22. A large concave mirror has a radius of curvature of 1.5 m. A person stands 10 m in front of the mirror. Where is the person's image ?
23. An object of 5.0 cm size is placed at a distance of 20.0 cm from a converging mirror of focal length 15.0 cm. At what distance from the mirror should a screen be placed to get the sharp image ? Also calculate the size of the image.
24. A concave mirror produces three times enlarged virtual image of an object placed at 10 cm in front of it. Calculate the radius of curvature of the mirror.
25. A bright object 50 mm high stands on the axis of a concave mirror of focal length 100 mm and at a distance of 300 mm from the concave mirror. How big will the image be ?
26. How far should an object be placed from the pole of a converging mirror of focal length 20 cm to form a real image of the size exactly $\frac{1}{4}$ th the size of the object ?
27. When an object is placed at a distance of 50 cm from a concave spherical mirror, the magnification produced is, $-\frac{1}{2}$. Where should the object be placed to get a magnification of, $-\frac{1}{5}$?
28. An object is placed (a) 20 cm, (b) 4 cm, in front of a concave mirror of focal length 12 cm. Find the nature and position of the image formed in each case.
29. A concave mirror produces a real image 1 cm tall of an object 2.5 mm tall placed 5 cm from the mirror. Find the position of the image and the focal length of the mirror.
30. A man holds a spherical shaving mirror of radius of curvature 60 cm, and focal length 30 cm, at a distance of 15 cm, from his nose. Find the position of image, and calculate the magnification.

Long Answer Type Question

31. (a) An object is placed just outside the principal focus of concave mirror. Draw a ray diagram to show how the image is formed, and describe its size, position and nature.
(b) If the object is moved further away from the mirror, what changes are there in the position and size of the image ?
(c) An object is 24 cm away from a concave mirror and its image is 16 cm from the mirror. Find the focal length and radius of curvature of the mirror, and the magnification of the image.

Multiple Choice Questions (MCQs)

32. Linear magnification produced by a concave mirror may be :
(a) less than 1 or equal to 1 (b) more than 1 or equal to 1
(c) less than 1, more than 1 or equal to 1 (d) less than 1 or more than 1
33. Magnification produced by a convex mirror is always :
(a) more than 1 (b) less than 1 (c) equal to 1 (d) more or less than 1
34. Magnification produced by a plane mirror is :
(a) less than one (b) greater than one (c) zero (d) equal to one
35. In order to obtain a magnification of, -2 (minus 2) with a concave mirror, the object should be placed :
(a) between pole and focus (b) between focus and centre of curvature
(c) at the centre of curvature (d) beyond the centre of curvature
36. A concave mirror produces a magnification of $+4$. The object is placed :
(a) at the focus (b) between focus and centre of curvature
(c) between focus and pole (d) beyond the centre of curvature

37. If a magnification of, -1 (minus one) is to be obtained by using a converging mirror, then the object has to be placed :
 (a) between pole and focus (b) at the centre of curvature
 (c) beyond the centre of curvature (d) at infinity
38. In order to obtain a magnification of, -0.6 (minus 0.6) with a concave mirror, the object must be placed :
 (a) at the focus (b) between pole and focus
 (c) between focus and centre of curvature (d) beyond the centre of curvature
39. An object is placed at a large distance in front of a concave mirror of radius of curvature 40 cm. The image will be formed in front of the mirror at a distance of :
 (a) 20 cm (b) 30 cm (c) 40 cm (d) 50 cm
40. In order to obtain a magnification of, -1.5 with a concave mirror of focal length 16 cm, the object will have to be placed at a distance :
 (a) between 6 cm and 16 cm (b) between 32 cm and 16 cm
 (c) between 48 cm and 32 cm (d) beyond 64 cm
41. Linear magnification (m) produced by a rear view mirror fitted in vehicles :
 (a) is equal to one (b) is less than one
 (c) is more than one (d) can be more or less than one depending on the position of object

Questions Based on High Order Thinking Skills (HOTS)

42. Between which two points of concave mirror should an object be placed to obtain a magnification of :
 (a) -3 (b) $+2.5$ (c) -0.4
43. At what distance from a concave mirror of focal length 10 cm should an object be placed so that :
 (a) its real image is formed 20 cm from the mirror ?
 (b) its virtual image is formed 20 cm from the mirror ?
44. If a concave mirror has a focal length of 10 cm, find the two positions where an object can be placed to give, in each case, an image twice the height of the object.
45. A mirror forms an image which is 30 cm from an object and twice its height.
 (a) Where must the mirror be situated ?
 (b) What is the radius of curvature ?
 (c) Is the mirror convex or concave ?

ANSWERS

1. 2 m 2. At infinity 3. (a) Virtual and erect (b) Real and inverted 6. Magnification
 9. Real and inverted 10. (a) virtual ; erect (b) real ; inverted 11. (b) 20 cm (c) Virtual and erect ; Magnified 12. $v = -18$ cm ; The position of image is 18 cm in front of concave mirror (to its left side); Real and inverted ; 5 cm 13. $u = -6.66$ cm ; The object should be placed at a distance of 6.66 cm on the left side of concave mirror 14. 6 cm 15. (i) 24 cm (ii) 12 cm behind the concave mirror (on its right side) 16. (i) 80 cm in front of the mirror (ii) 16 cm 17. $v = -54$ cm ; The screen should be placed at a distance of 54 cm in front of the concave mirror ; 14.0 cm; Real and inverted 18. $v = +20$ cm ; The image is formed at a distance of 20 cm behind the converging mirror; Virtual and erect ; 6 cm 19. Real and inverted ; $v = -7.2$ cm : The image is formed at a distance of 7.2 cm in front of concave mirror ; 1.6 cm 20. (a) 15 cm (b) 10 cm from the concave mirror ; 21. 1.2 cm 22. 0.81 m in front of the concave mirror 23. 60 cm in front of the converging mirror ; 15.0 cm 24. 30 cm 25. 25 mm 26. 100 cm 27. 100 cm 28. (a) $v = -30$ cm ; The image is formed at a distance of 30 cm in front of mirror (on its left side); Real and inverted (b) $v = +6$ cm ; The image is formed at a distance of 6 cm behind the mirror (on its right side); Virtual and erect 29. $v = -20$ cm ; The image is formed 20 cm in front of the mirror ; 4 cm 30. $v = +30$ cm ; The image is formed at a distance of 30 cm behind the mirror ; $m = +2$ 31. (c) 9.6 cm ; 19.2 cm ; 0.66 32. (c) 33. (b) 34. (d) 35. (b) 36. (c) 37. (b) 38. (d) 39. (a) 40. (b) 41. (b) 42. (a) Between focus and centre of curvature (b) Between pole and focus (c) Beyond the centre of curvature 43. (a) 20 cm (b) $\frac{20}{3}$ cm 44. 15 cm ; 5 cm 45. (a) 10 cm from the object (b) 40 cm (c) Concave mirror

RULES FOR OBTAINING IMAGES FORMED BY CONVEX MIRRORS

In order to construct ray-diagrams to find out the position, nature and size of the images formed by a convex mirror, we should remember the paths of the following rays of light. We can call them the rules for obtaining images in convex mirrors.

Rule 1. A ray of light which is parallel to the principal axis of a convex mirror, appears to be coming from its focus after reflection from the mirror. This is shown in Figure 45. In Figure 45, the ray of light AB is parallel to the principal axis XP of a convex mirror. The ray of light AB gets reflected at point B on the mirror and goes in the direction BD . To a person on the left side, the reflected ray BD appears to be coming from the focus F of the convex mirror situated behind the mirror (as shown by dotted line).

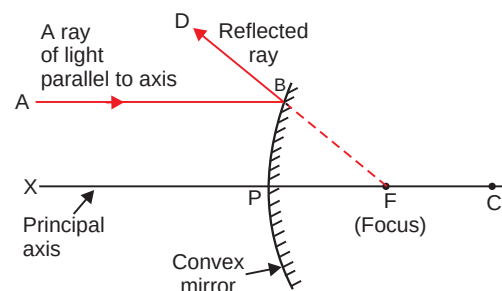


Figure 45.

Rule 2. A ray of light going towards the centre of curvature of a convex mirror is reflected back along the same path. This is shown in Figure 46. In Figure 46, the ray of light AD is going towards the centre of curvature C of a convex mirror. It strikes the mirror surface at point D and gets reflected back along the same path DA . To a person on the left side, the reflected ray DA appears to be coming from the centre of curvature C (as shown by dotted line behind the mirror). Please note that the ray AD gets reflected back along the same path because it falls normally (or perpendicularly) on the mirror surface at point D .

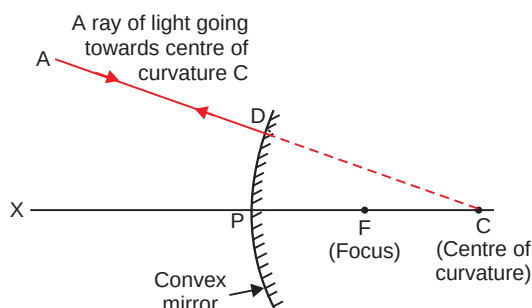


Figure 46.

Please note that if a ray of light is incident on a convex mirror along its principal axis, then it is reflected back along the same path (because it will be normal or perpendicular to the mirror surface). The angle of incidence as well as the angle of reflection for such a ray of light will be zero.

Rule 3. A ray of light going towards the focus of a convex mirror becomes parallel to the principal axis after reflection. This is just the reverse case of the first rule and it is shown in Figure 47. Here the incident ray of light AE is going towards the focus F of the convex mirror. It strikes the mirror surface at point E and gets reflected. After reflection, it becomes parallel to the principal axis and goes in the direction EG (see Figure 47).

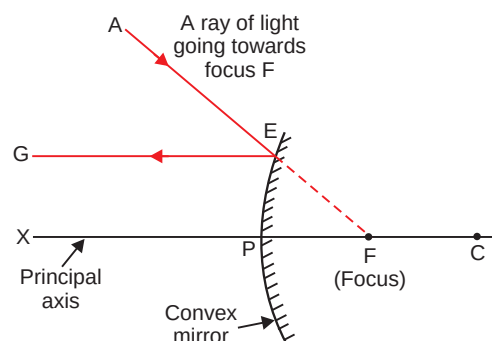


Figure 47.

Rule 4. A ray of light which is incident at the pole of a convex mirror is reflected back making the same angle with the principal axis. This is shown in Figure 48. Here a ray of light AP is incident on the pole P of the convex mirror making an angle of incidence i with the principal axis XP . It gets reflected along the direction PH making an equal angle of reflection r with the principal axis (see Figure 48).

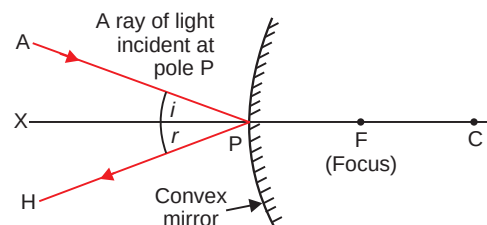


Figure 48.

It should be noted that a convex mirror has its focus and centre of curvature behind it. Since no real rays of light can go behind the convex mirror, all the rays shown behind the convex mirror are virtual (or unreal) and hence they have been represented by dotted lines. In fact, no actual rays can pass through the focus and centre of curvature of a convex mirror.

Please note that **whatever be the position of object in front of a convex mirror, the image formed by a convex mirror is always behind the mirror, it is virtual, erect and smaller than the object (or diminished).** When the distance of the object is changed from convex mirror, then only the position and size of the image changes. There are two main positions of an object in the case of a convex mirror from the point of view of position and size of image. The object can be :

- (i) anywhere between pole (P) and infinity, and
- (ii) at infinity.

We will discuss both these cases one by one. Let us first describe the formation of image by a convex mirror when the object is placed between pole (P) of the mirror and infinity.

FORMATION OF IMAGE BY A CONVEX MIRROR

In Figure 49, we have an object AB placed in front of a convex mirror M anywhere between pole P and infinity. A ray of light AD , parallel to the principal axis of the convex mirror, strikes the mirror at point D . Now, according to the first rule of image formation, this parallel ray of light should appear to be coming from focus F after reflection. So, we join the points D and F by a dotted line and produce the line FD towards the left in the direction DX . Now, DX gives us the reflected ray which appears to be coming from focus F of the convex mirror.

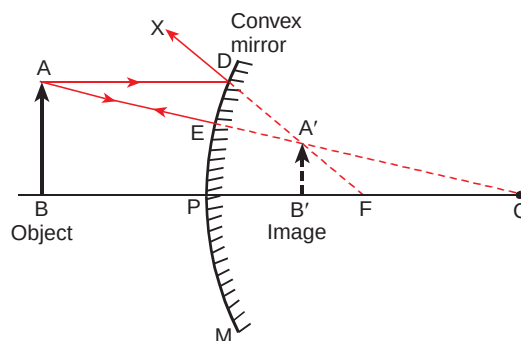


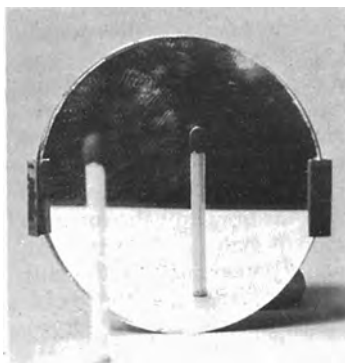
Figure 49. Formation of image by a convex mirror when the object is placed anywhere between the pole of mirror and infinity.

We have now to draw a second ray of light from the point A going towards the centre of curvature C of the convex mirror. For this we join the point A with point C by a line which cuts the mirror at point E . The line from A to E is a solid line and it represents a real ray of light but the line from E to C is a dotted line which represents a virtual ray of light. Now, AE represents a ray of light going towards the centre of curvature C of the convex mirror. According to the second rule of image formation, this ray is reflected along the same path EA but it appears to be coming from the centre of curvature C .

The two reflected rays DX and EA are diverging rays but they appear to intersect at point A' when produced backwards. Thus A' is the virtual image of point A of the object. To get the full image of the object, we draw the perpendicular $A'B'$ to the axis from point A' . Thus $A'B'$ is the virtual image of the object AB . It is clear from Figure 49 that the image is formed behind the convex mirror between the pole and the focus. It is virtual, erect and smaller than the object (or diminished). From the above discussion we conclude that : **When an object is placed anywhere between pole (P) and infinity in front of a convex mirror, the image formed is :**

- (i) **behind the mirror between pole (P) and focus (F),**
- (ii) **virtual and erect, and**
- (iii) **diminished (smaller than the object).**

If we hold a matchstick (as object) in front of a convex mirror, a virtual, erect and diminished (smaller) image of the matchstick is seen on looking into the convex mirror [see Figure 50(a)]. We know that the back side of a shining steel spoon (which is bulging outwards) is a kind of convex mirror. So, if we keep our face in front of the back side of a shining steel spoon we will see a virtual and erect image of the face which is smaller in size as compared to the face [see Figure 50 (b)].



(a) The image of matchstick in the convex mirror is virtual, erect and diminished (smaller in size).



(b) The image of face in the backside (convex surface) of a shining steel spoon is virtual, erect and diminished.

Figure 50.

If we move the object more and more away from the pole of the convex mirror, the image becomes smaller and smaller in size and moves away from the mirror towards its focus but it remains virtual and erect for all the positions of the object. And *when the object is at infinity, the image is formed at the focus*. This is discussed below.

When the Object is at Infinity

When the object is at a far-off distance, we say that the object is at infinity. In Figure 51, we have a convex mirror M . Suppose an object (an arrow pointing upwards) has been placed at infinity in front of the convex mirror (Since the object is very far-off, it cannot be shown in the diagram). Because the object AB is very far-off, the two rays AD and AP coming from its top point A are parallel to one another but at an angle to the principal axis as shown in Figure 51. The ray AD gets reflected in the direction DX and the ray AP gets reflected in the direction PY . When the diverging reflected rays DX and PY are produced backwards (as shown by dotted lines in Figure 51), they intersect at point A' in the focal plane of the convex mirror. Thus, A' is the virtual image of the top point A of the object. To get the full image of the object, we draw $A'B'$ perpendicular to the axis. So, $A'B'$ is the image of the object AB placed at infinity. We find that the image is formed at the focus (F) of the convex mirror behind the mirror. It is virtual, erect and highly diminished (much smaller than the object). From the above discussion we conclude that : **When an object is at infinity from a convex mirror, the image formed is :**

- (i) **behind the mirror at focus (F),**
- (ii) **virtual and erect, and**
- (iii) **highly diminished (much smaller than the object).**

Since the image of a distant object formed by a convex mirror is highly diminished, we can see the full image of a tall building or tree even in a small convex mirror. Please note that when the object kept at infinity in front of a convex mirror is assumed to be a big arrow pointing upwards, then its image is formed at focus according to the ray-diagram shown in Figure 51. If, however, the object kept at infinity in front of a convex mirror is round in shape, then its

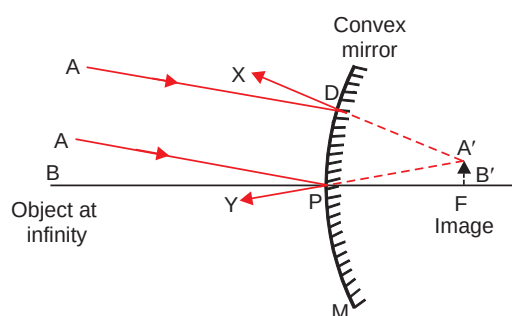


Figure 51. Formation of image by a convex mirror when the object is at infinity (large distance).



image is formed at the focus according to the ray diagram shown in Figure 27 on page 178. And before we conclude this discussion, here is a summary of the images formed by a convex mirror.

Summary of the Images Formed by a Convex Mirror

Position of object	Position of image	Size of image	Nature of image
1. Anywhere between pole P and infinity	Behind the mirror between P and F	Diminished	Virtual and erect
2. At infinity	Behind the mirror at focus (F)	Highly diminished	Virtual and erect

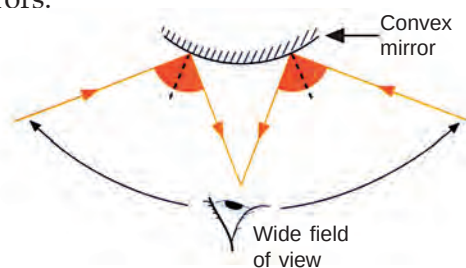
Uses of Convex Mirrors

(i) Convex mirrors are used as rear-view mirrors in vehicles (like cars, trucks and buses) to see the traffic at the rear side (or back side) (see Figure 52). A driver prefers to use a convex mirror as rear-view mirror because of two reasons :

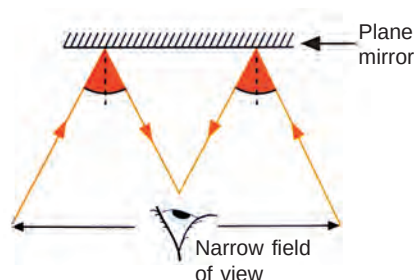
- A convex mirror always produces an erect (right side up) image of the objects.
- The image formed in a convex mirror is highly diminished or much smaller than the object, due to which a convex mirror gives a wide field of view (of the traffic behind) [see Figure 53(a)]. A convex mirror enables a driver to view much larger area of the traffic behind him than would be possible with a plane mirror. A plane mirror gives a narrow field of view [see Figure 53(b)]. Due to this if a plane mirror is used as a rear-view mirror in vehicles, it will give the driver a much smaller view of the road and traffic behind. Rear-view mirrors are also known as driving mirrors or side-view mirrors or wing mirrors.



Figure 52. Convex mirror is used as a rear view mirror (or side view mirror) in vehicles.



(a) A convex mirror gives a wide field of view (or large field of view).



(b) A plane mirror gives a narrow field of view (or small field of view).

Figure 53.

Please note that **we cannot use a concave mirror as a rear-view mirror in motor vehicles**. This is because a concave mirror produces inverted images (upside down images) of distant objects. So, if we use a concave mirror as a rear view mirror in a car (bus or truck, etc.) we will see in the mirror that all the vehicles on the road (at the back side) are running upside down with their wheels up in the air. This will be really a very funny situation to watch.

(ii) Big convex mirrors are used as 'shop security mirrors' (see Figure 54). By installing a big convex mirror at a strategic point in the shop, the shop owner can keep an eye on the customers to look for thieves and shoplifters among them.



Figure 54. This picture shows a big convex mirror installed as a security mirror in a shop. A large number of goods displayed in the shop can be seen in this convex mirror.

How to Distinguish Between a Plane Mirror, a Concave Mirror and a Convex Mirror Without Touching Them

We can distinguish between these mirrors just by looking into them, that is, by bringing our face close to each mirror, turn by turn. All of them will produce an image of our face but of different types. **A plane mirror will produce an image of the same size** as our face and we will look our normal self. **A concave mirror will produce a magnified image** and our face will look much bigger (like that of a giant !). **A convex mirror will produce a diminished image** and our face will look much smaller (like that of a small child !). Let us answer one question now.

Sample Problem. No matter how far you stand from a mirror, your image appears erect. The mirror may be :

- (i) plane (ii) concave (iii) convex (iv) either plane or convex

Choose the correct alternative.

(NCERT Book Question)

Answer. The correct alternative is : (iv) either plane or convex.

We are now in a position to **answer the following questions** :

Very Short Answer Type Questions

- What type of image/images are formed by :
(a) a convex mirror ?
(b) a concave mirror ?
- Which mirror has a wider field of view ?
- If you want to see an enlarged image of your face, state whether you will use a concave mirror or a convex mirror ?
- Which mirror always produces a virtual, erect and diminished image of an object ?
- An object is placed at a long distance in front of a convex mirror of radius of curvature 30 cm. State the position of its image.
- Name the spherical mirror which can produce a real and diminished image of an object.
- Name the spherical mirror which can produce a virtual and diminished image of an object.
- One wants to see a magnified image of an object in a mirror. What type of mirror should one use ?
- Name the mirror which can give :
(a) an erect and enlarged image of an object.
(b) an erect and diminished image of an object.
- State whether the following statement is true or false :
A diverging mirror is used as a rear-view mirror.
- What type of mirror could be used :
(a) as a shaving mirror ?
(b) as a shop security mirror ?
- Which type of mirror is usually used as a rear-view mirror in motor cars ?
- What kind of mirrors are used in big shopping centres to watch the activities of the customers ?
- A ray of light going towards the focus of a convex mirror becomes parallel to the principal axis after reflection from the mirror. Draw a labelled diagram to represent this situation.
- Fill in the following blank with a suitable word :
A ray of light which is parallel to the principal axis of a convex mirror, appears to be coming from.....after reflection from the mirror.

Short Answer Type Questions

- Why does a driver prefer to use a convex mirror as a rear-view mirror in a vehicle ?
- Why can you not use a concave mirror as a rear-view mirror in vehicles ?
- Where would the image be formed by a convex mirror if the object is placed :

(a) between infinity and pole of the mirror ?

(b) at infinity ?

Draw labelled ray-diagrams to show the formation of image in both the cases.

19. The shiny outer surface of a hollow sphere of aluminium of radius 50 cm is to be used as a mirror :

(a) What will be the focal length of this mirror ?

(b) Which type of spherical mirror will it provide ?

(c) State whether this spherical mirror will diverge or converge light rays.

20. What is the advantage of using a convex mirror as a rear-view mirror in vehicles as compared to a plane mirror ? Illustrate your answer with the help of labelled diagrams.

21. Give two uses of a convex mirror. Explain why you would choose convex mirror for these uses.

22. What would your image look like if you stood close to a large :

(a) convex mirror ?

(b) concave mirror ?

Give reasons for your answer.

23. Which of the following are concave mirrors and which convex mirrors ?

Shaving mirrors, Car headlight mirror, Searchlight mirror, Driving mirror, Dentist's inspection mirror, Torch mirror, Staircase mirror in a double-decker bus, Make-up mirror, Solar furnace mirror, Satellite TV dish, Shop security mirror.

24. How will you distinguish between a plane mirror, a concave mirror and a convex mirror without touching them ?

25. If a driver has one convex and one plane rear-view mirror, how would the images in each mirror appear different ?

Long Answer Type Questions

26. (a) Draw a labelled ray diagram to show the formation of image of an object by a convex mirror. Mark clearly the pole, focus and centre of curvature on the diagram.
 (b) What happens to the image when the object is moved away from the mirror gradually ?
 (c) State three characteristics of the image formed by a convex mirror.
27. (a) Draw a labelled ray diagram to show the formation of image in a convex mirror when the object is at infinity. Mark clearly the pole and focus of the mirror in the diagram.
 (b) State three characteristics of the image formed in this case.
 (c) Draw diagram to show how a convex mirror can be used to give a large field of view.

Multiple Choice Questions (MCQs)

28. The image formed by a spherical mirror is virtual. The mirror will be :
 (a) concave (b) convex (c) either concave or convex (d) metallic
29. Whatever be the position of the object, the image formed by a mirror is virtual, erect and smaller than the object. The mirror then must be :
 (a) plane (b) concave (c) convex (d) either concave or convex
30. The mirror used by a dentist to examine the teeth of a person is :
 (a) convex (b) concave (c) plane (d) any one of the above
31. If the image formed is always virtual, the mirror can be :
 (a) concave or convex (b) concave or plane
 (c) convex or plane (d) only convex
32. A concave mirror cannot be used as :
 (a) a magnifying mirror (b) a torch reflector
 (c) a dentist's mirror (d) a rear view mirror
33. A boy is standing in front of and close to a special mirror. He finds the image of his head bigger than normal, the middle part of his body of the same size, and his legs smaller than normal. The special mirror is made up of three types of mirrors in the following order from top downwards :

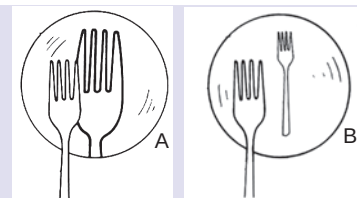
- (a) Convex, Plane, Concave (b) Plane, Convex, Concave
(c) Concave, Plane, Convex (d) Convex, Concave, Plane
34. The mirror which can form a magnified image of an object is :
(a) convex mirror (b) plane mirror
(c) concave mirror (d) both convex and concave mirrors
35. A real image of an object is to be obtained. The mirror required for this purpose is :
(a) convex (b) concave (c) plane (d) either convex or concave
36. Consider two statements A and B given below :
A : real image is always inverted
B : virtual image is always erect
Out of these two statements :
(a) only A is true (b) only B is true (c) both A and B are true (d) none is true

Questions Based on High Order Thinking Skills (HOTS)

37. The diagrams show the appearance of a fork when placed in front of and close to two mirrors A and B, turn by turn.

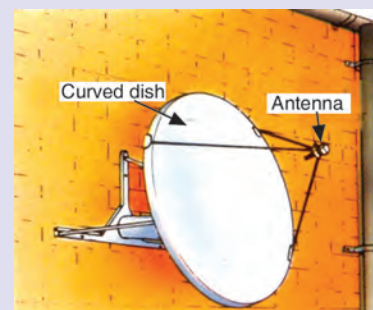
- (a) Which mirror is convex ?
(b) Which mirror is concave ?

Give reasons for your choice.



38. The diagram shows a dish antenna which is used to receive television signals from a satellite. The antenna (signal detector) is fixed in front of the curved dish.

- (a) What is the purpose of the dish ?
(b) Should it be concave or convex ?
(c) Where should the antenna be positioned to receive the strongest possible signals ?
(d) Explain what change you would expect in the signals if a larger dish was used.



39. A man standing in front of a special mirror finds his image having a very small head, a fat body and legs of normal size. What is the shape of :

- (a) top part of the mirror ?
(b) middle part of the mirror ?
(c) bottom part of the mirror ?

Give reasons for your choice.

40. Two big mirrors A and B are fitted side by side on a wall. A man is standing at such a distance from the wall that he can see the erect image of his face in both the mirrors. When the man starts walking towards the mirrors, he finds that the size of his face in mirror A goes on increasing but that in mirror B remains the same.

- (a) mirror A is concave and mirror B is convex
(b) mirror A is plane and mirror B is concave
(c) mirror A is concave and mirror B is plane
(d) mirror A is convex and mirror B is concave

ANSWERS

1. (a) Virtual and erect (b) Virtual and erect ; Real and inverted 5. At focus ; 15 cm behind convex mirror
9. (a) Concave mirror (b) Convex mirror 10. True 15. focus 19. (a) 25 cm (b) Convex mirror (c) Diverge light
28. (c) 29. (c) 30. (b) 31. (c) 32. (d) 33. (c) 34. (c) 35. (b) 36. (c) 37. (a) Mirror B is convex ; It forms a smaller image of fork (b) Mirror A is concave ; It forms a larger image of fork 38. (a) To collect a large amount of TV signals from the satellite (b) Concave (c) At the focus of the dish (d) Stronger signals will be received 39. (a) Convex (b) Concave (c) Plane 40. (c)

NUMERICAL PROBLEMS BASED ON CONVEX MIRRORS

In order to solve the numerical problems based on convex mirrors, we should remember that the mirror formula for a convex mirror is the same as that for a concave mirror, which is :

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

where v = image distance

u = object distance

and f = focal length

Please note that since a convex mirror always forms an image behind the mirror (on the right side of the mirror), therefore, **the image distance (v) in the case of convex mirrors is always positive**. The object is always placed to the left of mirror, so the object distance (u) is always taken with a negative sign. The focus of a convex mirror is behind the mirror on its right side, therefore, **the focal length of a convex mirror is always taken as positive**.

The two magnification formulae for convex mirror are also just the same as that for a concave mirror. That is :

$$m = \frac{h_2}{h_1} \quad \text{and} \quad m = -\frac{v}{u}$$

where the symbols have their usual meaning. Let us now solve one numerical problem based on convex mirror.

Sample Problem. An object 5 cm high is placed at a distance of 10 cm from a convex mirror of radius of curvature 30 cm. Find the nature, position and size of the image.

Solution. Here, Object distance, $u = -10$ cm (To left of mirror)
Image distance, $v = ?$ (To be calculated)

$$\begin{aligned} \text{And, Focal length, } f &= \frac{\text{Radius of curvature}}{2} \\ &= \frac{30}{2} \text{ cm} \\ &= +15 \text{ cm} \quad (\text{Convex mirror}) \end{aligned}$$

Putting these values in the mirror formula :

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

we get :
$$\frac{1}{v} + \frac{1}{-10} = \frac{1}{+15}$$

or
$$\frac{1}{v} - \frac{1}{10} = \frac{1}{15}$$

or
$$\frac{1}{v} = \frac{1}{15} + \frac{1}{10}$$

$$\frac{1}{v} = \frac{2+3}{30}$$

$$\frac{1}{v} = \frac{5}{30}$$

$$\frac{1}{v} = \frac{1}{6}$$

So, Image distance, $v = +6$ cm

Thus, the position of image is 6 cm behind the convex mirror. Since the image is formed behind the convex mirror, its nature will be virtual and erect.

To find the size of image, we will calculate the magnification first.

Now, Magnification, $m = -\frac{v}{u}$

Here, Image distance, $v = +6$ cm
and Object distance, $u = -10$ cm

$$\text{So, } m = -\frac{(+6)}{(-10)}$$

$$m = \frac{6}{10}$$

Magnification, $m = 0.6$

We also have another formula for magnification, which is :

$$m = \frac{h_2}{h_1}$$

Here, Magnification, $m = +0.6$

Height of image, $h_2 = ?$ (To be calculated)

and Height of object, $h_1 = +5$ cm

$$\text{So, } +0.6 = \frac{h_2}{+5}$$

$$h_2 = 5 \times 0.6$$

Height of image, $h_2 = 3$ cm (or $+3$ cm)

Thus, the size of image is 3 cm.

We are now in a position to **answer the following questions and problems :**

Short Answer Type Questions

1. An object is kept at a distance of 5 cm in front of a convex mirror of focal length 10 cm. Calculate the position and magnification of the image and state its nature.
2. An object is placed at a distance of 10 cm from a convex mirror of focal length 5 cm.
 - (i) Draw a ray-diagram showing the formation of image.
 - (ii) State two characteristics of the image formed.
 - (iii) Calculate the distance of the image from mirror.
3. An object is placed at a distance of 6 cm from a convex mirror of focal length 12 cm. Find the position and nature of the image.
4. An object placed 20 cm in front of a mirror is found to have an image 15 cm (a) in front of it, (b) behind the mirror. Find the focal length of the mirror and the kind of mirror in each case.
5. An arrow 2.5 cm high is placed at a distance of 25 cm from a diverging mirror of focal length 20 cm. Find the nature, position and size of the image formed.
6. A convex mirror used as a rear-view mirror in a car has a radius of curvature of 3 m. If a bus is located at a distance of 5 m from this mirror, find the position of image. What is the nature of the image ?
7. A diverging mirror of radius of curvature 40 cm forms an image which is half the height of the object. Find the object and image positions.
8. The radius of curvature of a convex mirror used as a rear view mirror in a moving car is 2.0 m. A truck is coming from behind it at a distance of 3.5 m. Calculate (a) position, and (b) size, of the image relative to the size of the truck. What will be the nature of the image ?

Long Answer Type Question

9. (a) Draw a diagram to represent a convex mirror. On this diagram mark principal axis, principal focus F and the centre of curvature C if the focal length of convex mirror is 3 cm.
(b) An object 1 cm tall is placed 30 cm in front of a convex mirror of focal length 20 cm. Find the size and position of the image formed by the convex mirror.

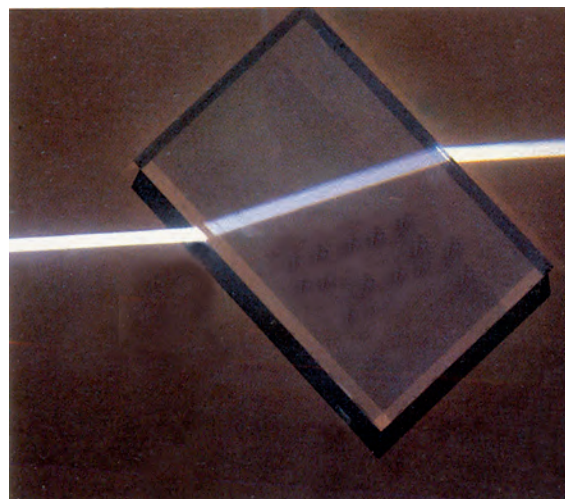
Questions Based on High Order Thinking Skills (HOTS)

10. A shop security mirror 5.0 m from certain items displayed in the shop produces one-tenth magnification.
 - (a) What is the type of mirror ?
 - (b) What is the radius of curvature of the mirror ?

11. An object is placed 15 cm from (a) a converging mirror, and (b) a diverging mirror, of radius of curvature 20 cm. Calculate the image position and magnification in each case.
12. An object 20 cm from a spherical mirror gives rise to a virtual image 15 cm behind the mirror. Determine the magnification of the image and the type of mirror used.

ANSWERS

1. 3.3 cm behind the convex mirror ; 0.66 ; Virtual and erect
2. (ii) Virtual and erect ; Diminished (Smaller than the object) (iii) 3.3 cm behind the convex mirror
3. 4 cm behind the mirror ; Virtual and erect
4. (a) Concave mirror of focal length $\frac{60}{7}$ cm (b) Convex mirror of focal length 60 cm
5. $v = 11.1$ cm ; The image is formed 11.1 cm behind the convex mirror ; Virtual and erect ; 1.1 cm tall
6. 1.15 m behind the mirror ; Virtual and erect
7. 20 cm ; 10 cm behind the mirror
8. (a) 0.77 m behind the mirror
- (b) $\frac{1}{4.5}$; Virtual and erect
9. 0.4 cm ; 12 cm behind the mirror
10. (a) Convex mirror (b) $\frac{10}{9}$ m
11. (a) $v = -30$ cm ; The image is formed 30 cm in front of converging mirror ; $m = -2$ (b) $v = +6$ cm ; The image is formed 6 cm behind the diverging mirror ; $m = +0.4$
12. $m = +0.75$; Convex mirror



Refraction of Light

We know that light travels in a straight line path. This is true as long as light rays are travelling in the same medium (or same substance) having the same density throughout. If, however, the light rays are made to go from one medium to another, the light rays change their direction (or bend) at the boundary between the two media. For example, when a ray of light travelling in 'air' goes *obliquely* into another medium 'glass', it changes the direction (or bends) on entering the glass block (see Figure 1). The change in direction of ray of light (or bending of ray of light) occurs again when it goes out from 'glass' into 'air' (see Figure 1). **The change in direction of light when it passes from one medium to another obliquely, is called refraction of light.** In other words, **the bending of light when it goes from one medium to another obliquely is called refraction of light.** The refraction of light takes place when light rays enter from *air* into *glass*; or from *glass* into *air*. The refraction of light takes place when light enters from *air* into *water*; or from *water* into *air*. And the refraction of light also takes place when light enters from *water* into *glass*; or from *glass* into *water*. The optical instruments like camera, microscope, and telescope work on the refraction of light through glass lenses. We will now understand the refraction of light more clearly with the help of a diagram.

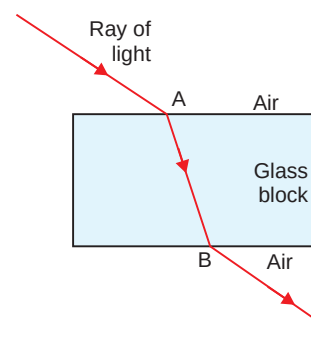


Figure 1. The ray of light changes direction (or refracts) at point A when it goes from air into glass. The ray of light changes direction (or refracts) again at point B when it goes from glass into air.

Consider a rectangular glass slab PQRS shown in Figure 2. Now, here we have two different media, one is air and the other is glass ('media' is the plural of 'medium'). Please note that glass is optically denser medium as compared to air (we will give the meaning of optically denser medium after a while). Now, a ray of light AO travelling in air is incident (or falls) obliquely on the glass slab at point O (see Figure 2). Since the glass slab is transparent, most of the incident light passes into the glass slab. Since glass is an optically denser medium than air, so when the ray of light AO passes from air into glass, its direction changes, it goes along the line OB inside the glass slab and we say that the light ray has been refracted (or bent). This is because AO and OB are not in the same straight line. Please note that **the refraction (or bending) of light takes place at the boundary between the two media.** For example, in Figure 2, the refraction of light (or bending of light) takes place at point O at the boundary of the two media : air and glass. We will now define the angle of incidence and the angle of refraction.

In Figure 2, for the light passing from air into glass, AO is the incident ray and OB is the refracted ray. Let us draw a normal NON' at the point of incidence O . Now, **the angle between incident ray and normal (at the point of incidence) is called the angle of incidence**. In Figure 2, the angle AON is the angle of incidence. The angle of incidence is denoted by the letter i . **The angle between the refracted ray and the normal (at the point of incidence) is called the angle of refraction**. In Figure 2, the angle $N'OB$ is the angle of refraction. The angle of refraction is denoted by the letter r .

Please note that though in the reflection of light, the angle of reflection is always equal to the angle of incidence, but **in the refraction of light, the angle of refraction is usually not equal to the angle of incidence**. The angle of refraction is either *smaller* than the angle of incidence or *greater* than the angle of incidence. For example, in Figure 2, the angle of refraction (r) is not equal to the angle of incidence (i). In this case the angle of refraction (r) is smaller than the angle of incidence (i). We will give the reasons for this later on in this chapter. At the moment we will discuss why the refraction of light takes place on going from one medium to another.

Cause of Refraction

The speed of light is different in different media (or substances). For example, the speed of light in air is 3×10^8 m/s whereas that in glass is 2×10^8 m/s. It is clear from these values that the speed of light is more in air but less in glass. In other words, light travels faster in air but slower in glass. **The refraction of light is due to the change in the speed of light on going from one medium to another**. Thus, when light goes from one medium to another, its speed changes. And this change in speed of light causes the refraction of light. The refraction of light on going from air to glass takes place because the speed of light is different in air and glass : being more in air and lesser in glass. So, when light enters from air into glass, its speed changes (it gets reduced). And this change in speed of light in going from air to glass causes the refraction of light (or bending of light). Please note that **greater the difference in the speeds of light in the two media, greater will be the amount of refraction (or bending) of light**.

Why a Change in Speed of Light Causes Refraction of Light (or Bending of Light)

The refraction of light or change in direction of light on going from one medium to another can be explained by using the *wave theory* of light. A beam of light is made up of tiny waves. When a beam of light consisting of light waves and travelling in a certain medium falls obliquely (at an angle) on the boundary of another medium, then one part of the light waves enters into the other medium first and its speed changes first but the rest of waves enter the other medium a little later and hence its speed changes a little later. **The fact that the speed of light waves on one side of a beam of light changes a little before the change in speed of light waves on its other side, causes a change in the direction of light**. And this change in direction of light is called refraction of light or bending of light. This will become more clear from the following example in which a beam of light is entering from air into a glass slab obliquely and then emerging into air from its other side.

Light waves travel faster in air but slower in glass. Now, when a beam of light consisting of light waves and travelling in air falls on a glass slab obliquely then the part of light waves on the left side of the beam of light reaches the glass slab first (at point A) and slows down first on entering the glass slab (see Figure 3). The rest of light waves on the right side of the beam of light are still in air and have to travel more distance in air before reaching the glass slab (at point B) and hence slow down a little later on entering the glass slab (see Figure 3). **Since the speed of left side of the beam of light is reduced a little before its**

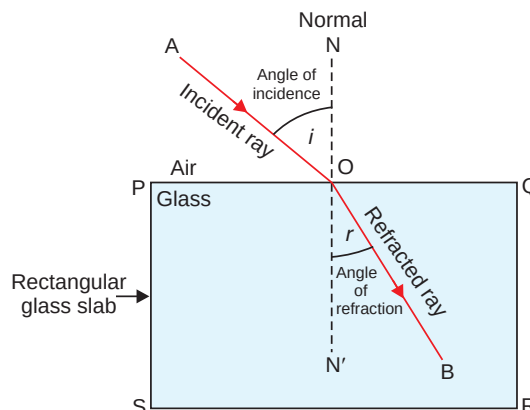


Figure 2. Diagram to show the refraction of light when it passes from air into glass. In this diagram, the light ray going along AO in air bends in the direction OB on entering the glass.

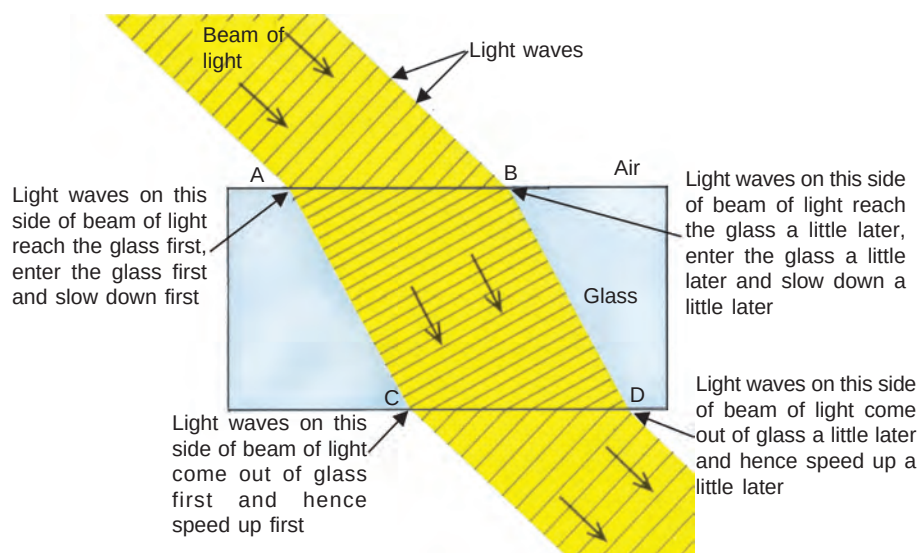


Figure 3. Diagram to explain the refraction of light on going from air into glass; and from glass into air.

right side, therefore, the direction of light changes (or bending of light occurs) on entering the glass slab (see Figure 3). Please note that a decrease in speed of light waves on going from air into glass causes bending of light towards left side (towards the normal) (see Figure 3).

When the beam of light travelling in glass slab comes out into air obliquely (at an angle), then the part of light waves on the left side of beam of light emerges out into air first (at point C) and speeds up first (see Figure 3). The part of light waves on the right side of beam of light emerges out into air (at point D) a little later and hence its speed increases a little later. **Since the speed of left side of the beam of light increases a little before its right side, therefore, the direction of light changes (or bending of light occurs) on coming out of glass slab into air** (see Figure 3). Please note that an increase in speed of light waves on coming out from glass into air causes bending of light towards the right side (away from the normal) (see Figure 3).

When light waves move from air into glass, their speed decreases and their wavelength also decreases (they become closer). On the other hand, when light waves come out from glass into air, their speed increases and their wavelength also increases (they become farther apart). Before we discuss the refraction of light in detail, we should know the meaning of the terms 'optically rarer medium' and 'optically denser medium'. This is described below.

Optically Rarer Medium and Optically Denser Medium

A transparent substance in which light travels is known as a medium. Air, glass, certain plastics, water, kerosene, alcohol, etc., are all examples of medium. Different media are said to have different optical densities. **A medium in which the speed of light is more is known as optically rarer medium** (or less dense medium). Air is an optically rarer medium as compared to glass and water. **A medium in which the speed of light is less, is known as optically denser medium.** Glass is an optically denser medium than air and water. Now, the speed of light in water is 2.25×10^8 m/s, which is less than that in air but more than that in glass. So, water is optically denser medium than air but it is optically rarer than glass. Please note that the refraction of light takes place whenever it goes from an optically rarer medium to an optically denser medium; or from a denser medium to a rarer medium. The optically rarer medium and optically denser medium can also be defined on the basis of their refractive index. We will learn this after a while. Keeping the above discussion in mind, we can now write down two rules which give the direction of bending of a ray of light when it goes from one medium to another.

It has been found that :

1. **When a ray of light goes from a rarer medium to a denser medium, it bends towards the normal** (at the point of incidence).



2. When a ray of light goes from a denser medium to a rarer medium, it bends away from the normal (at the point of incidence).

We will now understand these rules more clearly with the help of ray-diagrams.

Case 1 : Refraction of Light When it Goes From a Rarer Medium to a Denser Medium

Figure 4 shows a ray of light AO going from air (a rarer medium) into glass (which is a denser medium). In this case, the incident ray AO gets refracted at point O and bends towards the normal ON' and goes in the direction OB inside the glass slab. Thus, **when a ray of light goes from air into glass, it bends towards the normal** (at the point of incidence). In this case, the angle of refraction (r) is smaller than the angle of incidence (i).

Water is also an optically denser medium than air, so **when a ray of light goes from air into water, it bends towards the normal**. Thus, the refraction of light on going from air to water is similar to the refraction of light from air to glass (which has been shown in Figure 4). To show the path of a ray of light going from air into water, we can use Figure 4 but we will have to write 'water' in place of 'glass' and in place of glass slab we will have to show water by drawing some dotted lines. Please draw the diagram to show the refraction of a ray of light from air to water yourself.

Case 2 : Refraction of Light When it Goes From a Denser Medium to a Rarer Medium

Figure 5 shows a ray of light AO going from glass (a denser medium) into air (which is a rarer medium). In this case, the incident ray AO gets refracted at point O and bends away from the normal ON' in the direction OB . Thus, **when a ray of light goes from glass into air, it bends away from the normal** (at the point of incidence). In this case the angle of refraction (r) is greater than the angle of incidence (i).

Please note that water is also optically denser than air, so **when a beam of light travelling in water enters into air, it bends away from the normal**. We can show the refraction of light on going from water into air by using the ray-diagram given in Figure 5. All that we have to do is to write 'water' in place of 'glass' and show water by drawing some dotted lines. Please draw the diagram to show the refraction of a ray of light from water to air yourself. Please note that a **parallel-sided glass slab is also called rectangular glass block**.

When a ray of light goes from a rarer medium to a denser medium, its speed decreases or it slows down. On the other hand, when a ray of light goes from a denser medium to a rarer medium, then its speed increases or it speeds up. So, we can now say that a ray of light travelling from a rarer medium to a denser medium slows down and bends towards the normal but when a ray of light travels from a denser medium to a rarer medium, it speeds up and bends away from the normal.

The Case of Light Going From Air into Glass and Again into Air

We have just studied the refraction of light from air into glass (Figure 4), and from glass into air (Figure 5). We will now show these two types of refraction in the same diagram. In other words, we will now show the complete path of a ray of light when it passes from air into glass slab and again into air. This is shown in Figure 6.

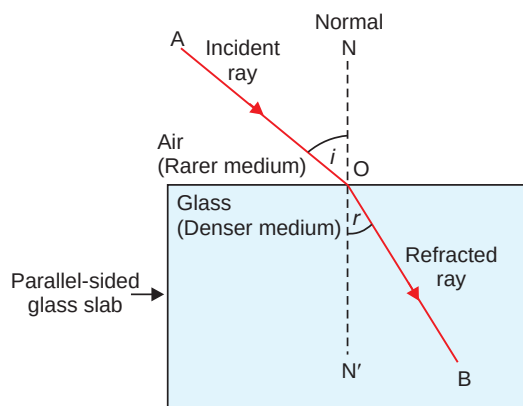


Figure 4.

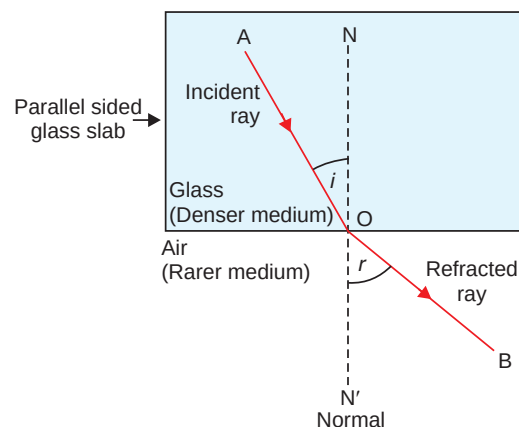


Figure 5.

A ray of light AO travelling in air is incident on a rectangular glass slab $PQRS$ at point O . On entering the glass slab, it gets refracted along OB and bends towards the normal ON' (see Figure 6). A second change of direction takes place when the refracted ray of light OB , travelling in glass emerges (or comes out) into air at point B . Since the ray of light OB now goes from a denser medium 'glass' into the rarer medium 'air', it bends away from the normal BN_1' and goes in the direction BC . Please note that the incident ray AO and the emergent ray BC are parallel to each other (though the emergent ray has been displaced parallel to the incident ray). The incident ray AO and emergent ray BC are parallel to each other because the extent of bending of the ray of light at points O and B on the opposite, parallel faces (PQ and SR) of the rectangular glass slab is equal and opposite. The incident ray AO bends towards the normal at point O whereas the refracted ray OB bends away from the normal at point B by an equal amount. Thus, **the light emerges from a parallel-sided glass slab in a direction parallel with that in which it enters the glass slab**. Though the emergent ray BC is parallel to the incident ray AO , but the emergent ray has been sideways displaced (or laterally displaced) from the original path of the incident ray by a perpendicular distance CD (see Figure 6). In Figure 6, the original path of incident light is AOD but the emergent light goes along BC , the lateral displacement between them being CD . **The perpendicular distance between the original path of incident ray and the emergent ray coming out of the glass slab is called lateral displacement of the emergent ray of light.** Lateral displacement depends mainly on three factors : angle of incidence, thickness of glass slab, and refractive index of glass slab. Actually, lateral displacement is directly proportional to (i) angle of incidence (ii) thickness of glass slab, and (iii) refractive index of glass slab. Higher the values of these factors, greater will be the lateral displacement. Another point to be noted is that in this case the refraction (or bending) of light takes place twice : first at point O (when light enters from air into glass), and then at point B (when light goes out from glass into air).

The angle which the emergent ray makes with the normal is called the angle of emergence. In Figure 6, the angle $N_1'BC$ is the angle of emergence. Since the incident ray AO and the emergent ray BC are parallel to one another, so the angle of emergence (e) is equal to the angle of incidence (i). Please note that if a beam of light travelling in air passes into water and then emerges into air, we will get a ray-diagram similar to that shown in Figure 6. Draw it yourself by writing 'water' in place of 'glass'.

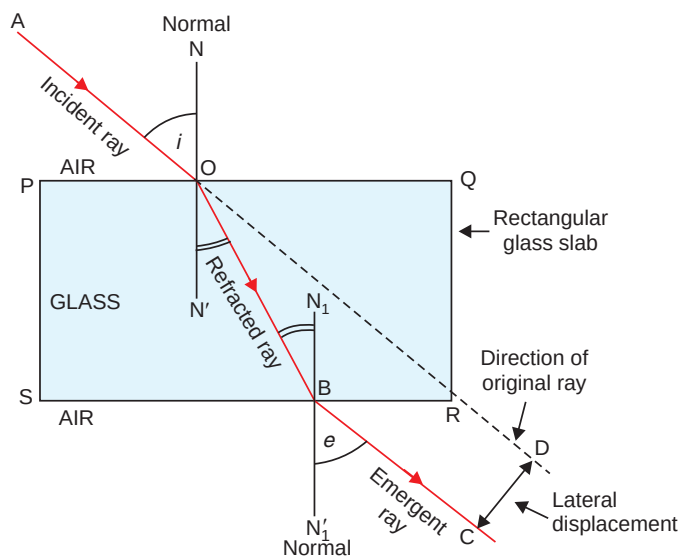


Figure 6.

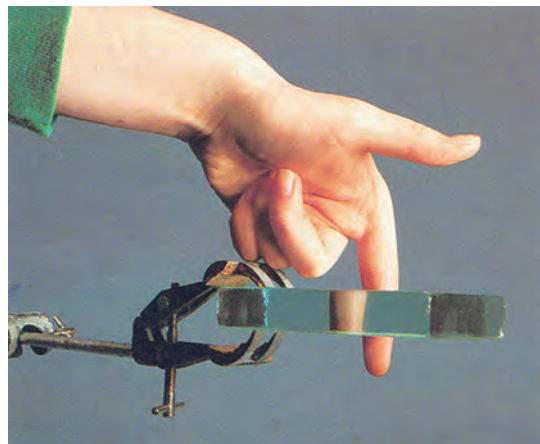


Figure 7. This picture shows the lateral displacement of a finger caused by the refraction of light through a glass slab.

The Case of Light Falling Normally (or Perpendicularly) on a Glass Slab

In all our discussions so far we have assumed that the incident ray of light falls *obliquely* to the surface of glass slab and bending of ray of light takes place. **If the incident ray falls normally (or perpendicularly) to the surface of a glass slab, then there is no bending of the ray of light, and it goes straight.** For example, in Figure 8, a ray of light AO travelling in air falls on a glass slab normally (or perpendicularly) at

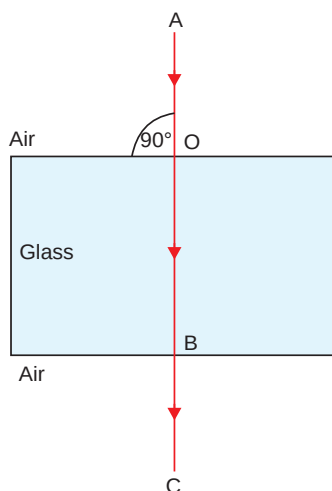


Figure 8.

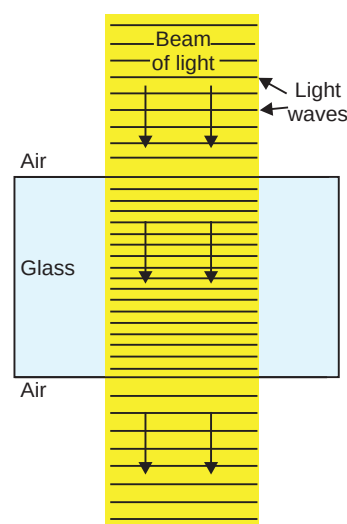


Figure 9.

point O , so it does not bend on entering the glass slab or on coming out of the glass slab. It goes straight in the direction $AOBC$. Since the incident ray goes along the normal to the surface, the angle of incidence in this case is zero (0) and the angle of refraction is also zero (0).

We can explain the case of *no refraction* (or *no bending*) of light on going perpendicularly from air to glass as follows : When a beam of light consisting of light waves and travelling in air falls perpendicularly (at right angles) to the surface of glass slab, then all the parts of light waves reach the glass slab at the same time, enter the glass slab at the same time and hence get slowed down at the same time (see Figure 9). Due to this no change in direction of light takes place. In other words, no refraction of light (no bending of light) takes place. Similarly, all the parts of the light waves of the beam of light travelling in glass slab come out of the glass slab into air at the same time and hence speed up at the same time. Due to this, no bending of light occurs when the light waves go out from glass slab into air (see Figure 9).

Please note that if a ray of light falls normally (or perpendicularly) to the surface of water, even then there is no bending of light ray, and it goes straight through water.

EFFECTS OF REFRACTION OF LIGHT

The refraction of light produces many effects which can be easily observed in our day to day life. We will now describe some of the important effects of the refraction of light. It is due to the refraction of light that :

- (i) a stick (or pencil) held obliquely and partly immersed in water appears to be bent at the water surface.
- (ii) an object placed under water appears to be raised.
- (iii) a pool of water appears to be less deep than it actually is.
- (iv) when a thick glass slab is placed over some printed matter, the letters appear raised when viewed from the top.
- (v) a lemon kept in water in a glass tumbler appears to be bigger than its actual size, when viewed from the sides.
- (vi) the stars appear to twinkle on a clear night.

All these effects are produced by the refraction of light (or change in direction of light) when it passes from one medium to another. Let us discuss some of these effects of refraction in detail.

1. A Stick Partly Immersed in Water Appears to be Bent at the Water Surface

When a straight stick is partly immersed in water and held obliquely to the surface, it appears to be bent at the point where it enters water (see Figure 10). This apparent bending of the stick is due to the

refraction of light when it passes from water into air. Let us understand it more clearly with the help of a ray-diagram.

Figure 10 shows a straight stick AO whose lower portion BO is immersed in water. Though the stick is actually straight but on immersing in water, it appears to be bent at point B , in the direction BI (see Figure 10). This can be explained as follows : A ray of light OC coming from the lower end O of the stick passes from water into air at point C and gets refracted away from normal in the direction CX (because it passes from a denser medium water into a rarer medium air). Another ray of light OD gets refracted in the direction DY . The two refracted rays CX and DY , when produced backwards, appear to meet at point I , nearer to the water surface than point O (see Figure 10). Thus, I is the virtual image of the end O of the stick which is formed by the refraction of light on going from water to air. Thus, an eye at position E sees the end O of the stick at position I which is nearer to the water surface.

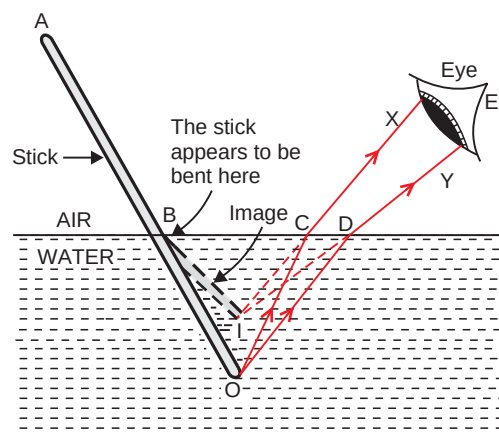


Figure 10. A straight stick AO appears bent at point B where it enters water.

We can extend this reasoning to all the points which make up part BO of the stick under water. Thus, due to the refraction of light, a virtual image of the part BO of the stick is formed at BI . So, what we see under water is actually the virtual image BI of the part BO of the stick under water. Since the part AB of the stick which is above water and the image BI under water are not in the same straight line, the stick AO appears to be bent at point B along BI . Thus, though the actual stick is ABO , it appears to be ABI . Please note that it is not the stick that is really bent. It is actually the light passing from water to air that is bent (or refracted) (see Figure 11).



Figure 11. The stick is partly immersed in water. Due to the refraction of light coming from the part of the stick that is under water, the stick appears bent.



Figure 12. The pencil is partly immersed in water. Because of the refraction of light coming from the part of the pencil that is under water, the pencil appears bent.

When a pencil is partly immersed in water and held obliquely to the surface, the pencil appears to bend at the water surface (when viewed from the side) (see Figure 12). This apparent bending of the pencil is due to the refraction of light when it passes from water into air. We can use Figure 10 for explaining the bending of pencil in water. Just make a pencil in place of stick.

2. An Object Placed Under Water Appears to be Raised

We will perform an experiment to show that an object placed under water appears to be raised. This can be done as follows : We place a coin at the bottom of an empty basin (a shallow vessel). Let us move our head away from the basin slowly, till the coin disappears from our view and we cannot see it [see

Figure 13(a)]. Now, without moving our head, we pour water into the basin. We will find that on adding water, the coin appears to rise and we are able to see it [see Figure 13(b)]. The coin under water appears to be raised (and becomes visible) due to the refraction of light which takes place when it goes from water into air. We will now understand all this more clearly with the help of a ray-diagram.

Figure 13(a) shows a coin O placed in an empty basin (having no water). In this case the rays of light OA and OB coming from the coin travel in straight line paths in air and do not enter our eye E (because the eye is at a lower level). Since the rays of light coming from the coin do not enter our eye, we cannot see the coin from this position of our eye.

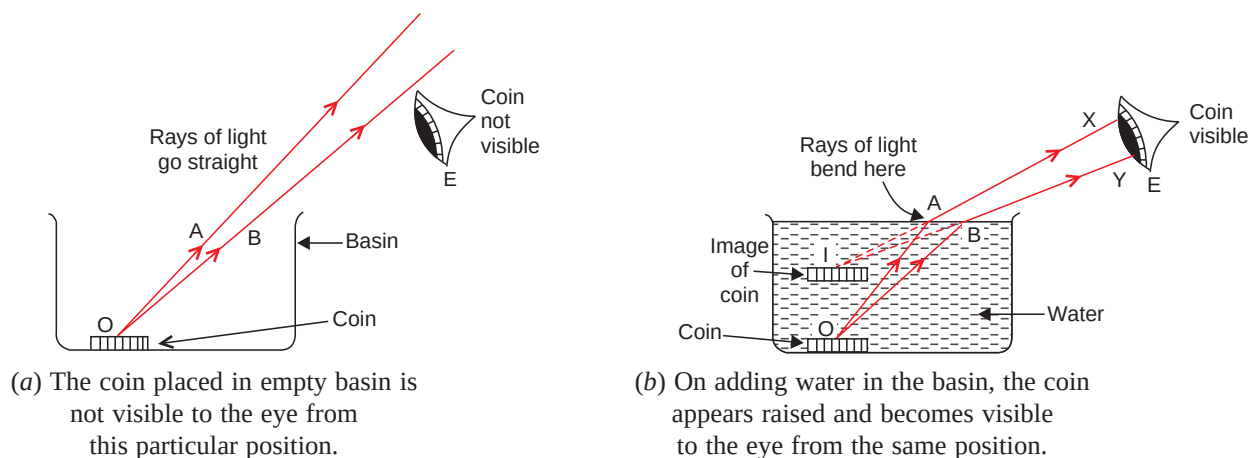


Figure 13.

When the coin is under water, then the rays OA and OB coming from the coin travel in water in straight line paths until they reach the surface of water. When the rays of light OA and OB travelling in water, go out into air, they get refracted (they change their directions). The ray of light OA gets refracted at point A , bends away from the normal, and goes in the direction AX [see Figure 13(b)]. Similarly, the ray of light OB gets refracted at point B , bends away from the normal, and goes in the direction BY .

If we extend the refracted rays AX and BY backwards (to the left side), then a virtual image of the coin is formed at point I , nearer to the surface of water [see Figure 13(b)]. The refracted rays AX and BY , which appear to be coming from the virtual image of the coin can enter our eye at position E due to which the coin becomes visible to us. Thus, **when the coin is under water then due to refraction of light, a virtual image of the coin is formed nearer to the water surface. And since the virtual image of coin which we see, is nearer to the water surface, so the coin appears to rise on adding water in the basin.** Thus, it is due to the refraction of light that a coin placed at the bottom of a container appears to rise as the container is slowly filled with water.

If instead of water, we take some other transparent liquid such as kerosene or turpentine in the above two experiments, then the bending of stick or raising of coin will appear to take place by different amounts than that in water. This is because light is refracted by different amounts in different liquids.

3. A Pool of Water Appears to be Less Deep than it Actually is

If we look into a pool of water (or tank of water), it appears to be less deep than it really is. This is due to the refraction of light which takes place when light rays pass from the pool of water into the air. Let us understand it more clearly with the help of a ray-diagram.

Figure 14 shows a pool of water (or a tank of water). Let us take any point O at the bottom of this pool. This point is under the surface of water. Our eye sees this point by the light rays coming from it. Now, a ray of light OA coming from the point O is travelling in water and it comes out into the air at point A . It gets refracted away from the normal in the direction AX (see Figure 14.) Similarly, another ray of light OB , coming from the point O gets refracted at point B and goes away from the normal in the direction BY . The

two refracted rays AX and BY , when produced backwards, meet at point I under water. In fact, when the rays AX and BY enter the eye E , they appear to be coming from point I . So, point I is the virtual image of point O (which is at the bottom of the pool). It is clear from Figure 14 that the image I is nearer to the surface of water than the point O . Thus, a point O at the bottom of a pool appears to be much nearer at position I .

This reasoning can be applied to all the points which make up the bottom PQ of the pool, so that due to refraction of light, the bottom PQ of the pool will appear to be much nearer at the position $P'Q'$ (see Figure 14). Due to this, the pool of water will appear less deep than it actually is. Please note that **when we look into a pool of water, we do not see the actual bottom of the pool, we see a virtual image of the bottom of the pool which is formed by the refraction of light coming from the pool water into the air.** And since the image of the bottom of the pool is formed nearer to us, we feel that the pool is less deep. From this discussion we conclude that the bottom of a pool of water appears raised due to refraction of light. And the pool of water appears to be less deep than it actually is. *We should be careful while entering a swimming pool because the water in it will be deeper than it appears to be.*

It is also due to the refraction of light that a thick glass slab appears to be less thick (when seen from above), than it actually is. Similarly, an ink mark or the writing on a piece of paper appears to be raised and much nearer than it actually is, when viewed by keeping a glass slab over it (see Figure 15). This also happens due to refraction of light. As we will study in the next Chapter, the stars appear to twinkle on a clear night due to the refraction of light in the atmosphere. Before we go further and discuss the laws of refraction of light, **please answer the following questions :**

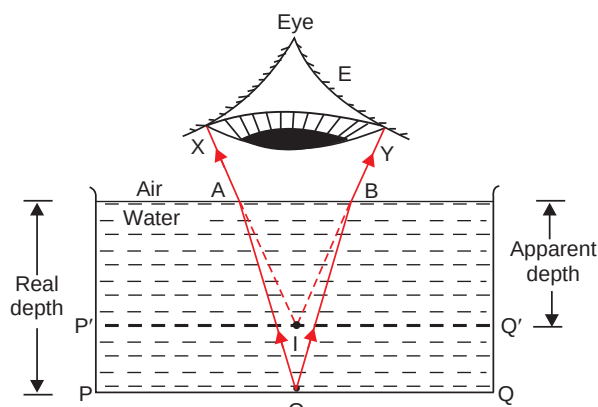


Figure 14. The bottom PQ of the pool of water appears to be raised at $P'Q'$ due to refraction of light. The pool thus appears to be less deep.

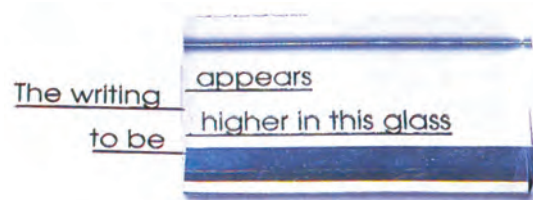


Figure 15. Those words of writing which are below the glass slab appear raised (or higher) due to refraction of light as it comes out of glass slab.

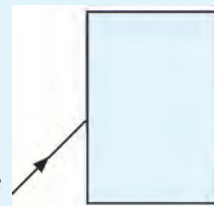
Very Short Answer Type Questions

1. If a ray of light goes from a rarer medium to a denser medium, will it bend towards the normal or away from it ?
2. If a ray of light goes from a denser medium to a rarer medium, will it bend towards the normal or away from the normal ?
3. A beam of light travelling in a rectangular glass slab emerges into air. Draw a ray-diagram indicating the change in its path.
4. A beam of light travelling in air is incident on water. Draw a ray-diagram indicating the change in its path in water.
5. A ray of light travelling in water emerges into air. Draw a ray-diagram indicating the change in its path.
6. A ray of light travelling in air is incident on a parallel-sided glass slab (or rectangular glass slab). Draw a ray-diagram indicating the change in its path in glass.
7. A ray of light travelling in glass emerges into air. State whether it will bend towards the normal or away from the normal.
8. A ray of light travelling in air enters obliquely into water. Does the ray of light bend towards the normal or away from the normal ? Why ?
9. A ray of light goes from water into air. Will it bend towards the normal or away from the normal ?
10. State two effects caused by the refraction of light.
11. Name the phenomenon due to which a swimming pool appears less deep than it really is.
12. When a ray of light passes from air into glass, is the angle of refraction greater than or less than the angle of incidence ?

13. A ray of light passes from air into a block of glass. Does it bend towards the normal or away from it ?
14. As light rays pass from water into glass, are they refracted towards the normal or away from the normal ?
15. In which material do you think light rays travel faster—glass or air ?
16. Which phenomenon of light makes the water to appear shallower than it really is ?
17. State whether the following statement is true or false :
Refraction occurs because light slows down in denser materials.
18. Why does a ray of light bend when it travels from one medium to another ?
19. Fill in the following blanks with suitable words :
(a) Light travelling along a normal isrefracted.
(b) Light bends when it passes from water into air. We say that it is.....

Short Answer Type Questions

20. What is meant by 'refraction of light' ? Draw a labelled ray diagram to show the refraction of light.
21. A ray of light travelling in air is incident on a rectangular glass block and emerges out into the air from the opposite face. Draw a labelled ray diagram to show the complete path of this ray of light. Mark the two points where the refraction of light takes place. What can you say about the final direction of ray of light ?
22. Draw a labelled ray diagram to show how a ray of light is refracted when it passes :
(a) from air into an optically denser medium.
(b) from an optically denser medium into air.
23. The diagram given alongside shows a ray of light entering a rectangular block of glass.
(a) Copy the diagram and draw the normal at the point of entry.
(b) Draw the approximate path of the ray of light through the glass block and out of the other side.
24. What is meant by the 'angle of incidence' and the 'angle of refraction' for a ray of light ? Draw a labelled ray diagram to show the angle of incidence and the angle of refraction for a refracted ray of light.
25. Light travels more quickly through water than through glass.
(a) Which is optically denser : water or glass ?
(b) If a ray of light passes from glass into water, which way will it bend : towards the normal or away from the normal ?
26. Draw a labelled ray diagram to show how a ray of light passes through a parallel sided glass block :
(a) if it hits the glass block at 90° (that is, perpendicular to the glass block)
(b) if it hits the glass block at an angle other than 90° (that is, obliquely to the glass block).
27. When a light ray passes from air into glass, what happens to its speed ? Draw a diagram to show which way the ray of light bends.



Long Answer Type Questions

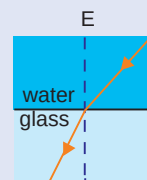
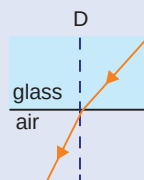
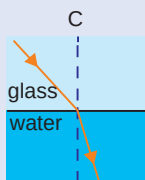
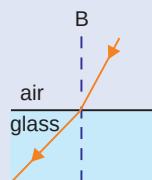
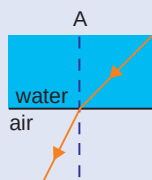
28. (a) Explain why, a stick half immersed in water appears to be bent at the surface. Draw a labelled diagram to illustrate your answer.
(b) A coin in a glass tumbler appears to rise as the glass tumbler is slowly filled with water. Name the phenomenon responsible for this effect.
29. (a) With the help of a labelled diagram, explain why a tank full of water appears less deep than it actually is.
(b) Name the phenomenon due to which a pencil partly immersed in water and held obliquely appears to be bent at the water surface.
30. (a) With the help of a diagram, show how when light falls obliquely on the side of a rectangular glass slab, the emergent ray is parallel to the incident ray.
(b) Show the lateral displacement of the ray on the diagram.
(c) State two factors on which the lateral displacement of the emergent ray depends.
31. Explain with the help of a labelled ray diagram, why a pencil partly immersed in water appears to be bent at the water surface. State whether the bending of pencil will increase or decrease if water is replaced by another liquid which is optically more dense than water. Give reason for your answer.

Multiple Choice Questions (MCQs)

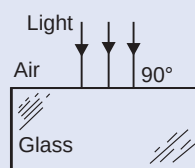
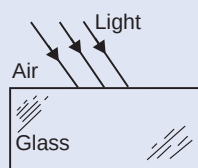
32. Light travelling from a denser medium to a rarer medium along a normal to the boundary :
 (a) is refracted towards the normal (b) is refracted away from the normal
 (c) goes along the boundary (d) is not refracted
33. A ray of light passes from glass into air. The angle of refraction will be :
 (a) equal to the angle of incidence (b) greater than the angle of incidence
 (c) smaller than the angle of incidence (d) 45°
34. A ray of light travelling in air goes into water. The angle of refraction will be :
 (a) 90° (b) smaller than the angle of incidence
 (c) equal to the angle of incidence (d) greater than the angle of incidence
35. The speed of light in air is :
 (a) 3×10^8 cm/s (b) 3×10^8 mm/s
 (c) 3×10^8 km/s (d) 3×10^8 m/s
36. When a ray of light travelling in glass enters into water obliquely :
 (a) it is refracted towards the normal (b) it is not refracted at all
 (c) it goes along the normal (d) it is refracted away from the normal
37. A ray of light travelling in water falls at right angles to the boundary of a parallel-sided glass block. The ray of light :
 (a) is refracted towards the normal (b) is refracted away from the normal
 (c) does not get refracted (d) is reflected along the same path.
38. A ray of light passes from a medium X to another medium Y. No refraction of light occurs if the ray of light hits the boundary of medium Y at an angle of :
 (a) 0° (b) 45° (c) 90° (d) 120°

Questions Based on High Order Thinking Skills (HOTS)

39. Which of the following diagrams shows the ray of light refracted correctly ?



40. A vertical ray of light strikes the horizontal surface of some water :
 (a) What is the angle of incidence ?
 (b) What is the angle of refraction ?
41. How is the reflection of light ray from a plane mirror different from the refraction of light ray as it enters a block of glass ?
42. How does the light have to enter the glass :
 (a) to produce a large amount of bending ?
 (b) for no refraction to happen ?
43. (a) How can you bend light away from the normal ?
 (b) How must light travel out of a substance if it is not going to be refracted ?
44. Draw and complete the following diagrams to show what happens to the beams of light as they enter the glass block and then leave it :



45. Why does a beam of light bend when it enters glass at an angle ? Why does it not bend if it enters the glass at right angles ?

ANSWERS

12. Angle of refraction is less than the angle of incidence 17. True 19. (a) not (b) refracted 28. (b) Refraction of light (as it comes out from water into air) 31. The bending of pencil will increase; The optically denser medium will cause more refraction (or more bending) of light rays 32. (d) 33. (b) 34. (b) 35. (d) 36. (d) 37. (c) 38. (c) 39. E 40. (a) 0° (b) 0° 41. The angle of reflection is equal to the angle of incidence but the angle of refraction is not equal to the angle of incidence 42. (a) Obliquely ; Making a large angle of incidence (b) Perpendicularly (at right angles) to the glass surface 43. (a) Make the light enter from a denser medium to a rarer medium (b) At right angles (90°) to the surface of substance.

LAWS OF REFRACTION OF LIGHT

The refraction of light on going from one medium to another takes place according to two laws which are known as the laws of refraction of light. These are given below.

1. According to the first law of refraction of light : **The incident ray, the refracted ray and the normal at the point of incidence, all lie in the same plane.** For example, in Figure 16, the incident ray AO , the refracted ray OB , and the normal ON , all lie in the same plane (which is the plane of the paper here).

2. The second law of refraction gives a relationship between the angle of incidence and the angle of refraction. This relationship was discovered by Snell experimentally in 1621, so the second law of refraction is called Snell's law of refraction. According to Snell's law of refraction of light : **The ratio of sine of angle of incidence to the sine of angle of refraction is constant for a given pair of media** (such as 'air and glass' or 'air and water'). That is :

$$\frac{\text{sine of angle of incidence}}{\text{sine of angle of refraction}} = \text{constant}$$

$$\text{or} \quad \frac{\sin i}{\sin r} = \text{constant}$$

This constant is called refractive index. We will now discuss the refractive index in somewhat detail.

Suppose a ray of light travelling in air enters into another medium and gets refracted. Let the angle of incidence in air be i and the angle of refraction in that medium be r (see Figure 16). **The value of the constant $\frac{\sin i}{\sin r}$ for a ray of light passing from air into a particular medium is called the refractive index of that medium.** The refractive index is usually denoted by the symbol n . So :

$$\text{Refractive index, } n = \frac{\sin i}{\sin r}$$

where $\sin i$ = sine of the angle of incidence (in air)

and $\sin r$ = sine of the angle of refraction (in medium)

Suppose the angle of incidence (i) for a ray of light in air is 37° and the angle of refraction (r) in glass be 24° . Then :

$$\text{Refractive index of glass, } n = \frac{\sin 37^\circ}{\sin 24^\circ}$$

Now, if we look up the sine tables we will find that the value of $\sin 37^\circ = 0.60$ and $\sin 24^\circ = 0.40$. Putting these values in the above relation, we get :

$$n = \frac{0.60}{0.40}$$

$$n = 1.50$$

Thus, the refractive index of this glass is 1.50.

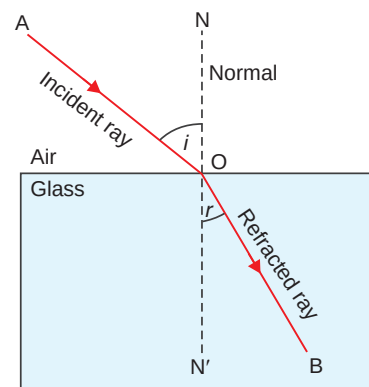


Figure 16.



Please note that **since the refractive index is a ratio of two similar quantities (the sines of angles), it has no units.** It is a pure number. The refractive index of a substance does not depend on the angle of incidence. When we talk of refractive index of a substance, say glass, we mean the value of $\frac{\sin i}{\sin r}$ for light passing from air to glass. Strictly speaking, it should mean the value of $\frac{\sin i}{\sin r}$ for light passing from vacuum to glass. But the difference in using air in place of vacuum is so small that it is ignored.

When light rays go from air (or vacuum) into another medium, then they bend (or refract) to some extent. Now, some media bend the light rays more than the others. **The refractive index of a medium gives an indication of the light-bending ability of that medium.** For example, the refractive index of glass is greater than the refractive index of water, therefore, the light rays bend more on passing from air into glass than from air into water. We will now solve one problem based on refractive index.

Sample Problem. A beam of light passes from air into a substance X. If the angle of incidence be 72° and the angle of refraction be 40° , calculate the refractive index of substance X. (Given : $\sin 72^\circ = 0.951$ and $\sin 40^\circ = 0.642$)

Solution. We know that :

$$\text{Refractive index} = \frac{\text{Sine of angle of incidence}}{\text{Sine of angle of refraction}}$$

$$\text{or} \quad n = \frac{\sin i}{\sin r}$$

Here, Angle of incidence, $i = 72^\circ$

And, Angle of refraction, $r = 40^\circ$

$$\text{So,} \quad n = \frac{\sin 72^\circ}{\sin 40^\circ}$$

We are given that $\sin 72^\circ = 0.951$ and $\sin 40^\circ = 0.642$. So, putting these values of $\sin 72^\circ$ and $\sin 40^\circ$ in the above relation, we get :

$$n = \frac{0.951}{0.642}$$

$$\text{or} \quad n = 1.48$$

Thus, the refractive index of substance X is 1.48.

So far we have denoted the refractive index of a substance just by the letter n . In its full form, the refractive index n has, however, two subscripts (lower words or letters) which show the two substances or media between which the light travels. For example, the refractive index for light going from *air* into *glass* is written as $_{\text{air}}n_{\text{glass}}$ (or ${}_a n_g$ where $a = \text{air}$ and $g = \text{glass}$). We will discuss this in more detail after a while.

Refractive Index and Speed of Light

The material (or substance) through which light travels is called medium. Light is refracted (or bent) in going from one medium to another because its speed changes (it slows down or speeds up). Due to this, **the refractive index (n) can also be written as a ratio of speeds of light in the two media.**

Look at Figure 17 in which a ray of light AO is going from medium 1 to medium 2 as OB . The speed of light in medium 2 is different from that in medium 1. Let the speed of light in medium 1 be v_1 and that in medium 2 be v_2 . Now, **the refractive index of medium 2 with respect to medium 1 is equal to the ratio of speed of light in medium 1 to the speed of light in medium 2.**

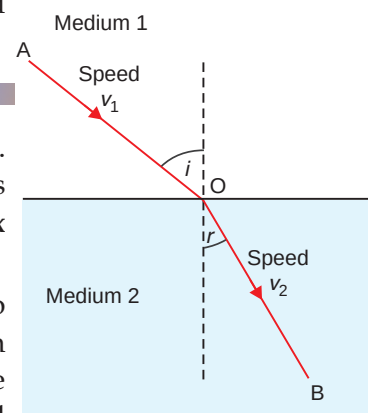


Figure 17.

This can be written as :

$$\begin{aligned} \text{medium 1 } n_{\text{medium 2}} &= \frac{\text{Speed of light in medium 1}}{\text{Speed of light in medium 2}} \\ \text{or } {}_1n_2 &= \frac{v_1}{v_2} \\ \text{where } {}_1n_2 &= \text{Refractive index of medium 2} \\ &\quad \text{with respect to medium 1} \\ v_1 &= \text{Speed of light in medium 1} \\ \text{and } v_2 &= \text{Speed of light in medium 2} \end{aligned}$$

Please note that the symbol $\text{medium 1 } n_{\text{medium 2}}$ or ${}_1n_2$ means that it is the refractive index of medium 2 with respect to medium 1 (for the light entering from medium 1 into medium 2). We have represented the refractive index of medium 2 with respect to medium 1 by the symbol ${}_1n_2$. In some books, however, the refractive index of medium 2 with respect to medium 1 is represented by the symbol n_{21} (read as n-two-one and not as n-twenty one). We have not used this notation because we find it a bit confusing.

When light is going from one medium (other than vacuum or air) to another medium, then the value of refractive index is called *relative refractive index*. For example, when the light is going from water into glass, then the value of refractive index will be the relative refractive index of glass with respect to water. The relative refractive index has always 'two subscripts' with its symbol n which indicate the two media in which the light travels. For example, for the light going from water to glass, the refractive index is written as $\text{water } n_{\text{glass}}$ (or ${}_w n_g$). The symbol $\text{water } n_{\text{glass}}$ means that it is the refractive index of glass with respect to water, that is, it is the refractive index of glass for light entering from water into glass.

When light is going from vacuum to another medium, then the value of refractive index is called the *absolute refractive index*. The absolute refractive index has only one subscript with its symbol n on its right side which indicates the name of the medium (the word vacuum is not written as a subscript). For example, for the light going from vacuum into glass, the absolute refractive index of glass is represented as n_{glass} (and not as $\text{vacuum } n_{\text{glass}}$). The symbol n_{glass} means that it is the refractive index of glass with respect to vacuum, that is, it is the refractive index of glass for light entering from vacuum into glass. Please note that the symbol n_{glass} is also written in short as n_g .

The exact speed of light in vacuum is 2.9979×10^8 m/s and that in air is 2.9970×10^8 m/s. We can see that the speed of light in air is almost the same as that in vacuum, so for the purpose of determining refractive index we can also treat air as if it were vacuum. So, **the refractive index of a medium (or substance) with respect to air is also considered to be its *absolute refractive index*.** Thus, $\text{air } n_{\text{glass}}$ can also be written as n_{glass} . The absolute refractive index of a medium (or substance) is just called its refractive index.

We will now write some simplified formulae for calculating the refractive index from the given values of the speed of light in the two media. These are given below.

The ratio of speed of light in vacuum to the speed of light in a medium, is called the refractive index of that medium. That is :

$$\text{Refractive index (of a medium)} = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$$

Since the speed of light in air is almost equal to the speed of light in vacuum, so for all practical purposes we can also say that : **The ratio of speed of light in air to the speed of light in a medium, is called refractive index of that medium.** That is :

$$\text{Refractive index (of a medium)} = \frac{\text{Speed of light in air}}{\text{Speed of light in medium}}$$

Let us take 'glass' as the medium and write a relation for its refractive index. Now, the speed of light in air is 3×10^8 m/s and the speed of light in common glass is 2×10^8 m/s. So :

$$\text{Refractive index of glass, } n_g = \frac{\text{Speed of light in air (or vacuum)}}{\text{Speed of light in glass}}$$

$$\text{or } n_g = \frac{3 \times 10^8}{2 \times 10^8}$$

$$\text{or } n_g = \frac{3}{2}$$

$$\text{or } n_g = 1.5$$

Thus, the refractive index of this glass is 1.5. **By saying that the refractive index of glass is 1.5 we mean that the ratio of the speed of light in air (or vacuum) to the speed of light in glass is equal to 1.5.** Let us solve one problem now.

Sample Problem. Light enters from air into a glass plate having refractive index 1.50. What is the speed of light in glass ? (The speed of light in vacuum is 3×10^8 m s⁻¹). **(NCERT Book Question)**

Solution. We know that :

$$\text{Refractive index of glass} = \frac{\text{Speed of light in air (or vacuum)}}{\text{Speed of light in glass}}$$

$$\text{So, } 1.50 = \frac{3 \times 10^8}{\text{Speed of light in glass}}$$

$$\begin{aligned} \text{or Speed of light in glass} &= \frac{3 \times 10^8}{1.50} \text{ m s}^{-1} \\ &= 2 \times 10^8 \text{ m s}^{-1} \end{aligned}$$

Thus, the speed of light in glass is 2×10^8 m s⁻¹ (or 2×10^8 m/s).

The refractive index depends on the nature of the material of the medium and on the wavelength (or colour) of the light used. The value of refractive index of a substance is a characteristic property of that substance which can be used to identify it. The refractive indices of some of the common substances are given below (indices is the plural of index). These refractive index values have been obtained by using yellow sodium light of wavelength 589 nm (or 5.89×10^{-7} m).

Refractive Index of Some Common Substances

(with respect to air or vacuum)

Substance (or Medium)	Refractive index (n)	Substance (or Medium)	Refractive index (n)
1. Air	1.0003	8. Benzene	1.50
2. Ice	1.31	9. Crown glass	1.52
3. Water	1.33	10. Carbon disulphide	1.63
4. Alcohol	1.36	11. Dense flint glass	1.65
5. Sulphuric acid	1.43	12. Ruby	1.71
6. Kerosene	1.44	13. Sapphire	1.77
7. Turpentine oil	1.47	14. Diamond	2.42

Please note that different types of glass have different chemical compositions due to which they have somewhat different values of refractive indices. Because of this reason no single value can be given for the refractive index of all types of glass. The refractive index of glass usually varies from 1.5 to 1.9. Another point to be noted is that diamond has a very high refractive index of 2.42.

The construction of lenses of the optical instruments like cameras, microscopes and telescopes, etc., depends on an accurate knowledge of the refractive index of glass used for making lenses. Please note that **if any two media are optically exactly the same, then no bending occurs when light passes from one medium to another.** In other words, if the refractive indices of two media are equal, then there will be no bending of light rays when they pass from one medium to another. The ability of a substance to refract light is also expressed in terms of its optical density. The optical density of a substance (or medium) is the degree to which it retards (or slows down) the rays of light passing through it. **A substance having higher refractive index is optically denser than another substance having lower refractive index.** For example, the refractive index of one type of glass is 1.52 and that of water is 1.33. Since glass has a higher refractive index than water, therefore, glass is optically denser than water, and more bending of light rays takes place in glass than in water. From this we conclude that **higher the refractive index of a substance, more it will change the direction of a beam of light passing through it.**



Figure 18. Diamonds sparkle partly because of their high refractive index. The shape to which they are cut also helps them to sparkle.

Please note that **the optical density of a substance is different from its mass density.** A substance may have a higher optical density than another substance but its mass density may be less. For example, kerosene having a higher refractive index than water is optically denser than water though its mass density is less than that of water. We have been using the terms 'rarer medium' and 'denser medium' in our discussions. It actually means 'optically rarer medium' and 'optically denser medium'.

We will now show that **the refractive index for light going from medium 1 to medium 2 is equal to the reciprocal of the refractive index for light going from medium 2 to medium 1.** Suppose we have two media : medium 1 and medium 2. We can find out the refractive index in two ways : one for the light going from medium 1 to medium 2 (as shown in Figure 19) and the other for light going in the reverse direction, from medium 2 to medium 1 (as shown in Figure 20).

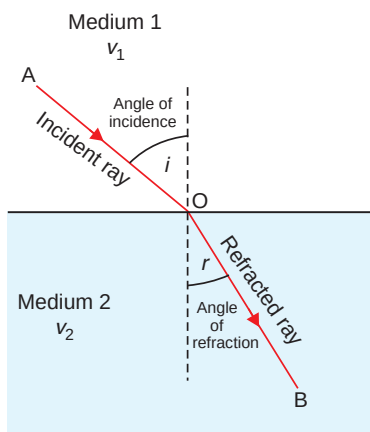


Figure 19. A ray of light going from medium 1 into medium 2

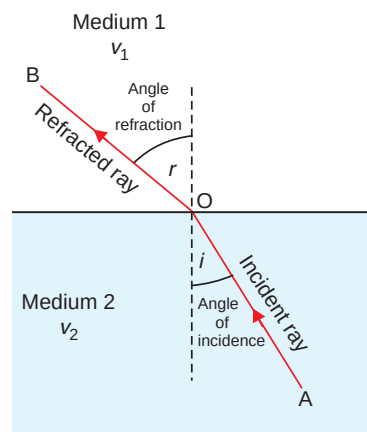


Figure 20. A ray of light going from medium 2 into medium 1

Suppose the speed of light in medium 1 is v_1 and that in medium 2 is v_2 . Now :

- (i) For the light going from medium 1 to medium 2 (Figure 19), the refractive index is given by :

$${}_1n_2 = \frac{v_1}{v_2} \quad \text{.....(1)}$$

- (ii) And for the light going from medium 2 to medium 1 (Figure 20), the refractive index is given by :

$${}_2n_1 = \frac{v_2}{v_1}$$

Let us take the reciprocal of this equation. This will give us :

$$\frac{1}{{}_2n_1} = \frac{v_1}{v_2} \quad \dots\dots(2)$$

Now, if we compare equations (1) and (2), we find that their right hand sides are equal, so their left hand sides should also be equal. Thus,

$${}_1n_2 = \frac{1}{{}_2n_1}$$

This can also be written as :

$$\text{medium 1 } n_{\text{medium 2}} = \frac{1}{\text{medium 2 } n_{\text{medium 1}}}$$

This means that the refractive index for light going from medium 1 to medium 2 is equal to the reciprocal of refractive index for light going from medium 2 to medium 1.

If medium 1 is air and medium 2 is glass, then the above relation can be written as :

$${}_{\text{air}}n_{\text{glass}} = \frac{1}{{}_{\text{glass}}n_{\text{air}}}$$

or

$${}_a n_g = \frac{1}{{}_g n_a}$$

Thus, the refractive index of glass for light going from *air* to *glass* is the reciprocal of the refractive index for light going from *glass* to *air*. Let us solve some problems now.

Sample Problem 1. If the refractive index of water for light going from air to water be 1.33, what will be the refractive index for light going from water to air ?

Solution. Here,

$${}_a n_w = 1.33$$

Now,

$$\begin{aligned} {}_w n_a &= \frac{1}{{}_a n_w} \\ &= \frac{1}{1.33} \\ &= 0.75 \end{aligned}$$

Sample Problem 2. The refractive indices of kerosene, turpentine and water are 1.44, 1.47 and 1.33, respectively. In which of these materials does light travel fastest ? **(NCERT Book Question)**

Solution. We know that :

$$\text{Refractive index} = \frac{\text{Speed of light in air}}{\text{Speed of light in medium}}$$

$$\text{So, Speed of light in medium} = \frac{\text{Speed of light in air}}{\text{Refractive index}}$$

It is obvious from the above relation that the speed of light will be the *maximum* in that medium (or substance) which has the *lowest* refractive index. Now, out of kerosene, turpentine and water, water has the lowest refractive index of 1.33. So, the light will have maximum speed in water or light will travel fastest in water.

We are now in a position to **answer the following questions :**

Very Short Answer Type Questions

1. What name is given to the ratio of sine of angle of incidence to the sine of angle of refraction ?
2. Write the relation between the angle of incidence and the angle of refraction for a medium.

- What is the unit of refractive index ?
- Which has higher refractive index : water or glass ?
- Refractive indices of carbon disulphide and ethyl alcohol are 1.63 and 1.36 respectively. Which is optically denser ?
- The refractive index of diamond is 2.42. What is the meaning of this statement in relation to the speed of light ?
- If the refractive index for light going from air to diamond be 2.42, what will be the refractive index for light going from diamond to air ?
- How is the refractive index of a material related to the speed of light in it ?
- Fill in the following blank with a suitable word :
When a ray of light goes from air into a clear material, you see the ray bend. How much the ray bends is determined by the.....of the material.

Short Answer Type Questions

- Give three examples of materials that refract light rays. What happens to the speed of light rays when they enter these materials ?
- Define Snell's law of refraction. A ray of light is incident on a glass slab at an angle of incidence of 60° . If the angle of refraction be 32.7° , calculate the refractive index of glass. (Given : $\sin 60^\circ = 0.866$, and $\sin 32.7^\circ = 0.540$).
- The speed of light in vacuum and in two different glasses is given in the table below :

Medium	Speed of light
Vacuum	$3.00 \times 10^8 \text{ m/s}$
Flint glass	$1.86 \times 10^8 \text{ m/s}$
Crown glass	$1.97 \times 10^8 \text{ m/s}$

- Calculate the absolute refractive indexes of flint glass and crown glass.
 - Calculate the relative refractive index for light going from crown glass to flint glass.
- The speed of light in air is $3 \times 10^8 \text{ m/s}$. In medium X its speed is $2 \times 10^8 \text{ m/s}$ and in medium Y the speed of light is $2.5 \times 10^8 \text{ m/s}$. Calculate:
(a) $_{\text{air}} n_X$ (b) $_{\text{air}} n_Y$ (c) ${}_X n_Y$
 - What is the speed of light in a medium of refractive index $\frac{6}{5}$ if its speed in air is 3,00,000 km/s ?
 - The refractive index of glass is 1.5. Calculate the speed of light in glass. The speed of light in air is $3.0 \times 10^8 \text{ ms}^{-1}$.
 - The speed of light in water is $2.25 \times 10^8 \text{ m/s}$. If the speed of light in vacuum be $3 \times 10^8 \text{ m/s}$, calculate the refractive index of water.
 - Light enters from air into diamond which has a refractive index of 2.42. Calculate the speed of light in diamond. The speed of light in air is $3.0 \times 10^8 \text{ ms}^{-1}$.

Long Answer Type Question

- State and explain the laws of refraction of light with the help of a labelled diagram.
- What is meant by the refractive index of a substance ?
- Light travels through air at 300 million ms^{-1} . On entering water it slows down to 225 million ms^{-1} . Calculate the refractive index of water.

Multiple Choice Questions (MCQs)

- The refractive indices of four substances P, Q, R and S are 1.50, 1.36, 1.77 and 1.31 respectively. The speed of light is the maximum in the substance :
(a) P (b) Q (c) R (d) S
- The refractive indices of four materials A, B, C and D are 1.33, 1.43, 1.71 and 1.52 respectively. When the light rays pass from air into these materials, they refract the maximum in :
(a) material A (b) material B (c) material C (d) material D

21. The refractive index of glass for light going from air to glass is $\frac{3}{2}$. The refractive index for light going from glass to air will be :
 (a) $\frac{1}{3}$ (b) $\frac{4}{5}$ (c) $\frac{4}{6}$ (d) $\frac{5}{2}$
22. The refractive indices of four media A, B, C and D are 1.44, 1.52, 1.65 and 1.36 respectively. When light travelling in air is incident in these media at equal angles, the angle of refraction will be the minimum :
 (a) in medium A (b) in medium B (c) in medium C (d) in medium D
23. The speed of light in substance X is 1.25×10^8 m/s and that in air is 3×10^8 m/s. The refractive index of this substance will be :
 (a) 2.4 (b) 0.4 (c) 4.2 (d) 3.75
24. The refractive indexes of four substances P, Q, R and S are 1.77, 1.50, 2.42 and 1.31 respectively. When light travelling in air is incident on these substances at equal angles, the angle of refraction will be the maximum in :
 (a) substance P (b) substance Q (c) substance R (d) substance S
25. The refractive index of water is :
 (a) 1.33 (b) 1.50 (c) 2.42 (d) 1.36
26. The refractive index of water with respect to air is $\frac{4}{3}$. The refractive index of air with respect to water will be :
 (a) 1.75 (b) 0.50 (c) 0.75 (d) 0.25
27. Refractive indices of water, sulphuric acid, glass and carbon disulphide are 1.33, 1.43, 1.53 and 1.63 respectively. The light travels slowest in :
 (a) sulphuric acid (b) glass (c) water (d) carbon disulphide
28. The refractive index of glass with respect to air is $\frac{3}{2}$ and the refractive index of water with respect to air is $\frac{4}{3}$. The refractive index of glass with respect to water will be :
 (a) 1.525 (b) 1.225 (c) 1.425 (d) 1.125

Questions Based on High Order Thinking Skills (HOTS)

29. The following table gives the refractive indices of a few media :

	1	2	3	4	5
Medium :	Water	Crown glass	Rock salt	Ruby	Diamond
Refractive index :	1.33	1.52	1.54	1.71	2.42

Use this table to give an example of :

- (i) a medium pair so that light speeds up when it goes from one of these medium to another.
 (ii) a medium pair so that light slows down when it goes from one of these medium to another.
30. Refractive indices of four media A, B, C and D are given below :

Medium	Refractive index
A	1.33
B	1.44
C	1.52
D	1.65

In which of these four media is the speed of light (i) maximum, and (ii) minimum ?

ANSWERS

7. 0.41 9. refractive index 11. 1.60 12. (a) Flint glass : 1.61 ; Crown glass : 1.52 (b) 1.059 13. (a) 1.5 (b) 1.2 (c) 0.8 14. 2,50,000 km/s 15. 2.0×10^8 ms⁻¹ 16. 1.33 17. 1.24×10^8 ms⁻¹ 18. (c) 1.33 19. (d) 20. (c) 21. (c) 22. (c) 23. (a) 24. (d) 25. (a) 26. (c) 27. (d) 28. (d) 29. (i) Crown glass to Water (ii) Water to Diamond 30. (i) A (ii) D

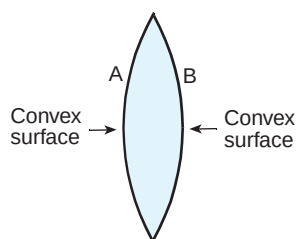
REFRACTION OF LIGHT BY SPHERICAL LENSES

We have all seen a palmist using a lens (called magnifying glass) for seeing the details of the lines of a person's palm. A watch maker also uses a lens to see the extremely small parts of a watch clearly. In fact, lenses play a very important role in our everyday life. Lenses are used in making spectacles, cameras, microscopes, telescopes, film projectors, and many, many other optical instruments. We have already studied that the working of a mirror is based on the reflection of light rays from its surface. **The working of a lens is based on the refraction of light rays when they pass through it.** We will now study the formation of images by lenses in detail. Before we do that, we should know the various types of spherical lenses and the terms like optical centre, principal axis, principal focus (or just focus), and focal length, etc., which are used in the study of refraction of light by lenses. These are discussed below.

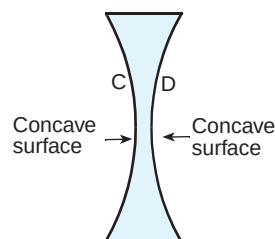


A lens is a piece of transparent glass bound by two spherical surfaces. **Figure 21.** This is a microscope. Lenses are used in making microscopes. There are two types of lenses : Convex lens and Concave lens.

(i) A convex lens is thick at the centre but thinner at the edges. Figure 22(a) shows a convex lens in which the two surfaces *A* and *B* are convex or bulging out at the centre.



(a) Convex lens



(b) Concave lens

Figure 22.

(ii) A concave lens is thin in the middle but thicker at the edges. Figure 22(b) shows a concave lens in which the two surfaces *C* and *D* are concave or bent inward.

Figure 22(a) shows the side view of a convex lens. When we look at a convex lens from the front side, it looks like a piece of transparent spherical glass (round glass) having a bulge in the middle [see Figure 23(a)]. We can feel the bulge in the middle of the convex lens by touching it. Similarly, Figure 22(b) shows the side view of a concave lens. When we look at a concave lens from the front side, it looks like a piece of transparent spherical glass (round glass) having a 'depression' in the middle [see Figure 23(b)]. We can feel the depression in the middle of a concave lens by touching it. Please note that the lenses (convex lens and concave lens) work on the refraction of light through them.



(a) Convex lens (b) Concave lens

Figure 23. Front view of spherical lenses.

Optical Centre and Principal Axis of a Lens

The centre point of a lens is known as its optical centre. The optical centre of a lens is usually denoted by the letter *C*. In Figure 24, *C* is the optical centre of the convex lens. The optical centre of a lens has a property that a ray of light passing through it does not suffer any deviation and goes straight. The optical centre of a lens is sometimes also denoted by the letter *O*.

The principal axis of a lens is a line passing through the optical centre of the lens and perpendicular

to both the faces of the lens. In Figure 24, the line $F'F$ is the principal axis of the convex lens and it passes through the optical centre C .

Principal Focus and Focal Length of a Convex Lens

Suppose a parallel beam of light rays falls on a convex lens as shown in Figure 24. These light rays are parallel to one another and also parallel to the axis of the lens. The incident rays pass through the convex lens and get refracted (or bent) according to the laws of refraction. All the rays, after passing through the convex lens, converge at the same point F on the other side (right side) of the lens. The point F is called principal focus (or just focus) of the convex lens. We can now say that : **The principal focus of a convex lens is a point on its principal axis to which light rays parallel to the principal axis converge after passing through the lens.** In Figure 24, point F is the principal focus for the light rays coming from the left side. If the incident light rays fall on the convex lens from the right hand side, they will converge to a point F' on the left side of the lens. Thus, F' is the second focus of the convex lens. From this discussion we conclude that **a lens has two foci. The two foci of a lens are at equal distances from the optical centre, one on either side of the lens** (The word '*foci*' is the plural form of '*focus*'). The two foci of a lens are usually denoted by the letters F and F' . Since all the light rays actually pass through the focus of a convex lens, therefore, **a convex lens has real focus.**

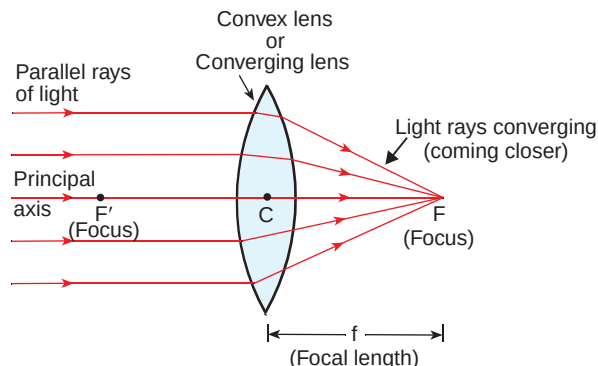


Figure 24. A convex lens converges (brings closer) a parallel beam of light rays to a point F on its other side (right side).

We are now in a position to define the focal length of a lens. **The focal length of a lens is the distance between optical centre and principal focus of the lens.** In Figure 24, the distance CF is the focal length of the convex lens. It should be noted that the distance CF' is also equal to the focal length of the lens. The focal length of a lens is denoted by the letter f . The focal length of a lens depends on the refractive index of the glass from which it is made, and the curvature of its two surfaces. Higher the refractive index, shorter will be the focal length. Similarly, more the curvature, shorter is the focal length.

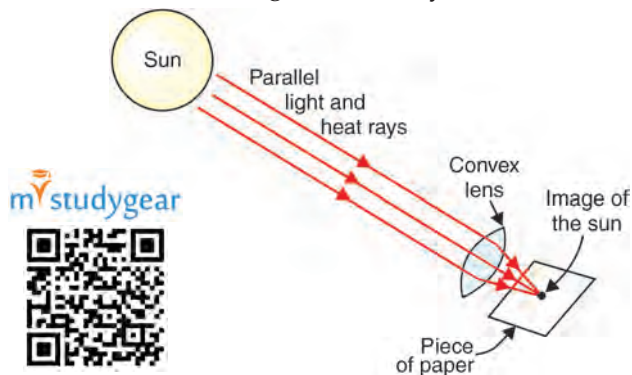


Figure 25. A convex lens forms a real image of the sun on a piece of paper.

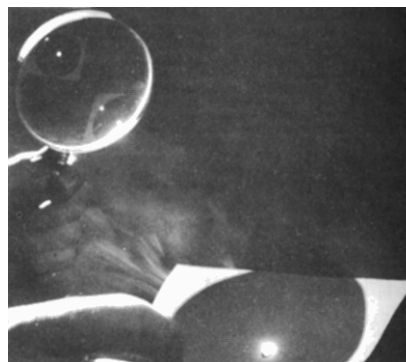


Figure 26. A picture showing the formation of image of the sun by a convex lens.

A convex lens is also known as a converging lens because it converges (brings to a point), a parallel beam of light rays passing through it (see Figure 24). The fact that a convex lens converges (or focusses) parallel rays of light to a single point can be shown as follows : Place a piece of paper on the ground in bright sunshine (see Figure 25). Hold a convex lens some distance above the piece of paper in such a way that a sharp image of the sun is formed on the piece of paper. Here the convex lens is converging the parallel rays of sunlight due to which the sun's rays get concentrated on a small part of the paper (where image is formed). The heat energy of focussed sunlight rays burns a hole in the piece of paper (where sun's

image is formed). Please note that we should never look at the sun through a convex lens. It can damage our eyes permanently by focussing a lot of sunlight energy into our eyes.

Principal Focus and Focal Length of a Concave Lens

We have just studied that a **convex lens converges a parallel beam of light rays**. A concave lens has just the opposite effect on such rays of light. **A concave lens diverges a parallel beam of light rays**. The action of a concave lens on a parallel beam of light rays is shown in Figure 27. When a parallel beam of light rays falls on a concave lens, the rays will spread out (or diverge) after passing through the lens. Since the refracted rays are diverging away from one another, they do not actually meet at a point. The diverging rays when produced backwards (as shown by dotted lines in Figure 27) appear to meet at a point F on the left side of the lens. To a person on the right hand side, the refracted rays appear to be diverging (or coming) from a point F on the principal axis of the concave lens. This point is the principal focus of the concave lens. Thus, **the principal focus of a concave lens is a point on its principal axis from which light rays, originally parallel to the axis, appear to diverge after passing through the concave lens**. In Figure 27, the parallel rays of light appear to be diverging from point F after refraction. So, F is the principal focus of the concave lens for the light rays coming from the left side. Like a convex lens, a concave lens also has two foci, one on each side of the concave lens. For example, if the parallel rays fall on the concave lens from the right side, then they will appear to diverge from a point F' . Thus, F' is the second focus of the concave lens. **A concave lens is also known as a diverging lens because it diverges a parallel beam of light rays** (see Figure 27). Since the light rays do not actually pass through the focus of a concave lens, **a concave lens has a virtual focus**. In Figure 27, the distance CF is the focal length of concave lens. The distance CF' is also equal to the focal length.

A yet another term which is used in the study of spherical lenses is 'aperture'. The aperture of a spherical lens (convex lens or concave lens) is the surface from which refraction of light takes place through the lens. The aperture of spherical lens is represented by its diameter. In most simple words, aperture tells us the size of the lens.

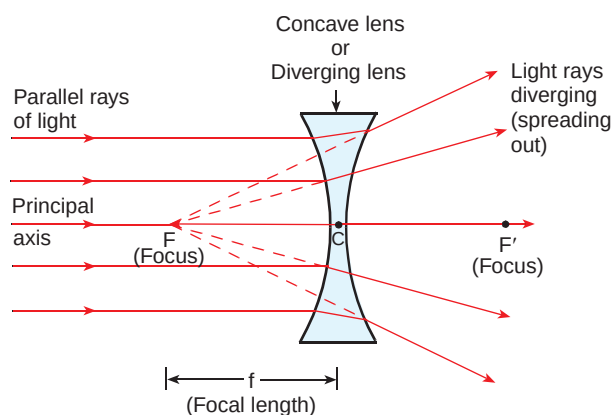


Figure 27. A concave lens diverges (spreads out) a parallel beam of light rays.

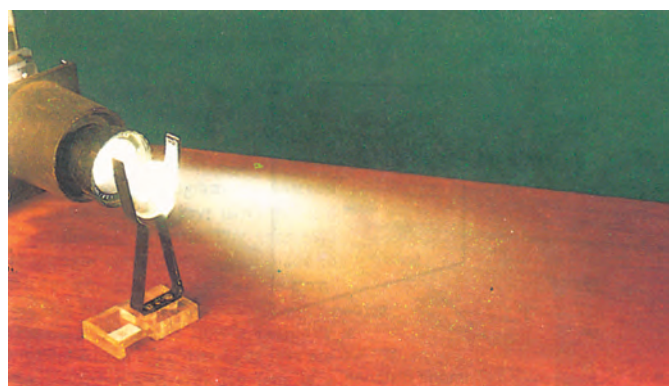


Figure 28. This picture shows that light diverges (or spreads) after passing through a concave lens.

Rules for Obtaining Images Formed by Convex Lenses

When an object is placed in front of a convex lens, an image is formed. **The image is formed at that point where at least two refracted light rays meet (or appear to meet)**. To find out the position and nature of the image formed by a convex lens, we will use only those two rays of light (coming from the top of the object) whose paths after refraction from the lens are known to us and easy to draw (the bottom of the object is always assumed to be on the principal axis of the lens). Any two of the following rays of light are usually used to locate the images formed by convex lenses. We call them rules for obtaining images in convex lenses.

Rule 1. A ray of light which is parallel to the principal axis of a convex lens, passes through its focus after refraction through the lens. This is shown in Figure 29. Here we have a convex lens L and its principal

axis is $X'Y'$. Now, a ray of light AD (coming from an object) is parallel to the principal axis of the convex lens. It enters the convex lens and gets refracted (or bends) at point D inside it. After refraction its path changes, it passes through focus F and goes in the direction DX (see Figure 29).

Rule 2. A ray of light passing through the optical centre of a convex lens goes straight after refraction through the lens. It does not get deviated (or bent). This is shown in Figure 30. A ray of light AC is passing through the optical centre C of a convex lens. It goes straight in the direction CY after passing through the lens. It does not get deviated (or bent) from its original path (see Figure 30). Please note that a ray of light going along the principal axis of a convex lens also passes straight through the lens without any deviation. In Figure 30, the principal axis of the convex lens is $X'Y'$. So, a ray of light going along the principal axis $X'Y'$ of this convex lens will also go straight (without bending). We should keep this point in mind because in drawing ray-diagrams, an object is always placed above the principal axis of a lens so that a ray of light coming from its bottom always goes straight through the lens (along the principal axis).

Rule 3. A ray of light passing through the focus of a convex lens becomes parallel to its principal axis after refraction through the lens. This rule is just the reverse case of the first rule and it is shown in Figure 31. Here a ray of light AD (coming from the object) is passing through the focus F' of the convex lens. It enters the convex lens and gets refracted (or bends) at point D inside it. After passing through the convex lens, it becomes parallel to the principal axis of the lens and goes in the direction DX (see Figure 31). Please note that in this case the ray of light has to pass through the second focus F' of the convex lens which lies on its left side.

We should remember the paths of the three rays of light described above because these will be used to construct ray-diagrams for finding the position and nature of images formed by convex lenses. At any given time, we will use only two of the above three types of light rays to find the position of image formed by a convex lens. We will now discuss the various positions at which an object can be placed in front of a convex lens to form images.

FORMATION OF DIFFERENT TYPES OF IMAGES BY A CONVEX LENS

The type of image formed by a convex lens depends on the position of the object in front of the lens. We can place the object at different positions (or distances) from a convex lens to get different types of images. We can place the object :

- (i) between the optical centre (C) and focus (F') (see Figure 32)
- (ii) at the focus (F')
- (iii) between F' and $2F'$
- (iv) at $2F'$
- (v) beyond $2F'$, and
- (vi) at infinity.

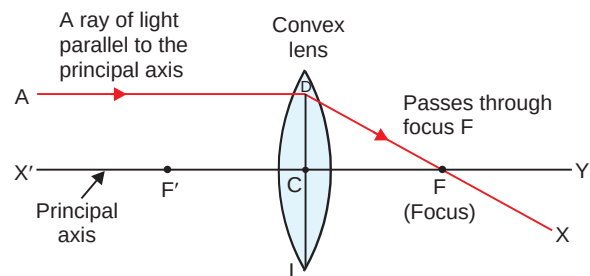


Figure 29.

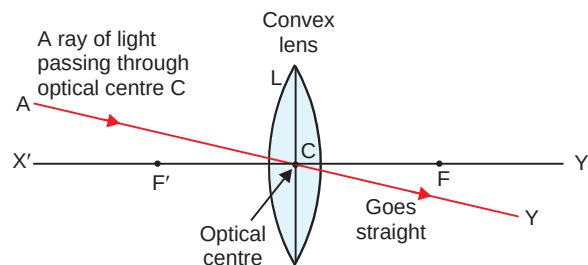


Figure 30.

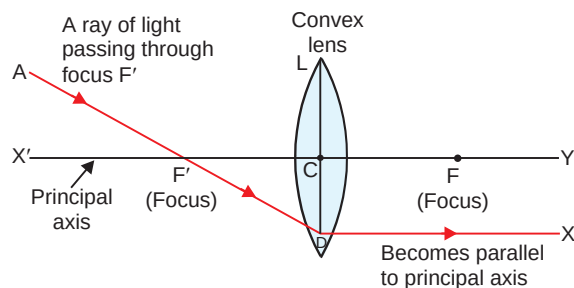


Figure 31.

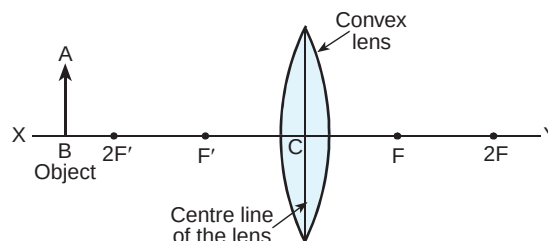


Figure 32.

We will consider all these six positions one by one.

Case 1. Image formed by a convex lens when the object is placed between optical centre and focus (Object between C and F')

In Figure 33 we have a convex lens with optical centre C and two foci F and F' . The object AB has been placed between the optical centre C and focus F' so that it is at a distance less than the focal length f of the convex lens. In other words, the object AB is within the focus of the convex lens. A ray of light AD starting from the top of the object and parallel to the axis, passes through the other focus F after refraction through the lens and goes in the direction DX . Another ray of light AC passing through the optical centre C of the lens goes straight in the direction CY . The two refracted rays DX and CY diverge away from one another and, therefore, they cannot meet at a point on the right side to form a real image. We produce the refracted rays DX and CY backwards (to the left side) by dotted lines. They appear to meet at point A' on the left side. Thus, A' is the virtual image of point A of the object. To get the complete image of the object, we draw $A'B'$ perpendicular to the axis from point A' . Thus, $A'B'$ is the complete image of object AB . When our eye looks into the lens from the right side, it appears as if the light rays are coming from A' and B' instead of A and B . Thus, $A'B'$ is the virtual image. It is erect, larger than the object and can be seen only by looking through the lens. It is formed on the same side of the lens as the object. From the above discussion we conclude that **when the object is placed within the focus of a convex lens, the image formed is :**

- (i) behind the object (on the left side of lens),
- (ii) virtual and erect, and
- (iii) larger than the object (enlarged or magnified).

Figure 33 explains the use of a convex lens as a magnifying glass (or simple microscope). It should be noted that to use a convex lens as a magnifying glass, the object to be viewed should be placed within its focus, so that a magnified and erect image of the object is obtained. That is, the object should be placed at

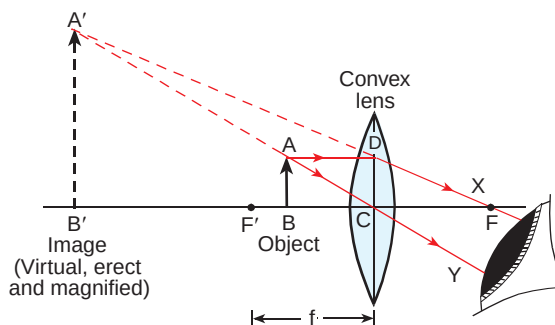


Figure 33. Formation of image by a convex lens when the object is placed between its optical centre (C) and focus (F').

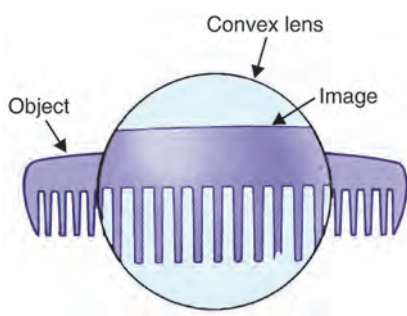


Figure 34. A convex lens is producing a magnified image of the middle part of the comb.



Figure 35. A convex lens being used as a magnifying glass to read the small print of a book.

a distance less than the focal length of the convex lens. For example, if the focal length of a convex lens is 5 centimetres, then the object to be magnified with it should be kept at a distance of less than 5 centimetres from this convex lens. So, if we look at a comb through this convex lens (by keeping the comb within 5 centimetres from it), then the part of comb seen through the convex lens appears much bigger in size than it actually is (see Figure 34). We actually see the magnified image of the comb through the convex lens. Figure 35 shows a convex lens being used as a magnifying glass to read the small print of a book.

Please note that **smaller the focal length of a convex lens, greater will be its magnifying power.** So,

we should use a convex lens of short focal length as a magnifying glass. It will produce large magnification and hence the object will appear much bigger when seen through it. For example, a convex lens of 5 cm focal length will have a greater magnifying power than a convex lens of 50 cm focal length. So, we should use the convex lens having a shorter focal length of 5 cm as the magnifying glass. Thus, if we are given two convex lenses of focal lengths 5 cm and 50 cm respectively, then we should prefer to use the convex lens having 5 cm focal length as a magnifying glass (to see small things or read small letters of a dictionary).

Case 2. When the object is placed at the focus of a convex lens (Object at F')

By saying that the object is at the focus of a convex lens, we mean that the object is at a distance equal to the focal length f of the lens. Figure 36 shows an object AB placed at the focus F' of a convex lens. A parallel ray of light AD passes through the focus F after refraction and goes along the path DX . Another ray AC passing through the centre C of the lens goes straight in the direction CY . The two refracted rays DX and CY are parallel to one another. These parallel rays will intersect (or meet) at a far off distance to form an image 'at infinity'. And since the image is formed at infinity, it is not possible to show it in our diagram. It will be real and inverted, and highly magnified or highly enlarged. From this discussion we conclude that

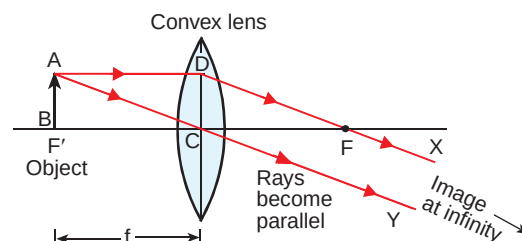


Figure 36. Formation of image by a convex lens when the object is placed at its focus (F')

when an object is placed at the focus of a convex lens, the image formed is :

- (i) at infinity,
- (ii) real and inverted, and
- (iii) highly enlarged.

Suppose we have a convex lens of focal length 5 cm, then its focus (F') will be at a distance of 5 cm from it. So, by saying that the object is placed at the focus of this convex lens, we mean that the object is placed at a distance of 5 cm from the convex lens on its left side. In this case the convex lens converts the diverging rays of light coming from the object into a parallel beam of light rays which form an image at infinity (or very large distance).

Case 3. When the object is between F' and $2F'$ (Object between f and $2f$)

By saying that the object is between F' and $2F'$, we mean that the object is at a distance greater than focal length f but less than twice the focal length $2f$. So, whether we say that the object is between F' and $2F'$ or between f and $2f$, it means the same thing. Figure 37 shows an object AB placed between F' and $2F'$ of a convex lens. A ray of light AD which is parallel to the principal axis passes through the focus F after refraction and goes in the direction DF (see Figure 37). Another ray of light AC passing through the centre C of the lens goes straight and meets the first refracted ray at point A' on the right side of the lens. Thus, A' is the real image of point A of the object. Draw $A'B'$ perpendicular to the axis to get the complete image $A'B'$. If we place a screen at B' , we can receive the image $A'B'$ on the screen. It should be noted

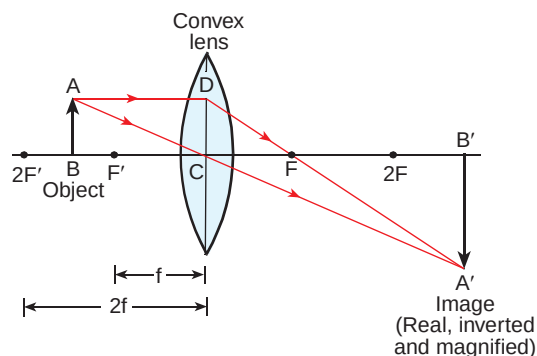


Figure 37. Formation of image by a convex lens when the object is placed between F' and $2F'$ (or between f and $2f$)

that the image is larger than the object (or magnified) and it is formed beyond $2F$ on the right side of the convex lens. From this discussion we conclude that **when an object is placed between F' and $2F'$ in front of a convex lens, the image formed is :**

- (i) beyond $2F$,
- (ii) real and inverted, and
- (iii) larger than the object (or magnified).

Figure 37 shows how a convex lens is used as a projection lens for the purpose of projecting a magnified real image of a slide (or a film) on a screen. The slide (or film) is kept between F' and $2F'$ of a convex lens and illuminated by a source of light (such as an electric lamp). A magnified image of the picture on the slide (or film) is produced on a screen placed on the other side of the convex lens.

Suppose we have a convex lens of 5 cm focal length, then its focus (F') will be at a distance of 5 cm from it. So, by saying that the object is between F' and $2F'$ from this convex lens, we will mean that the object is between 5 cm and $2 \times 5 = 10$ cm from the convex lens on its left side. And by saying that the image is formed beyond $2F$ on the other side, we mean that the image is formed at a distance of more than 10 cm from convex lens on its right side.

Case 4. When the object is at $2F'$ (Object at $2f$)

By saying that the object is placed at $2F'$, we mean that the object is at a distance equal to twice the focal length of the convex lens. In other words, the object is placed at a distance $2f$ from the convex lens. Figure 38 shows an object AB placed at a distance $2f$ (twice the focal length), from a convex lens. A ray of light AD which is parallel to the principal axis passes through the focus F after refraction and goes in the direction DF (see Figure 38). Another ray of light AC passing through the optical centre C of the lens, goes straight and meets the first refracted ray at point A' , on the right side of the lens. Thus, A' is the real image of point A of the object. We draw $A'B'$ perpendicular to the axis to get the complete image $A'B'$. We find that the image $A'B'$ is formed at $2F$, at a distance $2f$ on the right side of the convex lens. So, in this case, the object and its image are at equal distance ($2f$ each) from the convex lens, but they are on opposite sides of the lens. Thus, **when an object is placed at a distance $2f$ in front of a convex lens, then the image formed is :**

- (i) at a distance $2f$ on the other side of the lens,
- (ii) real and inverted, and
- (iii) of the same size as the object.

Suppose we have a convex lens of focal length 5 cm, then its focus (F') will be at a distance of 5 cm from it. So, by saying that the object is at $2F'$ of this convex lens, we will mean that the object is at a distance of $2 \times 5 = 10$ cm from this convex lens on its left side. And by saying that the image is formed at $2F$ on the other side, we will mean that the image is formed at a distance of $2 \times 5 = 10$ cm from the convex lens on its right side.

Case 5. When the object is beyond $2F'$ (Object beyond $2f$)

Figure 39 shows an object AB placed beyond $2f$ in front of a convex lens. To construct the ray diagram, we first take a ray of light AD parallel to the principal axis. After refraction, it will pass through focus F and go in the direction DF . A second ray of light AC , passing through the optical centre C will go straight in the direction CA' . The two refracted rays meet at point A' , so A' is the real image of point A of the object. To get the complete image, we draw $A'B'$ perpendicular to the axis. Thus, $A'B'$ is the complete image of the object AB . And it is formed between f and $2f$ on the other side of the lens. The image is real, inverted and smaller than the object. From the above discussion we conclude that **when an object is placed beyond $2f$ in front of a convex lens, then the image formed is :**

- (i) between f and $2f$ on the other side of the lens,
- (ii) real and inverted, and

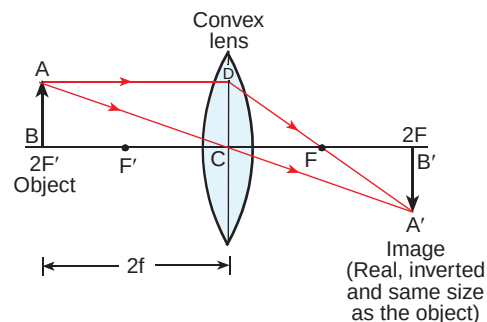


Figure 38. Formation of image by a convex lens when the object is placed at $2F'$ (at a distance $2f$ from the lens).

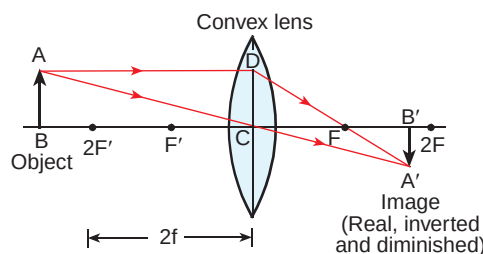


Figure 39. Formation of image by a convex lens when the object is placed beyond $2F'$ (at a distance more than $2f$).

(iii) smaller than the object (or diminished).

Figure 39 shows the action of a simple camera lens in producing a small, real and inverted image of an object on the film. A simple camera has a convex lens in it. The object to be photographed is at a distance of more than twice the focal length of the convex lens. The convex lens forms a real, inverted and small image (diminished image) of the object on the film.

Suppose we have a convex lens of focal length 5 cm. Then its focus (F') will be at a distance of 5 cm on its left side, and its $2F'$ will be at a distance of $2 \times 5 = 10$ cm on the left side. So, by saying that the object is beyond $2F'$ of this convex lens, we will mean that the object is beyond 10 cm from convex lens on the left side. And by saying that the image is formed between F and $2F$ on the other side, we will mean that the image is formed at a distance between 5 cm and 10 cm from the convex lens on the right hand side of the lens. Figure 40 shows an actual picture of the formation of a real, inverted and diminished image by a convex lens of the filament of a lighted bulb on a screen.

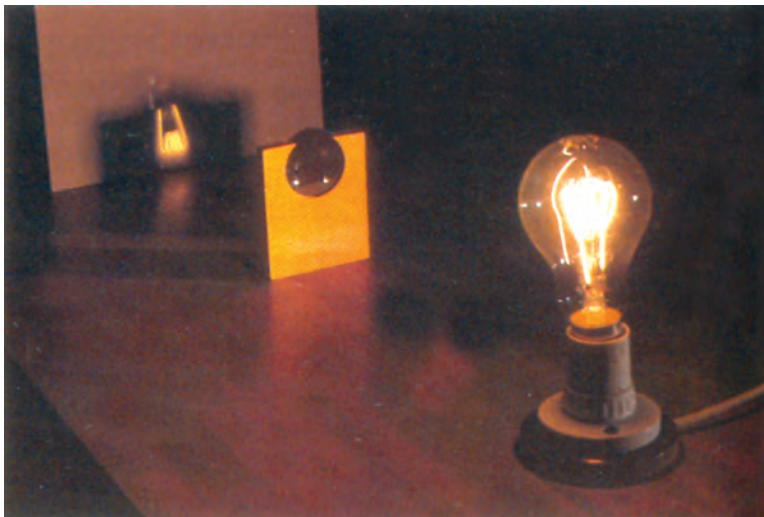


Figure 40. This picture shows a convex lens forming a real, inverted and diminished image of the filament of a light bulb. The bulb is placed at a distance more than twice the focal length from the convex lens.

Case 6. When the object is at infinity

When the object is at a considerable distance, we say that the object is at infinity. Suppose an object AB (in the form of an arrow pointing upwards) is at a considerable distance from a convex lens. Since the object is very far off from the lens, it has not been shown in Figure 41. Now, when the object is at infinity, then all the rays from a given point of the object, which are diverging in the beginning, become parallel to one another when they reach the lens after travelling a long distance. So, in Figure 41, we have two rays AD and AC coming from the same point A of the object. The incident rays AD and AC are parallel to one another but at an angle to the principal axis. The incident ray AD gets refracted along DX . The second ray AC passing through the optical centre C of the lens goes straight along CY and meets the first refracted ray at A' . Thus, A' is the real image of the top point A of the object. We draw $A'B'$ perpendicular to the axis from A' . Thus, $A'B'$ is the complete image of the object AB placed at infinity. It is clear from Figure 41, that the image is formed at the focus of the lens. It is real, inverted and much smaller than the object. From this discussion we conclude that **when an object is at infinity from a convex lens, then the image formed is :**

- (i) at the focus,**
- (ii) real and inverted, and**
- (iii) much smaller than the object (or highly diminished).**

Figure 41 represents the action of the objective lens of a telescope. Because in a telescope, the objective lens is a convex lens which forms a real, inverted and diminished image of the distant object at its focus. Please note that when the object kept at infinity in front of a convex lens is assumed to be a big arrow pointing upwards, then its image is formed at focus according to the ray-diagram shown in Figure 41. If, however, the object kept at infinity in front of a convex lens is round in shape (like the sun), then its image is formed at the focus according to the ray-diagram shown in Figure 24 on page 231.

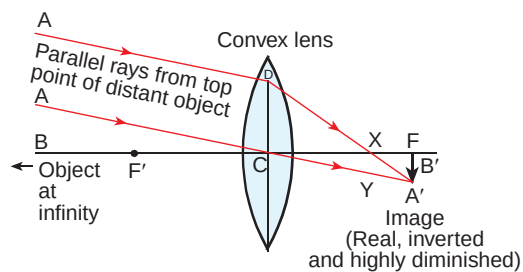


Figure 41. Formation of image by a convex lens when the object is placed at infinity (very large distance) from the lens.

To Determine the Focal Length of a Convex Lens Quickly but Approximately

When the object is at infinity, the distance of image from the lens will be equal to the focal length of the lens. This fact is used to find out the focal length of a convex lens quickly but approximately. Let us see how this is done.

To determine the focal length of a convex lens, we put the convex lens in a holder (or stand) and keep it in front of a distant object like a window (or a tree), so that the rays coming from the window pass through it. A cardboard screen is put behind the lens. We change the distance of the screen from the convex lens until a clear inverted image of the window is formed on the screen. Measure the distance of the screen from the lens with a scale. This distance will be the focal length of convex lens. For example, if the image of a distant window is formed at a distance of 20 cm from a convex lens, then the focal length of this convex lens will be 20 cm. And before we end this discussion, here is a summary of the images formed by a convex lens.

Summary of the Images Formed by a Convex Lens

Position of object	Position of image	Size of image	Nature of image
1. Between f and lens	On the same side as object	Enlarged	Virtual and erect
2. At f (at focus)	At infinity	Highly enlarged	Real and inverted
3. Between f and $2f$	Beyond $2f$	Enlarged	Real and inverted
4. At $2f$	At $2f$	Same size as object	Real and inverted
5. Beyond $2f$	Between f and $2f$	Diminished	Real and inverted
6. At infinity	At f (at focus)	Highly diminished	Real and inverted

Uses of Convex Lenses

- Convex lenses are used in spectacles to correct the defect of vision called hypermetropia (or long-sightedness).
- Convex lens is used for making a simple camera.
- Convex lens is used as a magnifying glass (or magnifying lens) (by palmists, watchmakers, etc.).
- Convex lenses are used in making microscopes, telescopes and slide projectors (or film projectors).

We will now solve some problems based on the formation of images by a convex lens.

Sample Problem 1. Where should an object be placed so that a real and inverted image of the same size is obtained by a convex lens ?

- at the focus of the lens.
- at twice the focal length.
- at infinity.
- between the optical centre of lens and its focus.

(NCERT Book Question)

Answer. (b) at twice the focal length.

Sample Problem 2. A convex lens has a focal length of 20 cm. Where should an object be placed in front of this convex lens so as to obtain an image which is real, inverted and same size as the object ? Draw the ray diagram to show the formation of image in this case.

(NCERT Book Question)

Solution. When the image formed by a convex lens is real, inverted and *same size* as the object, then the distance of the object from the lens is $2f$ (twice the focal length). Here,

$$\begin{aligned} \text{Focal length, } f &= 20 \text{ cm} \\ \text{So, } 2f &= 2 \times 20 \text{ cm} \\ &= 40 \text{ cm} \end{aligned}$$

Thus, the object should be placed at a distance of 40 cm in front of the convex lens. (Please draw the ray diagram yourself).

Sample Problem 3. An object is placed at the following distances from a convex lens of focal length 10 cm :

- (a) 8 cm (b) 15 cm (c) 20 cm (d) 25 cm

Which position of the object will produce :

- (i) a diminished real image ?
- (ii) a magnified real image ?
- (iii) a magnified virtual image ?
- (iv) an image of the same size as the object ?

Solution. The focal length of this convex lens is 10 cm. This means that $f = 10$ cm.

(i) A diminished real image is formed by a convex lens when the object is beyond $2f$, that is beyond 2×10 cm or beyond 20 cm. In this problem, the distance which is beyond 20 cm (or more than 20 cm) is 25 cm. Thus, 25 cm position of the object will produce a diminished real image.

(ii) A magnified real image is formed by a convex lens when the object is between f and $2f$, that is between 10 cm and 20 cm. In this problem, the distance which is between 10 cm and 20 cm is 15 cm. So, 15 cm position of the object will produce a magnified real image.

(iii) A magnified virtual image is formed by a convex lens when the object is within focus, at a distance less than the focal length f or less than 10 cm. In this problem, the distance which is less than the focal length of 10 cm is 8 cm. Thus, 8 cm position of object will produce a magnified virtual image.

(iv) An image of the same size as the object is formed by a convex lens when the object is at $2f$ or twice the focal length from lens. Here $2f = 2 \times 10 = 20$ cm. So, the 20 cm position of the object will produce an image of the same size as the object.

If this question is asked in the examination, then the answer can just be written as :

- (i) 25 cm (ii) 15 cm (iii) 8 cm (iv) 20 cm

Sample Problem 4. Which of the following lens would you prefer to use while reading small letters found in a dictionary ?

- (a) A convex lens of focal length 50 cm.
- (b) A concave lens of focal length 50 cm.
- (c) A convex lens of focal length 5 cm.
- (d) A concave lens of focal length 5 cm.

(NCERT Book Question)

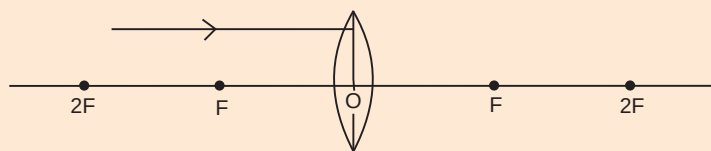
Answer. (c) A convex lens of focal length 5 cm.

Before we go further and discuss sign convention for lenses and the lens formula, please answer the following questions :

Very Short Answer Type Questions

- Name the lens which can concentrate sun's rays to a point and burn a hole in a piece of paper.
- Give the usual name for the following :
A point inside a lens through which the light passes undeviated.
- A 1 cm high object is placed at a distance of $2f$ from a convex lens. What is the height of the image formed ?
- If the image formed by a convex lens is of the same size as that of the object, what is the position of the image with respect to the lens ?
- If an object is placed at the focus of a convex lens, where is the image formed ?
- Where should an object be placed in order to use a convex lens as a magnifying glass ?
- Where should an object be placed in front of a convex lens so as to obtain its virtual, erect and magnified image ?
- Where should an object be placed in front of a convex lens so as to obtain its real, inverted and magnified image ?

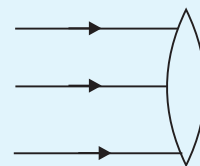
9. For what position of an object a real, diminished image is formed by a convex lens ?
10. If an object is at a considerable distance (or infinity) in front of a convex lens, where is the image formed ?
11. Draw the given diagram in your answer book and complete it for the path of a ray of light after passing through the lens.



12. What type of lens would you use as a magnifying glass ? How close must the object be to the lens ?
13. Name two factors on which the focal length of a lens depends.
14. State any two uses of convex lenses.
15. Fill in the following blanks with suitable words :
 - (a) Parallel rays of light are refracted by a convex lens to a point called the.....
 - (b) The image in a convex lens depends upon the distance of the.....from the lens.

Short Answer Type Questions

16. What is a lens ? Distinguish between a convex lens and a concave lens. Which of the two is a converging lens : convex lens or concave lens ?
17. (a) Explain with the help of a diagram, why the convex lens is also called a converging lens.
(b) Define principal axis, principal focus and focal length of a convex lens.
18. (a) Explain with the help of a diagram, why the concave lens is also called a diverging lens.
(b) Define the principal focus of a concave lens.
19. Draw a ray diagram to show the formation of a real magnified image by a convex lens. (In your sketch the position of object and image with respect to the principal focus of lens should be shown clearly).
20. Describe with the help of a ray-diagram, the formation of image of a finite object placed in front of a convex lens between f and $2f$. Give two characteristics of the image so formed.
21. Describe with the help of a ray diagram the nature, size and position of the image formed when an object is placed in front of a convex lens between focus and optical centre. State three characteristics of the image formed.
22. An object is placed at a distance equal to $2f$ in front of a convex lens. Draw a labelled ray diagram to show the formation of image. State two characteristics of the image formed.
23. Describe with the help of a ray-diagram, the size, nature and position of the image formed by a convex lens when an object is placed beyond $2f$ in front of the lens.
24. Describe with the help of a ray diagram the nature, size and position of the image formed when an object is placed at infinity (considerable distance) in front of a convex lens. State three characteristics of the image so formed.
25. (a) What type of lens is shown in the diagram on the right ? What will happen to the parallel rays of light ? Show by completing the ray diagram.
(b) Your eye contains a convex lens. Why is it unwise to look at the sun ?
26. Where must the object be placed for the image formed by a converging lens to be :
 - (a) real, inverted and smaller than the object ?
 - (b) real, inverted and same size as the object ?
 - (c) real, inverted and larger than the object ?
 - (d) virtual, upright and larger than the object ?
27. Draw a diagram to show how a converging lens held close to the eye acts as a magnifying glass. Why is it usual to choose a lens of short focal length for this purpose rather than one of long focal length ?
28. How could you find the focal length of a convex lens rapidly but approximately ?



Long Answer Type Questions

29. (a) With the help of a labelled diagram explain how a convex lens converges a beam of parallel light rays. Mark the principal axis, optical centre, principal focus and focal length of the convex lens on the diagram.

- (b) State whether convex lens has a real focus or a virtual focus.
 (c) List some things that convex lens and concave mirror have in common.
30. (a) With the help of a labelled diagram, explain how a concave lens diverges a beam of parallel light rays. Mark the principal axis, optical centre, principal focus and focal length of the concave lens on the diagram.
 (b) State whether concave lens has a real focus or a virtual focus.
 (c) List some things that concave lens and convex mirror have in common.
31. Draw ray diagrams to represent the nature, position and relative size of the image formed by a convex lens for the object placed :
 (a) at $2F_1$,
 (b) between F_1 and the optical centre O of the lens.
 Which of the above two cases shows the use of convex lens as a magnifying glass ? Give reasons for your choice.
32. (a) An object is placed well outside the principal focus of a convex lens. Draw a ray diagram to show how the image is formed, and say whether the image is real or virtual.
 (b) What is the effect on the size and position of the image of moving the object (i) towards the lens, and (ii) away from the lens ?
33. (a) Explain what is meant by a virtual, magnified image.
 (b) Draw a ray diagram to show the formation of a virtual magnified image of an object by a convex lens. In your diagram, the position of object and image with respect to the principal focus should be shown clearly.
 (c) Three convex lenses are available having focal lengths of 4 cm, 40 cm and 4 m respectively. Which one would you choose as a magnifying glass and why ?
34. (a) Explain why, a real image can be projected on a screen but a virtual image cannot.
 (b) Draw a ray diagram to show the formation of a real diminished image of an object by a convex lens. In your diagram, the position of object and image with respect to the principal focus should be shown clearly.
 (c) Name one simple optical instrument in which the above arrangement of convex lens is used.

Multiple Choice Questions (MCQs)

35. A convex lens has a focal length of 10 cm. At which of the following position should an object be placed so that this convex lens may act as a magnifying glass ?
 (a) 15 cm (b) 7 cm (c) 20 cm (d) 25 cm
36. Which one of the following materials cannot be used to make a lens ?
 (a) Water (b) Glass (c) Plastic (d) Clay
37. A small bulb is placed at the focal point of a converging lens. When the bulb is switched on, the lens produces :
 (a) a convergent beam of light (b) a divergent beam of light
 (c) a parallel beam of light (d) a patch of coloured light
38. An illuminated object is placed at a distance of 20 cm from a converging lens of focal length 15 cm. The image obtained on the screen is :
 (a) upright and magnified (b) inverted and magnified
 (c) inverted and diminished (d) upright and diminished
39. An object is placed between f and $2f$ of a convex lens. Which of the following statements correctly describes its image ?
 (a) real, larger than the object (b) erect, smaller than the object
 (c) inverted, same size as object (d) virtual, larger than the object
40. Which of the following can make a parallel beam of light when light from a bulb falls on it ?
 (a) concave mirror as well as concave lens (b) convex mirror as well as convex lens
 (c) concave mirror as well as convex lens (d) convex mirror as well as concave lens
41. In order to obtain a real image twice the size of the object with a convex lens of focal length 15 cm, the object distance should be :
 (a) more than 5 cm but less than 10 cm (b) more than 10 cm but less than 15 cm
 (c) more than 15 cm but less than 30 cm (d) more than 30 cm but less than 60 cm

42. A converging lens is used to produce an image of an object on a screen. What change is needed for the image to be formed nearer to the lens ?
 (a) increase the focal length of the lens
 (b) insert a diverging lens between the lens and the screen
 (c) increase the distance of the object from the lens
 (d) move the object closer to the lens
43. A convex lens of focal length 8 cm forms a real image of the same size as the object. The distance between object and its image will be :
 (a) 8 cm (b) 16 cm (c) 24 cm (d) 32 cm
44. A virtual, erect and magnified image of an object is to be obtained with a convex lens. For this purpose, the object should be placed :
 (a) between 2F and infinity (b) between F and optical centre
 (c) between F and 2F (d) at F
45. A burning candle whose flame is 1.5 cm tall is placed at a certain distance in front of a convex lens. An image of candle flame is received on a white screen kept behind the lens. The image of flame also measures 1.5 cm. If f is the focal length of convex lens, the candle is placed :
 (a) at f (b) between f and $2f$ (c) at $2f$ (d) beyond $2f$

Questions Based on High Order Thinking Skills (HOTS)

46. A lens of focal length 12 cm forms an erect image three times the size of the object. The distance between the object and image is :
 (a) 8 cm (b) 16 cm (c) 24 cm (d) 36 cm
47. If an object is placed 21 cm from a converging lens, the image formed is slightly smaller than the object. If the object is placed 19 cm from the lens, the image formed is slightly larger than the object. The approximate focal length of the lens is :
 (a) 5 cm (b) 10 cm (c) 18 cm (d) 20 cm
48. An object is placed at the following distances from a convex lens of focal length 15 cm :
 (a) 35 cm (b) 30 cm (c) 20 cm (d) 10 cm
 Which position of the object will produce :
 (i) a magnified real image ?
 (ii) a magnified virtual image ?
 (iii) a diminished real image ?
 (iv) an image of same size as the object ?
49. When an object is placed at a distance of 36 cm from a convex lens, an image of the same size as the object is formed. What will be the nature of image formed when the object is placed at a distance of :
 (a) 10 cm from the lens ? (b) 20 cm from the lens ?
50. (a) Draw a diagram to show how a converging lens focusses parallel rays of light.
 (b) How would you alter the above diagram to show how a converging lens can produce a beam of parallel rays of light.

ANSWERS

2. Optical centre 3. 1 cm 4. At 2F (at twice the focal length) 5. At infinity (very large distance)
 6. At a distance less than focal length 7. Within focus 8. Between F and 2F (or Between f and $2f$)
 9. Beyond 2F 10. At focus (F) 15. (a) focus (b) object 26. (a) Beyond 2F (b) At 2F (c) Between F and 2F (d) Between F and Optical centre 33. (c) Convex lens having 4 cm focal length ; It will produce greatest magnification 34. (c) A simple camera 35. (b) 36. (d) 37. (c) 38. (b) 39. (a) 40. (c) 41. (c)
 42. (c) 43. (d) 44. (b) 45. (c) 46. (b) 47. (b) 48. (i) 20 cm (ii) 10 cm (iii) 35 cm (iv) 30 cm
 49. (a) Virtual, erect and magnified (b) Real, inverted and magnified 50. (b) Place a source of light (say, a lighted bulb) at the focus of the converging lens.

SIGN CONVENTION FOR SPHERICAL LENSES

These days New Cartesian Sign Convention is used for measuring the various distances in the ray diagrams of spherical lenses (convex lenses and concave lenses). According to the New Cartesian Sign Convention :

- (i) All the distances are measured from the *optical centre* of the lens.
- (ii) The distances measured in the *same* direction as that of incident light are taken as *positive*.
- (iii) The distances measured *against* the direction of incident light are taken as *negative*.
- (iv) The distances measured *upward* and perpendicular to the principal axis are taken as *positive*.
- (v) The distances measured *downward* and perpendicular to the principal axis are taken as *negative*.

The New Cartesian Sign Convention for lenses is shown in Figure 42. **The object is always placed on the left side of the lens** (as shown in Figure 42), so that the direction of incident light is from left to right. All the distances measured from the optical centre (C) of the lens to the right side will be considered positive (because they will be in the same direction as the incident light). On the other hand, all the distances measured from the optical centre (C) of the lens to the left side are considered negative (because they are measured against the direction of incident light). On the basis of New Cartesian Sign Convention, **the focal length of a convex lens is considered positive** (and written with a plus sign). On the other hand, **the focal length of a concave lens is considered negative** (and written with a minus sign). Please note that the sign convention for *spherical lenses* is very similar to the sign convention for *spherical mirrors* which we have already studied.

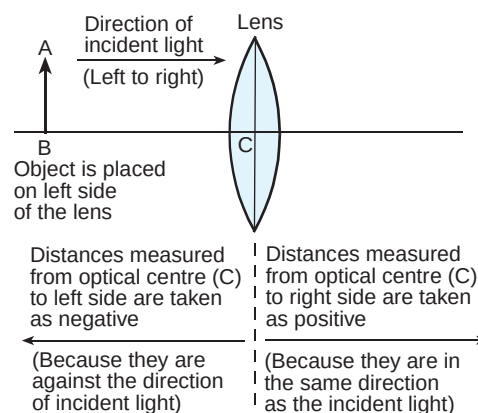


Figure 42. Sign convention for lenses.

Lens Formula

A formula which gives the relationship between image distance (v), object distance (u), and focal length (f) of a lens is known as the lens formula. The lens formula can be written as :

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

where v = image distance

u = object distance

and f = focal length

This lens formula applies to both types of spherical lenses : convex lenses as well as concave lenses. Please note that the lens formula differs from the mirror formula only in the sign between $\frac{1}{v}$ and $\frac{1}{u}$. The mirror formula is $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ whereas the lens formula is $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$. It is clear that the mirror formula has a plus sign (+) between $\frac{1}{v}$ and $\frac{1}{u}$ whereas the lens formula has a minus sign (-) between $\frac{1}{v}$ and $\frac{1}{u}$. The values of v and u should be substituted in the lens formula with their proper signs.

Magnification Produced by Lenses

The size of image formed by a lens depends on the position of the object from the lens. For example, the image formed by a convex lens can be smaller than the object, equal to the object or bigger than the object. The size of the image relative to the object is given by the linear magnification. **The linear magnification is the ratio of the height of the image to the height of the object.** That is :

$$\text{Magnification} = \frac{\text{height of image}}{\text{height of object}}$$



$$\text{or } m = \frac{h_2}{h_1}$$

where m = magnification
 h_2 = height of image
 and h_1 = height of object

We will now write another formula for the magnification produced by a lens in terms of the image distance and object distance.

The linear magnification produced by a lens is equal to the ratio of image distance to the object distance. That is :

$$\text{Magnification} = \frac{\text{Image distance}}{\text{Object distance}}$$

$$\text{or } m = \frac{v}{u}$$

where m = magnification
 v = image distance
 and u = object distance

It should be noted that this second magnification formula for lenses differs only in sign from the magnification formula for mirrors. The magnification formula for the mirrors is, $m = -\frac{v}{u}$ whereas that for lenses is, $m = \frac{v}{u}$. It is clear that the magnification formula for mirrors has a minus (-) sign but the magnification formula for lenses has no minus sign.

If the magnification m has a positive value, the image is virtual and erect. And if the magnification m has a negative value, the image will be real and inverted. Since a convex lens can form virtual images as well as real images, therefore, the magnification produced by a convex lens can be either positive or negative. A concave lens, however, forms only virtual images, so the magnification produced by a concave lens is always positive. A convex lens can form images which are smaller than the object, equal to the object or bigger than the object, therefore, the magnification (m) produced by a convex lens can be less than 1, equal to 1 or more than 1. On the other hand, a concave lens forms images which are always smaller than the object, so the magnification (m) produced by a concave lens is always less than 1.

Numerical Problems Based On Convex Lenses

We will now solve some numerical problems based on convex lenses by using the lens formula and the magnification formulae. Here are some examples.

Sample Problem 1. A convex lens of focal length 10 cm is placed at a distance of 12 cm from a wall. How far from the lens should an object be placed so as to form its real image on the wall ?

Solution. Here, the real image is formed on the wall which is at a distance of 12 cm from the convex lens. This means that the distance of image from the convex lens or image distance will be 12 cm. Since a real image is formed on the right side of the lens, so this image distance will be positive.

Now, Image distance, $v = +12$ cm (A real image)
 Object distance, $u = ?$ (To be calculated)
 Focal length, $f = +10$ cm (It is a convex lens)

Putting these values in the lens formula :

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{we get : } \frac{1}{12} - \frac{1}{u} = \frac{1}{10}$$

$$\begin{aligned}\frac{1}{12} - \frac{1}{10} &= \frac{1}{u} \\ \frac{5-6}{60} &= \frac{1}{u} \\ -\frac{1}{60} &= \frac{1}{u}\end{aligned}$$

So, Object distance, $u = -60$ cm

Thus, the object should be placed at a distance of 60 cm in front of the convex lens. The minus sign shows that the object is on the left side of the lens.

Sample Problem 2. If an object of 7 cm height is placed at a distance of 12 cm from a convex lens of focal length 8 cm, find the position, nature and height of the image.

Solution. First of all we will find out the position of the image. By the position of image we mean the distance of image from the lens.

Here, Object distance, $u = -12$ cm (It is to the left of lens)

Image distance, $v = ?$ (To be calculated)

Focal length, $f = +8$ cm (It is a convex lens)

Putting these values in the lens formula :

$$\begin{aligned}\frac{1}{v} - \frac{1}{u} &= \frac{1}{f} \\ \text{we get : } \frac{1}{v} - \frac{1}{-12} &= \frac{1}{8} \\ \text{or } \frac{1}{v} + \frac{1}{12} &= \frac{1}{8} \\ \frac{1}{v} &= \frac{1}{8} - \frac{1}{12} \\ \frac{1}{v} &= \frac{3-2}{24} \\ \frac{1}{v} &= \frac{1}{24}\end{aligned}$$

So, Image distance, $v = +24$ cm

Thus, the image is formed at a distance of 24 cm from the convex lens. The plus sign for image distance shows that the image is formed on the right side of the convex lens. Only a real and inverted image is formed on the right side of a convex lens, so the image formed is real and inverted.

Let us calculate the magnification now. We know that for a lens :

$$\text{Magnification, } m = \frac{v}{u}$$

Here, Image distance, $v = 24$ cm

Object distance, $u = -12$ cm

$$\text{So, } m = \frac{24}{-12}$$

$$\text{or } m = -2$$

Since the value of magnification is more than 1 (it is 2), so the image is larger than the object or magnified. The minus sign for magnification shows that the image is formed below the principal axis. Hence the image is real and inverted. Let us calculate the size of the image by using the formula :

$$m = \frac{h_2}{h_1}$$

Here, Magnification, $m = -2$ (Found above)
 Height of object, $h_1 = +7$ cm (Measured upwards)
 Height of image, $h_2 = ?$ (To be calculated)

Now, putting these values in the above formula, we get :

$$-2 = \frac{h_2}{7}$$

or $h_2 = -2 \times 7$

Thus, Height of image, $h_2 = -14$ cm

Thus, the height or size of the image is 14 cm. The minus sign shows that this height is in the downward direction, that is, the image is formed below the axis. Thus, the image is real and inverted.

Sample Problem 3. The magnification produced by a spherical lens is +2.5. What is the :

(a) nature of image ?

(b) nature of lens ?

Answer. (a) When the magnification is positive, then the image is virtual and erect. In this case, the magnification has a positive sign, so the nature of image is virtual and erect.

(b) The value of magnification given here is 2.5 (which is more than 1). So, the image is larger than the object or magnified. A virtual, erect and magnified image can be formed only by a convex lens, therefore, the nature of lens is convex.

We are now in a position to **answer the following questions and problems :**

Very Short Answer Type Questions

1. Write the formula for a lens connecting image distance (v), object distance (u) and the focal length (f). How does the lens formula differ from the mirror formula ?
2. Write down the magnification formula for a lens in terms of object distance and image distance. How does this magnification formula for a lens differ from the corresponding formula for a mirror ?
3. What is the nature of the image formed by a convex lens if the magnification produced by the lens is +3 ?
4. What is the nature of the image formed by a convex lens if the magnification produced by the lens is, -0.5 ?
5. What is the position of image when an object is placed at a distance of 10 cm from a convex lens of focal length 10 cm ?
6. Describe the nature of image formed when an object is placed at a distance of 30 cm from a convex lens of focal length 15 cm.
7. At what distance from a converging lens of focal length 12 cm must an object be placed in order that an image of magnification 1 will be produced ?

Short Answer Type Questions

8. State and explain the New Cartesian Sign Convention for spherical lenses.
9. An object 4 cm high is placed at a distance of 10 cm from a convex lens of focal length 20 cm. Find the position, nature and size of the image.
10. A small object is so placed in front of a convex lens of 5 cm focal length that a virtual image is formed at a distance of 25 cm. Find the magnification.
11. Find the position and nature of the image of an object 5 cm high and 10 cm in front of a convex lens of focal length 6 cm.
12. Calculate the focal length of a convex lens which produces a virtual image at a distance of 50 cm of an object placed 20 cm in front of it.
13. An object is placed at a distance of 100 cm from a converging lens of focal length 40 cm.
 - (i) What is the nature of image ?
 - (ii) What is the position of image ?
14. A convex lens produces an inverted image magnified three times of an object placed at a distance of 15 cm from it. Calculate focal length of the lens.

15. A converging lens of focal length 5 cm is placed at a distance of 20 cm from a screen. How far from the lens should an object be placed so as to form its real image on the screen ?
16. An object 5 cm high is held 25 cm away from a converging lens of focal length 10 cm. Find the position, size and nature of the image formed. Also draw the ray diagram.
17. At what distance should an object be placed from a convex lens of focal length 18 cm to obtain an image at 24 cm from it on the other side ? What will be the magnification produced in this case ?
18. An object 2 cm tall is placed on the axis of a convex lens of focal length 5 cm at a distance of 10 m from the optical centre of the lens. Find the nature, position and size of the image formed. Which case of image formation by convex lenses is illustrated by this example ?
19. The filament of a lamp is 80 cm from a screen and a converging lens forms an image of it on a screen, magnified three times. Find the distance of the lens from the filament and the focal length of the lens.
20. An erect image 2.0 cm high is formed 12 cm from a lens, the object being 0.5 cm high. Find the focal length of the lens.
21. A convex lens of focal length 0.10 m is used to form a magnified image of an object of height 5 mm placed at a distance of 0.08 m from the lens. Calculate the position, nature and size of the image.
22. A convex lens of focal length 6 cm is held 4 cm from a newspaper which has print 0.5 cm high. By calculation, determine the size and nature of the image produced.
23. Determine how far an object must be placed in front of a converging lens of focal length 10 cm in order to produce an erect (upright) image of linear magnification 4.
24. A lens of focal length 20 cm is used to produce a ten times magnified image of a film slide on a screen. How far must the slide be placed from the lens ?
25. An object placed 4 cm in front of a converging lens produces a real image 12 cm from the lens.
 - (a) What is the magnification of the image ?
 - (b) What is the focal length of the lens ?
 - (c) Draw a ray diagram to show the formation of image. Mark clearly F and 2F in the diagram.

Long Answer Type Questions

26. (a) An object 2 cm tall stands on the principal axis of a converging lens of focal length 8 cm. Find the position, nature and size of the image formed if the object is :
 - (i) 12 cm from the lens
 - (ii) 6 cm from the lens
 (b) State one practical application each of the use of such a lens with the object in position (i) and (ii).
27. (a) An object 3 cm high is placed 24 cm away from a convex lens of focal length 8 cm. Find by calculations, the position, height and nature of the image.
 - (b) If the object is moved to a point only 3 cm away from the lens, what is the new position, height and nature of the image ?
 - (c) Which of the above two cases illustrates the working of a magnifying glass ?
28. (a) Find the nature, position and magnification of the images formed by a convex lens of focal length 0.20 m if the object is placed at a distance of :
 - (i) 0.50 m (ii) 0.25 m (iii) 0.15 m
 (b) Which of the above cases represents the use of convex lens in a film projector, in a camera, and as a magnifying glass ?

Multiple Choice Questions (MCQs)

29. A spherical mirror and a spherical lens each have a focal length of, -15 cm. The mirror and the lens are likely to be :
 - (a) both concave.
 - (b) both convex.
 - (c) the mirror is concave but the lens is convex.
 - (d) the mirror is convex but the lens is concave.
30. Linear magnification produced by a convex lens can be :
 - (a) less than 1 or more than 1

- (b) less than 1 or equal to 1
 (c) more than 1 or equal to 1
 (d) less than 1, equal to 1 or more than 1
31. Magnification produced by a concave lens is always :
 (a) more than 1 (b) equal to 1 (c) less than 1 (d) more than 1 or less than 1
32. In order to obtain a magnification of, -3 (minus 3) with a convex lens, the object should be placed :
 (a) between optical centre and F (b) between F and $2F$
 (c) at $2F$ (d) beyond $2F$
33. A convex lens produces a magnification of $+5$. The object is placed :
 (a) at focus (b) between f and $2f$
 (c) at less than f (d) beyond $2f$
34. If a magnification of, -1 (minus 1) is obtained by using a converging lens, then the object has to be placed :
 (a) within f (b) at $2f$
 (c) beyond $2f$ (d) at infinity
35. To obtain a magnification of, -0.5 with a convex lens, the object should be placed :
 (a) at F (b) between optical centre and F
 (c) between F and $2F$ (d) beyond $2F$
36. An object is 0.09 m from a magnifying lens and the image is formed 36 cm from the lens. The magnification produced is :
 (a) 0.4 (b) 1.4 (c) 4.0 (d) 4.5
37. To obtain a magnification of, -2 with a convex lens of focal length 10 cm, the object should be placed :
 (a) between 5 cm and 10 cm (b) between 10 cm and 20 cm
 (c) at 20 cm (d) beyond 20 cm
38. A convex lens of focal length 15 cm produces a magnification of $+4$. The object is placed :
 (a) at a distance of 15 cm (b) between 15 cm and 30 cm
 (c) at less than 15 cm (d) beyond 30 cm
39. If a magnification of, -1 is to be obtained by using a converging lens of focal length 12 cm, then the object must be placed :
 (a) within 12 cm (b) at 24 cm (c) at 6 cm (d) beyond 24 cm
40. In order to obtain a magnification of, -0.75 with a convex lens of focal length 8 cm, the object should be placed :
 (a) at less than 8 cm (b) between 8 cm and 16 cm
 (c) beyond 16 cm (d) at 16 cm

Questions Based on High Order Thinking Skills (HOTS)

41. A student did an experiment with a convex lens. He put an object at different distances 25 cm, 30 cm, 40 cm, 60 cm and 120 cm from the lens. In each case he measured the distance of the image from the lens. His results were 100 cm, 24 cm, 60 cm, 30 cm and 40 cm, respectively. Unfortunately his results are written in wrong order.
 (a) Rewrite the image distances in the correct order.
 (b) What would be the image distance if the object distance was 90 cm ?
 (c) Which of the object distances gives the biggest image ?
 (d) What is the focal length of this lens ?
42. A magnifying lens has a focal length of 100 mm. An object whose size is 16 mm is placed at some distance from the lens so that an image is formed at a distance of 25 cm in front of the lens.
 (a) What is the distance between the object and the lens ?
 (b) Where should the object be placed if the image is to form at infinity ?
43. A lens forms a real image 3 cm high of an object 1 cm high. If the separation of object and image is 15 cm, find the focal length of the lens.

44. An object 50 cm tall is placed on the principal axis of a convex lens. Its 20 cm tall image is formed on the screen placed at a distance of 10 cm from the lens. Calculate the focal length of the lens.

ANSWERS

3. Virtual and erect 4. Real and inverted 5. At infinity 6. Real and inverted. 7. 24 cm 9. $v = -20$ cm ; The image is 20 cm in front of convex lens (on its left side) ; Virtual and erect; 8 cm 10. +6 11. $v = +15$ cm ; The image is formed 15 cm behind the convex lens (on its right side); Real and inverted 12. 33.3 cm 13. (i) Real and inverted (ii) $v = +66.6$ cm ; The image is formed 66.6 cm behind the convex lens (on its right side) 14. 11.2 cm 15. 6.6 cm 16. $v = +16.6$ cm ; The image is 16.6 cm behind the convex lens ; 3.3 cm ; Real and inverted 17. 72 cm from convex lens ; $-\frac{1}{3}$ 18. Real and inverted ; $v = +5$ cm ; The image is formed 5 cm behind the convex lens ; 0.01 cm 19. 20 cm ; + 15 cm 20. + 4.0 cm 21. $v = -0.40$ m ; The position of image is 0.40 m from the lens on the same side as the object (on the left of lens) ; The nature of image is virtual and erect ; The size of image is 25 mm 22. 1.5 cm high ; Virtual, erect and magnified (3 times) 23. 7.5 cm 24. 220 cm behind the lens 25. (a) 3 (b) + 3 cm 26. (a) (i) $v = +24$ cm ; The image is 24 cm behind the lens ; Real and inverted ; 4 cm (ii) $v = -24$ cm ; The image is 24 cm in front of the lens ; Virtual and erect ; 8 cm (b) (i) Used in film projector (ii) Used as a magnifying glass 27. (a) $v = +12$ cm ; The image is formed 12 cm behind the lens ; 1.5 cm high ; Real and inverted (b) $v = -4.8$ cm ; The image is formed 4.8 cm in front of the lens (on its left side); 4.8 cm high ; Virtual and erect (c) Case b 28. (a) (i) $v = +0.33$ m ; The image is formed 0.33 m behind the lens ; $m = -0.66$; Real and inverted (ii) $v = +1.00$ m ; The image is formed 1.00 m behind the lens ; $m = -4.0$; Real and inverted (iii) $v = -0.60$ m ; The image is formed 0.60 m in front of the lens ; $m = +4.0$; Virtual and erect (b) Film projector : Case (ii) ; Camera : Case (i) ; Magnifying glass : case (iii) 29. (a) 30. (d) 31. (c) 32. (b) 33. (c) 34. (b) 35. (d) 36. (c) 37. (b) 38. (c) 39. (b) 40. (c) 41. (a) 100 cm ; 60 cm ; 40 cm ; 30 cm ; 24 cm (b) 25.7 cm (c) 25 cm (d) 20 cm 42. (a) 7.14 cm (b) 10 cm 43. + 2.81 cm 44. 7.14 cm

RULES FOR OBTAINING IMAGES FORMED BY CONCAVE LENSES

We have already studied some rules for the formation of images by convex lenses. There are similar rules for constructing ray-diagrams for obtaining images with concave lenses. These are given below.

Rule 1. A ray of light which is parallel to the principal axis of a concave lens, appears to be coming from its focus after refraction through the lens. This is shown in Figure 43. Here we have a concave lens L and its principal axis is $X'Y'$. A ray of light AD parallel to the principal axis enters the concave lens, gets refracted at point D and goes in the direction DX . To a person looking into the concave lens from the right side, the refracted ray of light DX appears to be coming from the focus F of concave lens situated on its left side (as shown by dotted line) (see Figure 43).

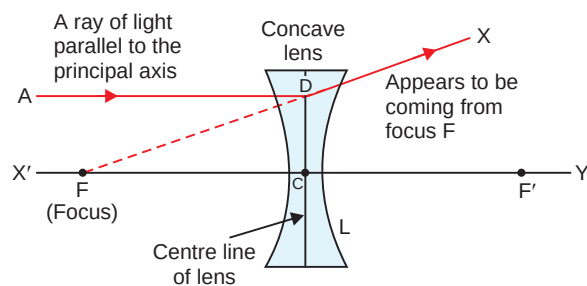


Figure 43.

Rule 2. A ray of light passing through the optical centre of a concave lens goes straight after passing through the lens. This is shown in Figure 44. The ray of light AC is passing through the optical centre C of concave lens. This ray of light goes straight in the direction CY after passing through the concave lens. It does not bend at all. Please note that a ray of light along the principal axis of a concave lens also goes straight.

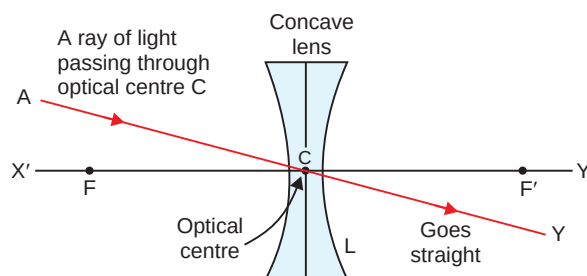


Figure 44.

Rule 3. A ray of light going towards the focus of a concave lens, becomes parallel to its principal axis after refraction through the lens. This rule is just the reverse case of the first rule and it is shown in Figure 45. Here a ray of light AD (coming from the object) is going

towards the focus F' of the concave lens. It enters the concave lens and gets refracted (or bends) at point D . After passing through the concave lens, it becomes parallel to the principal axis of the lens and goes in the direction DX (see Figure 45).

At any given time, we will use only two of the above three types of light rays to draw the ray-diagrams to find the position of image formed by a concave lens.

Please note that **no matter where the object is placed in front of a concave lens, the concave lens always forms a virtual, erect and diminished image of the object**. When the distance of an object from a concave lens is changed, then only the position and size of the image changes. There are two main positions of an object in the case of a concave lens from the point of view of position and size of image. The object can be :

- (i) anywhere between optical centre (C) and infinity, and
- (ii) at infinity.

We will discuss both these cases one by one. Let us first describe the formation of image by a concave lens when the object is anywhere between optical centre (C) and infinity.

Formation of Image by a Concave Lens

In Figure 46, we have a concave lens with optical centre C . And F is the focus of this lens. The arrow AB pointing upwards represents the object. Now, we want to find out the position, nature and size of the image of this object which will be formed by the concave lens. The method used for obtaining the images with concave lenses is very similar to the one we have used in the case of convex lenses. This is as follows.

Starting from the upper point A of the object AB , we draw a line AD parallel to the principal axis. Now, according to the first rule of image formation, this parallel ray AD should appear to be coming from focus F after refraction. So, we join DF by a dotted line and produce FD upwards in the direction DX by a solid line. Now, DX is the first refracted ray which appears to be coming from focus F of the concave lens.

We have now to draw a second ray of light starting from the same point A and passing through the optical centre C . For this we join the points A and C by a line. Thus, the line AC represents a ray of light passing through the optical centre C of the lens. Now, according to the second rule of image formation, this ray should go straight. So, we extend the line AC further in the direction CY . Thus, CY is the second refracted ray. The two refracted rays DX and CY are diverging rays and appear to intersect at point A' on the left side of the lens, only when they are produced backwards. So, to an eye placed on the right side of the concave lens (as shown in Figure 46), the top of arrow appears to be at A' . Thus, A' is the virtual image of the top point A of the object. To get the complete image of the object, we draw $A'B'$ perpendicular to the axis from point A' . Thus, $A'B'$ is the image of the object AB . The image is virtual, erect and smaller than the object. It is situated between

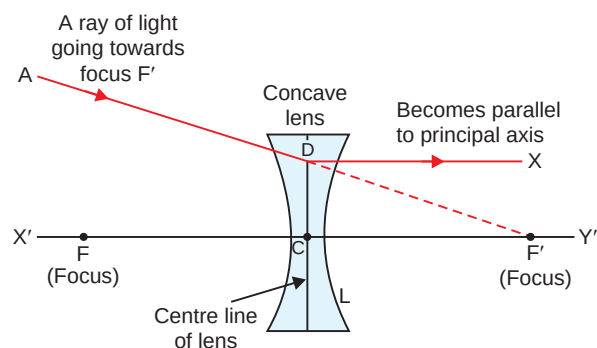


Figure 45.

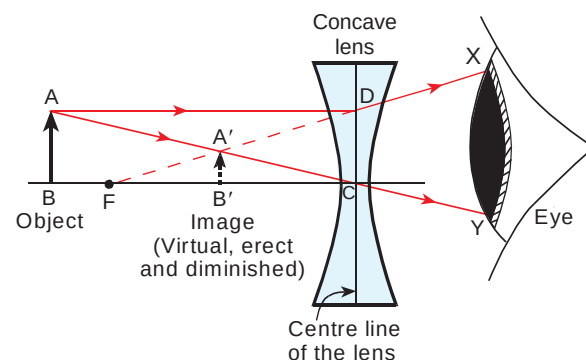


Figure 46. Diagram to show the formation of image by a concave lens.

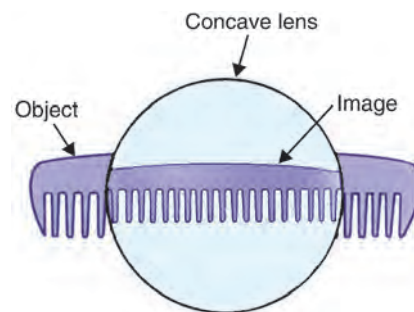


Figure 47. A concave lens always produces a smaller image than the object. Here a concave lens is producing a smaller image of the middle part of the comb.

the optical centre and focus (in front of the concave lens). From the above discussion we conclude that : **When an object is placed anywhere between optical centre (C) and infinity in front of a concave lens, the image formed is :**



- (i) between optical centre (C) and focus (F),
- (ii) virtual and erect, and
- (iii) diminished (smaller than the object).

If we move the object more and more away from the optical centre of the concave lens, the image becomes smaller and smaller in size and moves away from the lens towards its focus. And when the object is at infinity, the image is formed at the focus (see Figure 27 on page 232). From this discussion we conclude that : **When an object is at infinity from a concave lens, the image formed is :**

- (i) at focus (F),
- (ii) virtual and erect, and
- (iii) highly diminished (much smaller than the object).

And before we conclude this discussion, here is a summary of the images formed by a concave lens.

Summary of the Images Formed by a Concave Lens

Position of object	Position of image	Size of image	Nature of image
1. Anywhere between optical centre (C) and infinity.	Between optical centre (C) and focus (F)	Diminished	Virtual and erect
2. At infinity	At focus (F)	Highly diminished	Virtual and erect

How to Distinguish Between a Convex Lens and a Concave Lens Without Touching Them

We keep the lens close to the page of a book and see the image of the writing of the book through it. If the letters of the book appear enlarged, then it is a convex lens; and if the letters appear diminished, then it is a concave lens. This is due to the fact that when an object is within the focus of a convex lens, it produces an enlarged image. But a concave lens produces a diminished image for all positions of the object. So, in this case the object (the book) has been placed within focus of the lens.

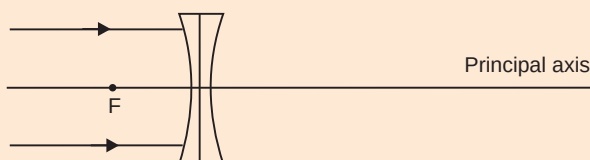
Uses of Concave Lenses

- (i) Concave lenses are used in spectacles to correct the defect of vision called myopia (or short-sightedness).
- (ii) Concave lens is used as eye-lens in Galilean telescope.
- (iii) Concave lenses are used in combination with convex lenses to make high quality lens systems for optical instruments.
- (iv) Concave lens is used in wide-angle spyhole in doors.

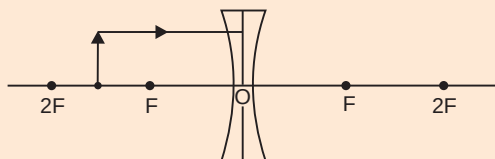
Before we go further and discuss the mirror formula and magnification formulae for a concave lens, please answer the following questions :

Very Short Answer Type Questions

- If the image formed by a lens is always diminished and erect, what is the nature of the lens ?
- Copy and complete the diagram below to show what happens to the rays of light when they pass through the concave lens :



3. Which type of lenses are :
 - (a) thinner in the middle than at the edges ?
 - (b) thicker in the middle than at the edges ?
4. A ray of light is going towards the focus of a concave lens. Draw a ray diagram to show the path of this ray of light after refraction through the lens.
5. (a) What type of images can a convex lens make ?
(b) What type of image is always made by a concave lens ?
6. Take down this figure into your answer book and complete the path of the ray.



7. Fill in the following blanks with suitable words :
 - (a) A convex lens.....rays of light, whereas a concave lens.....rays of light.
 - (b) Lenses refract light to form images : a.....lens can form both real and virtual images, but a diverging lens forms only.....images.
8. Things always look small on viewing through a lens. What is the nature of the lens ?

Short Answer Type Questions

9. An object lies at a distance of $2f$ from a concave lens of focal length f . Draw a ray-diagram to illustrate the image formation.
10. Show by drawing a ray-diagram that the image of an object formed by a concave lens is virtual, erect and diminished.
11. Give the position, size and nature of image formed by a concave lens when the object is placed :
 - (a) anywhere between optical centre and infinity.
 - (b) at infinity.
12. Which type of lens is : (a) a converging lens, and which is (b) a diverging lens ? Explain your answer with diagrams.
13. With the help of a diagram, explain why the image of an object viewed through a concave lens appears smaller and closer than the object.
14. How would a pencil look like if you saw it through (a) a concave lens, and (b) a convex lens ? (Assume the pencil is close to the lens). Is the image real or virtual ?

Long Answer Type Questions

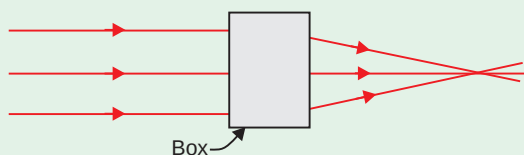
15. (a) An object is placed 10 cm from a lens of focal length 5 cm. Draw the ray diagrams to show the formation of image if the lens is (i) converging, and (ii) diverging.
(b) State one practical use each of convex mirror, concave mirror, convex lens and concave lens.
16. (a) Construct ray diagrams to illustrate the formation of a virtual image using (i) a converging lens, and (ii) a diverging lens.
(b) What is the difference between the two images formed above ?

Multiple Choice Questions (MCQs)

17. A diverging lens is used in :
 - (a) a magnifying glass
 - (b) a car to see objects on rear side
 - (c) spectacles for the correction of short sight
 - (d) a simple camera
18. When an object is kept at any distance in front of a concave lens, the image formed is always :
 - (a) virtual, erect and magnified
 - (b) virtual, inverted and diminished.
 - (c) virtual, erect and diminished
 - (d) virtual, erect and same size as object
19. When sunlight is concentrated on a piece of paper by a spherical mirror or lens, then a hole can be burnt in it. For doing this, the paper must be placed at the focus of :

- (a) either a convex mirror or convex lens
 (b) either a concave mirror or concave lens
 (c) either a concave mirror or convex lens
 (d) either a convex mirror or concave lens

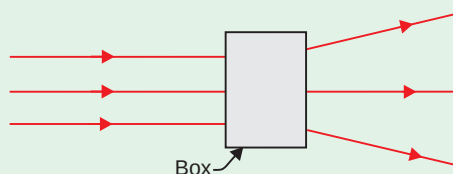
20. A beam of parallel light rays is incident through the holes on one side of a box and emerges out through the holes on its opposite side as shown in the diagram below :



Which of the following could be inside the box ?

- (a) a rectangular glass block
 (b) a concave lens
 (c) a convex lens
 (d) a glass prism

21. A beam of light is incident through the holes on one side of a box and emerges out through the holes on its opposite side as shown in the following figure :



The box contains :

- (a) a glass prism
 (b) a concave lens
 (c) a convex lens
 (d) a parallel-sided glass slab

22. Which of the following can form a virtual image which is always smaller than the object ?

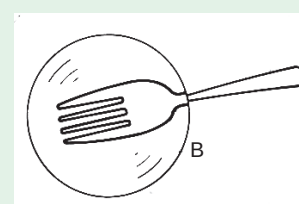
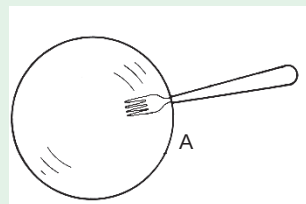
- (a) a plane mirror
 (b) a convex lens
 (c) a concave lens
 (d) a concave mirror

Questions Based on High Order Thinking Skills (HOTS)

23. When an object is placed 10 cm in front of lens A, the image is real, inverted, magnified and formed at a great distance. When the same object is placed 10 cm in front of lens B, the image formed is real, inverted and same size as the object.

- (a) What is the focal length of lens A ?
 (b) What is the focal length of lens B ?
 (c) What is the nature of lens A ?
 (d) What is the nature of lens B ?

24. When a fork is seen through lenses A and B one by one, it appears as shown in the diagrams. What is the nature of (i) lens A, and (ii) lens B ? Give reason for your answer.



25. What kind of lens can form :
 (a) an inverted magnified image ?
 (b) an erect magnified image ?
 (c) an inverted diminished image ?
 (d) an erect diminished image ?

ANSWERS

1. Concave lens 3. (a) Concave lenses (b) Convex lenses 5. (a) Real and Virtual (b) Virtual
 7. (a) converges; diverges (b) converging ; virtual 8. Concave lens 14. (a) Smaller (b) Bigger ; Virtual
 16. (b) The virtual image formed by a converging lens is magnified whereas that formed by a diverging lens is diminished 17. (c) 18. (c) 19. (c) 20. (c) 21. (b) 22. (c) 23. (a) 10 cm (b) 5 cm (c) Convex lens
 (d) Convex lens 24. (i) Concave lens (ii) Convex lens 25. (a) Convex lens (b) Convex lens (c) Convex lens (d) Concave lens.

LENS FORMULA AND MAGNIFICATION FORMULAE FOR A CONCAVE LENS

The lens formula for a concave lens is the same as that for a convex lens, which is :

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

The magnification formulae for a concave lens are also just the same as that for convex lens, which are :

$$m = \frac{h_2}{h_1} \quad \text{and} \quad m = \frac{v}{u}$$

Numerical Problems Based on Concave Lenses

We will now solve some numerical problems based on concave lenses by using the lens formula and the magnification formulae. Here are some examples.

Sample Problem 1. An object is placed at a distance of 50 cm from a concave lens of focal length 20 cm. Find the nature and position of the image.

Solution. First of all we will find out the position of image which is given by the image distance v .

Here, Object distance, $u = -50$ cm (It is to the left of lens)

Image distance, $v = ?$ (To be calculated)

Focal length, $f = -20$ cm (It is a concave lens)

Putting these values in the lens formula :

$$\begin{aligned} & \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \\ \text{we get : } & \frac{1}{v} - \frac{1}{-50} = \frac{1}{-20} \\ \text{or } & \frac{1}{v} + \frac{1}{50} = -\frac{1}{20} \\ \text{or } & \frac{1}{v} = -\frac{1}{20} - \frac{1}{50} \\ \text{or } & \frac{1}{v} = \frac{-5-2}{100} \\ \text{or } & \frac{1}{v} = \frac{-7}{100} \\ \text{or } & v = -\frac{100}{7} \end{aligned}$$

So, Image distance, $v = -14.3$ cm

Thus, the image is formed at a distance of 14.3 cm from the concave lens. The minus sign for image distance shows that the image is formed on the left side of the concave lens. We know that a concave lens always forms a virtual and erect image, so the nature of image is virtual and erect.

Sample Problem 2. An object placed 50 cm from a lens produces a virtual image at a distance of 10 cm in front of the lens. Draw a diagram to show the formation of image. Calculate focal length of the lens and magnification produced.

Solution. First of all we will find out the focal length of the lens. We know that the object is always placed in front of the lens on the left side, so the object distance is always taken as negative. Here the image is also formed in front of the lens on the left side, so the image distance will also be negative. Thus,

Object distance, $u = -50$ cm (To the left of lens)

Image distance, $v = -10$ cm (To the left of lens)

Focal length, $f = ?$ (To be calculated)

Putting these values in the lens formula :

$$\begin{aligned} \frac{1}{v} - \frac{1}{u} &= \frac{1}{f} \\ \text{we get : } \frac{1}{-10} - \frac{1}{-50} &= \frac{1}{f} \\ -\frac{1}{10} + \frac{1}{50} &= \frac{1}{f} \\ \frac{-5+1}{50} &= \frac{1}{f} \\ -\frac{4}{50} &= \frac{1}{f} \\ f &= -\frac{50}{4} \end{aligned}$$

So, Focal length, $f = -12.5$ cm

The minus sign for focal length shows that it is a concave lens. Please draw the ray diagram yourself.

We will now calculate the magnification produced by concave lens. We know that for a lens :

$$\text{Magnification, } m = \frac{v}{u}$$

Here, Image distance, $v = -10$ cm

And, Object distance, $u = -50$ cm

$$\text{So, } m = \frac{-10}{-50}$$

$$m = +\frac{1}{5}$$

$$m = +0.2$$

Thus, the magnification produced by this concave lens is $+0.2$. Since the value of magnification is less than 1 (it is 0.2), therefore, the image is smaller than the object (or diminished). The plus sign for the magnification shows that the image is virtual and erect.

Sample Problem 3. The magnification produced by a spherical lens is $+0.75$. What is the :

(a) nature of image ?

(b) nature of lens ?

Answer. (a) When the magnification is positive, the nature of image is virtual and erect. In this case, the magnification is positive, so the nature of image is virtual and erect.

(b) The value of magnification given here is 0.75 (which is less than 1), so the image is smaller than the object or diminished. A virtual, erect and diminished image can be formed only by a concave lens, so the nature of lens is concave.

We are now in position to **answer the following questions and problems :**

Very Short Answer Type Questions

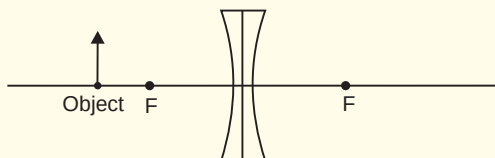
- The lens A produces a magnification of, -0.6 whereas lens B produces a magnification of $+0.6$.
 (a) What is the nature of lens A ?
 (b) What is the nature of lens B ?
- A 50 cm tall object is at a very large distance from a diverging lens. A virtual, erect and diminished image of the object is formed at a distance of 20 cm in front of the lens. How much is the focal length of the lens ?

Short Answer Type Questions

- An object is placed at a distance of 4 cm from a concave lens of focal length 12 cm. Find the position and nature of the image.
- A concave lens of focal length 15 cm forms an image 10 cm from the lens. How far is the object placed from the lens? Draw the ray-diagram.
- An object 60 cm from a lens gives a virtual image at a distance of 20 cm in front of the lens. What is the focal length of the lens? Is the lens converging or diverging? Give reasons for your answer.
- A concave lens of 20 cm focal length forms an image 15 cm from the lens. Compute the object distance.
- A concave lens has focal length 15 cm. At what distance should the object from the lens be placed so that it forms an image at 10 cm from the lens? Also find the magnification produced by the lens.
- Calculate the image distance for an object of height 12 mm at a distance of 0.20 m from a concave lens of focal length 0.30 m, and state the nature and size of the image.
- A concave lens has a focal length of 20 cm. At what distance from the lens a 5 cm tall object be placed so that it forms an image at 15 cm from the lens? Also calculate the size of the image formed.
- An object is placed 20 cm from (a) a converging lens, and (b) a diverging lens, of focal length 15 cm. Calculate the image position and magnification in each case.
- A 2.0 cm tall object is placed 40 cm from a diverging lens of focal length 15 cm. Find the position and size of the image.

Long Answer Type Questions

- (a) Find the position and size of the virtual image formed when an object 2 cm tall is placed 20 cm from :
 (i) a diverging lens of focal length 40 cm.
 (ii) a converging lens of focal length 40 cm.
 (b) Draw labelled ray diagrams to show the formation of images in cases (i) and (ii) above (The diagrams may not be according to scale).
- (a) A small object is placed 150 mm away from a diverging lens of focal length 100 mm.
 (i) Copy the figure below and draw rays to show how an image is formed by the lens.



- Calculate the distance of the image from the lens by using the lens formula.
- The diverging lens in part (a) is replaced by a converging lens also of focal length 100 mm. The object remains in the same position and an image is formed by the converging lens. Compare two properties of this image with those of the image formed by the diverging lens in part (a).

Multiple Choice Questions (MCQs)

- A concave lens produces an image 20 cm from the lens of an object placed 30 cm from the lens. The focal length of the lens is :
 (a) 50 cm (b) 40 cm (c) 60 cm (d) 30 cm
- Only one of the following applies to a concave lens. This is :
 (a) focal length is positive (b) image distance can be positive or negative
 (c) height of image can be positive or negative (d) image distance is always negative
- The magnification produced by a spherical mirror and a spherical lens is + 0.8.
 (a) The mirror and lens are both convex (b) The mirror and lens are both concave
 (c) The mirror is concave but the lens is convex (d) The mirror is convex but the lens is concave
- The magnification produced by a spherical lens and a spherical mirror is + 2.0.
 (a) The lens and mirror are both concave (b) The lens and mirror are both convex
 (c) The lens is convex but the mirror is concave (d) The lens is concave but the mirror is convex

Questions Based on High Order Thinking Skills (HOTS)

18. A camera fitted with a lens of focal length 50 mm is being used to photograph a flower that is 5 cm in diameter. The flower is placed 20 cm in front of the camera lens.
 - (a) At what distance from the film should the lens be adjusted to obtain a sharp image of the flower ?
 - (b) What would be the diameter of the image of the flower on the film ?
 - (c) What is the nature of camera lens ?
19. An object is 2 m from a lens which forms an erect image one-fourth (exactly) the size of the object. Determine the focal length of the lens. What type of lens is this ?
20. An image formed on a screen is three times the size of the object. The object and screen are 80 cm apart when the image is sharply focussed.
 - (a) State which type of lens is used.
 - (b) Calculate focal length of the lens.

ANSWERS

1. (a) Convex lens (b) Concave lens 2. 20 cm 3. $v = -3$ cm ; The image is formed at 3 cm in front of the concave lens (on its left side) ; Virtual and erect 4. $u = -30$ cm ; The object is at 30 cm from the concave lens on its left side 5. $f = -30$ cm ; Diverging lens : Negative sign of focal length 6. 60 cm on left side of lens 7. $u = -30$ cm ; $m = +0.33$ 8. $v = -0.12$ m ; Virtual and erect ; 7.2 mm tall 9. $u = -60$ cm ; 1.25 cm 10. (a) $v = +60$ cm ; $m = -3$ (b) $v = -8.5$ cm ; $m = +0.42$; 11. $v = -10.90$ cm ; 0.54 cm tall 12. (a) (i) $v = -13.3$ cm ; 1.33 cm tall (ii) $v = -40$ cm ; 4 cm tall 13. (a) (ii) $v = -60$ mm (b) $v = +300$ mm ; The image formed by converging lens is real, inverted and magnified (2 times). It is formed behind the converging lens. On the other hand, the image formed by diverging lens is virtual, erect and diminished. It is formed in front of the diverging lens. 14. (c) 15. (d) 16. (d) 17. (c) 18. (a) $v = +6.66$ cm ; The film should be at a distance of 6.66 cm behind the camera lens (b) 1.66 cm (c) Convex lens 19. $f = -66.7$ cm ; Concave lens 20. (a) Convex lens (b) + 15 cm

POWER OF A LENS

A convex lens converges the light rays falling on it whereas a concave lens diverges the light rays falling on it. **The power of a lens is a measure of the degree of convergence or divergence of light rays falling on it.** If a convex lens converges a beam of parallel light rays more strongly by focussing them closer to the optical centre, it is said to have greater power (than another convex lens which focusses the same parallel light rays at a greater distance from the optical centre). Similarly, a concave lens which diverges a parallel beam of light rays more strongly is said to have a greater diverging power than another concave lens which diverges the light rays less strongly. *The power of a lens depends on its focal length.* We can define the power of a lens as follows :

The power of a lens is defined as the reciprocal of its focal length in metres.

Thus, Power of a lens = $\frac{1}{\text{focal length of the lens (in metres)}}$

$$\text{or } P = \frac{1}{f}$$

where P = Power of the lens
and f = focal length of the lens (in metres)

Since the power of a lens is *inversely proportional* to its focal length, therefore, **a lens of short focal length has more power whereas a lens of long focal length has less power.** For example, a lens of 5 cm focal length will have more power than a lens of 20 cm focal length. A more powerful lens is one that bends the light rays more ; and has a shorter focal length (see Figure 48).

The unit of the power of a lens is dioptre, which is denoted by the letter D. **One dioptre is the power of a lens whose focal length is 1 metre.** The power of a lens can be measured directly by using an instrument called dioptrimeter. It is used by opticians to measure the power of spectacle lenses (see Figure 49). A convex lens has a positive focal length, so **the power of a convex lens is positive** (and written with a

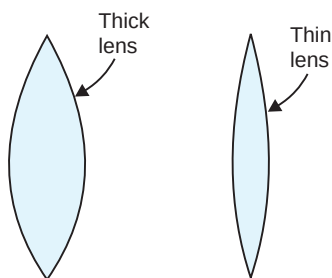


Figure 48. A thick lens has short focal length but more power. A thin lens has large focal length but less power.



Figure 49. An optician measuring the power of a spectacle lens by using an instrument called dioptrimeter.

+ sign). A concave lens has a negative focal length, so **the power of a concave lens is negative** [and written with a minus (–) sign]. Since a convex lens is also known as a converging lens and a concave lens is also known as a diverging lens, we can also say that the power of a converging lens is positive whereas the power of a diverging lens is negative. When an optician, after testing the eyes of a person, prescribes the corrective lenses of say, + 2.0 D and + 1.5 D for the left eye and right eye, respectively, he actually refers to the power of convex lenses required for making eye-glasses or spectacles to make him see clearly.

In order to calculate the power of a lens, we need its focal length in metres. In many problems, the focal length of a lens is usually given to us in 'centimetres'. So, to calculate the power of such a lens, we should first convert the focal length of the lens into 'metres' by dividing the given 'centimetre' value by 100. If, however, the focal length is already in metres, then there is no need to change it. We will now solve some problems based on the calculation of power of lenses.

Sample Problem 1. A convex lens is of focal length 10 cm. What is its power ?

Solution. Here, the focal length of the lens is given in 'centimetres' so to calculate the power of this lens, we should first convert the focal length into 'metres' because our formula uses the focal length in 'metres'.

$$\begin{aligned}\text{Now, } 10 \text{ cm} &= \frac{10}{100} \text{ m} \\ &= 0.1 \text{ m}\end{aligned}$$

$$\text{So, Focal length, } f = 0.1 \text{ m} \quad (\text{A convex lens has positive focal length})$$

Now, putting this value of focal length in the formula for the power of a lens :

$$P = \frac{1}{f \text{ (in metres)}}$$

$$\text{we get : } P = \frac{1}{0.1}$$

$$\text{or } P = \frac{1 \times 10}{1}$$

$$\text{Thus, Power, } P = +10 \text{ dioptres} \quad (\text{or } +10 \text{ D})$$

Thus, the power of this convex lens is +10 dioptres which is also written as +10 D. The plus sign with the power indicates that it is a converging lens or convex lens.

Sample Problem 2. A person having a myopic eye uses a concave lens of focal length 50 cm. What is the power of the lens ?

Solution. Here we have a concave lens. Now, the focal length of a concave lens is considered negative, so it is to be written with a minus sign.

Thus, Focal length, $f = -50$ cm

$$= -\frac{50}{100} \text{ m}$$

$$= -0.5 \text{ m}$$

Now, Power, $P = \frac{1}{f \text{ (in metres)}}$

$$P = \frac{1}{-0.5}$$

$$P = -\frac{1 \times 10}{5}$$

$$P = -2 \text{ dioptres (or } -2 \text{ D)}$$

Thus, the power of this concave lens is, -2 dioptres which can also be written as, -2 D. The minus sign with the power indicates that it is a diverging lens or concave lens.

Sample Problem 3. A thin lens has a focal length of, -25 cm. What is the power of the lens and what is its nature ?

Solution. Since the focal length is negative, it is a concave lens or diverging lens. Calculate the power yourself as shown in the above question. The power will be, -4 D.

Sample Problem 4. The power of a lens is $+2.5$ D. What kind of lens it is and what is its focal length ?

Solution. The power of this lens has positive sign, so it is a convex lens. Now,

$$\text{Power, } P = \frac{1}{f \text{ (in metres)}}$$

$$\text{So, } +2.5 = \frac{1}{f}$$

$$\text{and } f = \frac{1}{2.5} \text{ m}$$

$$= \frac{1}{2.5} \times 100 \text{ cm}$$

$$\text{So, Focal length, } f = 40 \text{ cm (or } +40 \text{ cm)}$$

Sample Problem 5. A lens has a power of, -2.5 D. What is the focal length and nature of the lens ?

Solution. The power of this lens has minus sign, so it is a concave lens. Calculate the focal length yourself as shown in the above question. The focal length will be, -40 cm.

Sample Problem 6. Find the power of a concave lens of focal length 2 m. (NCERT Book Question)

Solution. A concave lens has negative focal length, so it is to be written with a minus sign. Thus,

$$\text{Focal length, } f = -2 \text{ m (It is in metres)}$$

$$\text{Now, Power, } P = \frac{1}{f \text{ (in metres)}}$$

$$P = \frac{1}{-2}$$

$$P = -0.5 \text{ D}$$

Thus, the power of this concave lens is, -0.5 dioptre.

Sample Problem 7. A convex lens forms a real and inverted image of a needle at a distance of 50 cm from the lens. If the image is of the same size as the needle, where is the needle placed in front of the lens? Also, find the power of the lens. (NCERT Book Question)

Solution. (i) In this case needle is the object. Since the image is real, inverted and of same size as the needle (or object), the needle must be at the same distance (50 cm) in front of lens, as the image is behind the lens. Thus, the needle is placed at a distance of 50 cm from lens in the front.

(ii) When the image formed by a convex lens is of the same size as the needle (or object), then the distance of needle from the lens is $2f$ (twice the focal length). In this case :

$$2f = 50 \text{ cm}$$

$$\text{So, } f = \frac{50}{2} \text{ cm}$$

$$f = 25 \text{ cm}$$

Thus, the focal length of this convex lens is +25 cm. This is equal to $\frac{+25}{100}$ m or +0.25 m. Now,

$$\text{Power, } P = \frac{1}{f \text{ (in metres)}}$$

$$= \frac{1}{+0.25} = +4.0 \text{ D}$$

So, the power of this convex lens is +4.0 dioptres.

Power of a Combination of Lenses

If a number of lenses are placed in close contact, then the power of the combination of lenses is equal to the algebraic sum of the powers of individual lenses. Thus, if two lenses of powers p_1 and p_2 are placed in contact with each other, then their resultant power P is given by :

$$P = p_1 + p_2$$

For example, if a convex lens of power, +4 D and a concave lens of power, -10 D are placed in contact with each other, then their resultant power will be :

$$P = p_1 + p_2$$

$$= +4 + (-10)$$

$$= 4 - 10$$

$$= -6 \text{ D}$$

This shows that a combination of convex lens of power +4 D and a concave lens of power, -10 D has a resultant power of, -6 D. So, this combination of convex lens and concave lens behaves like a concave lens (of power, -6 dioptres). This is shown clearly in the Figure given below :

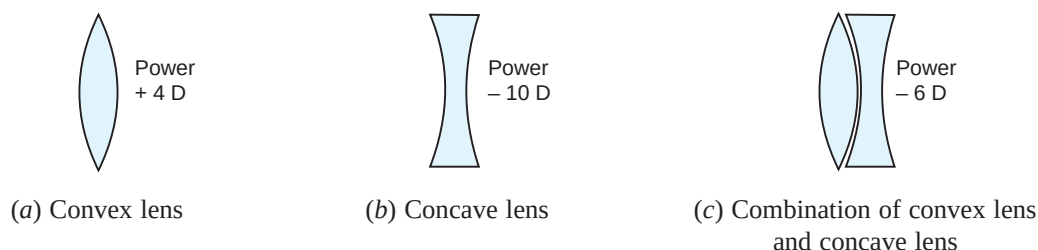


Figure 50. A combination of convex lens of power +4 D and a concave lens of power, -10 D kept in close contact has a resultant power of, -6 D.

In general, if a number of thin lenses having powers $p_1, p_2, p_3, \dots$ etc., are placed in close contact with one another, then their resultant power P is given by :

$$P = p_1 + p_2 + p_3 + \dots$$

Please note that the individual powers p_1, p_2, p_3 , etc., of the lenses should be put in the above formula with their proper signs.

The use of powers of lenses (instead of their focal lengths) makes the work of opticians very simple and straightforward. For example, when an optician places two convex lenses of powers +2.0 D and +0.25 D in

front of a person's eye during eye-testing, he immediately knows that this convex lens combination is equivalent to a single convex lens of power +2.25 D.

The lens systems consisting of several lenses in contact are used in designing the optical instruments like cameras, microscopes and telescopes, etc. The use of a combination of lenses increases the sharpness of the image. The image produced by using a combination of lenses is also free from many defects which otherwise occur while using a single lens. We will now solve a problem based on combination of lenses.

Sample Problem. Two thin lenses of power, +3.5 D and, -2.5 D are placed in contact. Find the power and focal length of the lens combination. **(NCERT Book Question)**

Solution. We know that :

Power of combination of lenses, $P = p_1 + p_2$

$$\text{So, } P = +3.5 + (-2.5)$$

$$P = +3.5 - 2.5$$

$$P = +1.0 \text{ D}$$

Thus, the power of this combination of lenses is, +1.0 dioptre.

We will now calculate the focal length of this combination of lenses. We know that :

$$\text{Power, } P = \frac{1}{f}$$

$$\text{or } +1 = \frac{1}{f}$$

$$\text{And, } f = +1 \text{ m}$$

So, the focal length of this combination of lenses is, +1 metre.

We are now in a position to **answer the following questions and problems :**

Very Short Answer Type Questions

1. The lens A has a focal length of 25 cm whereas another lens B has a focal length of 60 cm. Giving reason state, which lens has more power : A or B.
2. Which causes more bending (or more refraction) of light rays passing through it : a convex lens of long focal length or a convex lens of short focal length ?
3. Name the physical quantity whose unit is dioptre.
4. Define 1 dioptre power of a lens.
5. Which type of lens has (a) a positive power, and (b) a negative power ?
6. Which of the two has a greater power : a lens of short focal length or a lens of large focal length ?
7. How is the power of a lens related to its focal length ?
8. Which has more power : a thick convex lens or a thin convex lens, made of the same glass ? Give reason for your choice.
9. The focal length of a convex lens is 25 cm. What is its power ?
10. What is the power of a convex lens of focal length 0.5 m ?
11. A converging lens has a focal length of 50 mm. What is the power of the lens ?
12. What is the power of a convex lens whose focal length is 80 cm ?
13. A diverging lens has a focal length of 3 cm. Calculate the power.
14. The power of a lens is + 0.2 D. Calculate its focal length.
15. The power of a lens is, - 2 D. What is its focal length ?
16. What is the nature of a lens having a power of + 0.5 D ?
17. What is the nature of a lens whose power is, - 4 D ?
18. The optician's prescription for a spectacle lens is marked + 0.5 D. What is the :
(a) nature of spectacle lens ?
(b) focal length of spectacle lens ?
19. A doctor has prescribed a corrective lens of power, -1.5 D. Find the focal length of the lens. Is the prescribed lens diverging or converging ?

20. A lens has a focal length of, -10 cm. What is the power of the lens and what is its nature ?
21. The focal length of a lens is $+150$ mm. What kind of lens is it and what is its power ?
22. Fill in the following blanks with suitable words :
 - (a) The reciprocal of the focal length in metres gives you theof the lens, which is measured in.....
 - (b) For converging lenses, the power is.....while for diverging lenses, the power is.....

Short Answer Type Questions

23. An object of height 4 cm is placed at a distance of 15 cm in front of a concave lens of power, -10 dioptres. Find the size of the image.
24. An object of height 4.25 mm is placed at a distance of 10 cm from a convex lens of power $+5$ D. Find (i) focal length of the lens, and (ii) size of the image.
25. A convex lens of power 5 D and a concave lens of power 7.5 D are placed in contact with each other. What is the :
 - (a) power of this combination of lenses ?
 - (b) focal length of this combination of lenses ?
26. A convex lens of focal length 25 cm and a concave lens of focal length 10 cm are placed in close contact with one another.
 - (a) What is the power of this combination ?
 - (b) What is the focal length of this combination ?
 - (c) Is this combination converging or diverging ?
27. The power of a combination of two lenses X and Y is 5 D. If the focal length of lens X be 15 cm :
 - (a) calculate the focal length of lens Y .
 - (b) state the nature of lens Y .
28. Two lenses A and B have focal lengths of $+20$ cm and, -10 cm, respectively.
 - (a) What is the nature of lens A and lens B ?
 - (b) What is the power of lens A and lens B ?
 - (c) What is the power of combination if lenses A and B are held close together ?

Long Answer Type Questions

29. (a) What do you understand by the power of a lens ? Name one factor on which the power of a lens depends.
 (b) What is the unit of power of a lens ? Define the unit of power of a lens.
 (c) A combination of lenses for a camera contains two converging lenses of focal lengths 20 cm and 40 cm and a diverging lens of focal length 50 cm. Find the power and focal length of the combination.
30. (a) Two lenses A and B have power of (i) $+2$ D and (ii) -4 D respectively. What is the nature and focal length of each lens ?
 (b) An object is placed at a distance of 100 cm from each of the above lenses A and B . Calculate (i) image distance, and (ii) magnification, in each of the two cases.

Multiple Choice Questions (MCQs)

31. The focal lengths of four convex lenses P , Q , R and S are 20 cm, 15 cm, 5 cm and 10 cm, respectively. The lens having greatest power is :
 - (a) P (b) Q (c) R (d) S
32. A converging lens has a focal length of 50 cm. The power of this lens is :
 - (a) $+0.2$ D (b) -2.0 D (c) $+2.0$ D (d) -0.2 D
33. A diverging lens has a focal length of 0.10 m. The power of this lens will be :
 - (a) $+10.0$ D (b) $+1.0$ D (c) -1.0 D (d) -10.0 D
34. The power of a lens is $+2.0$ D. Its focal length should be :
 - (a) 100 cm (b) 50 cm (c) 25 cm (d) 40 cm
35. If a spherical lens has a power of, -0.25 D, the focal length of this lens will be :
 - (a) -4 cm (b) -400 mm (c) -4 m (d) -40 m

36. The power of a concave lens is 10 D and that of a convex lens is 6 D. When these two lenses are placed in contact with each other, the power of their combination will be :
 (a) + 16 D (b) + 4 D (c) -16 D (d) - 4 D
37. The power of a converging lens is 4.5 D and that of a diverging lens is 3 D. The power of this combination of lenses placed close together is :
 (a) + 1.5 D (b) + 7.5 D (c) - 7.5 D (d) - 1.5 D
38. A convex lens of focal length 10 cm is placed in contact with a concave lens of focal length 20 cm. The focal length of this combination of lenses will be :
 (a) + 10 cm (b) + 20 cm (c) - 10 cm (d) - 20 cm

Questions Based on High Order Thinking Skills (HOTS)

39. The optical prescription for a pair of spectacles is :
 Right eye : - 3.50 D Left eye : - 4.00 D
 (a) Are these lenses thinner at the middle or at the edges ?
 (b) Which lens has a greater focal length ?
 (c) Which is the weaker eye ?
40. A person got his eyes tested by an optician. The prescription for the spectacle lenses to be made reads :
 Left eye : + 2.50 D Right eye : + 2.00 D
 (a) State whether these lenses are thicker in the middle or at the edges.
 (b) Which lens bends the light rays more strongly ?
 (c) State whether these spectacle lenses will converge light rays or diverge light rays.

ANSWERS

1. Lens A has more power ; It has shorter focal length. 2. Convex lens of short focal length 6. Lens of short focal length 8. Thick convex lens ; It has shorter focal length 9. + 4 D 10. + 2 D
 11. + 20 D 12. + 1.25 D 13. - 33.3 D 14. + 5 m 15. - 50 cm 16. Convex lens 17. Concave lens
 18. (a) Convex lens (b) 2 m 19. - 66.6 cm ; Diverging lens 20. -10 D ; Concave lens 21. Convex lens ; + 6.6 D 22. (a) power ; dioptres (b) positive ; negative 23. 1.6 cm 24. (i) 20 cm (ii) 8.50 mm
 25. (a) -2.5 D (b) - 40 cm 26. (a) - 6 D (b) -16.66 cm (c) Diverging 27. (a) - 60 cm (b) Concave lens
 28. (a) Lens A is convex ; Lens B is concave (b) Power of lens A = + 5 D ; Power of lens B = - 10 D (c) - 5 D
 29. (c) $P = + 5.5 \text{ D}$; $f = + 18.18 \text{ cm}$ 30. (a) Lens A is convex ; $f = + 50 \text{ cm}$; Lens B is concave ; $f = - 25 \text{ cm}$
 (b) Lens A : $v = +100 \text{ cm}$; $m = - 1$; Lens B : $v = -20 \text{ cm}$; $m = 0.2$ 31. (c)
 32. (c) 33. (d) 34. (b) 35. (c) 36. (d) 37. (a) 38. (b) 39. (a) Thinner at the middle (b) Lens of lower power : - 3.50 D (c) Left eye 40. (a) Thicker in the middle (b) Lens having greater power of + 2.50 D (c) Converge light rays.